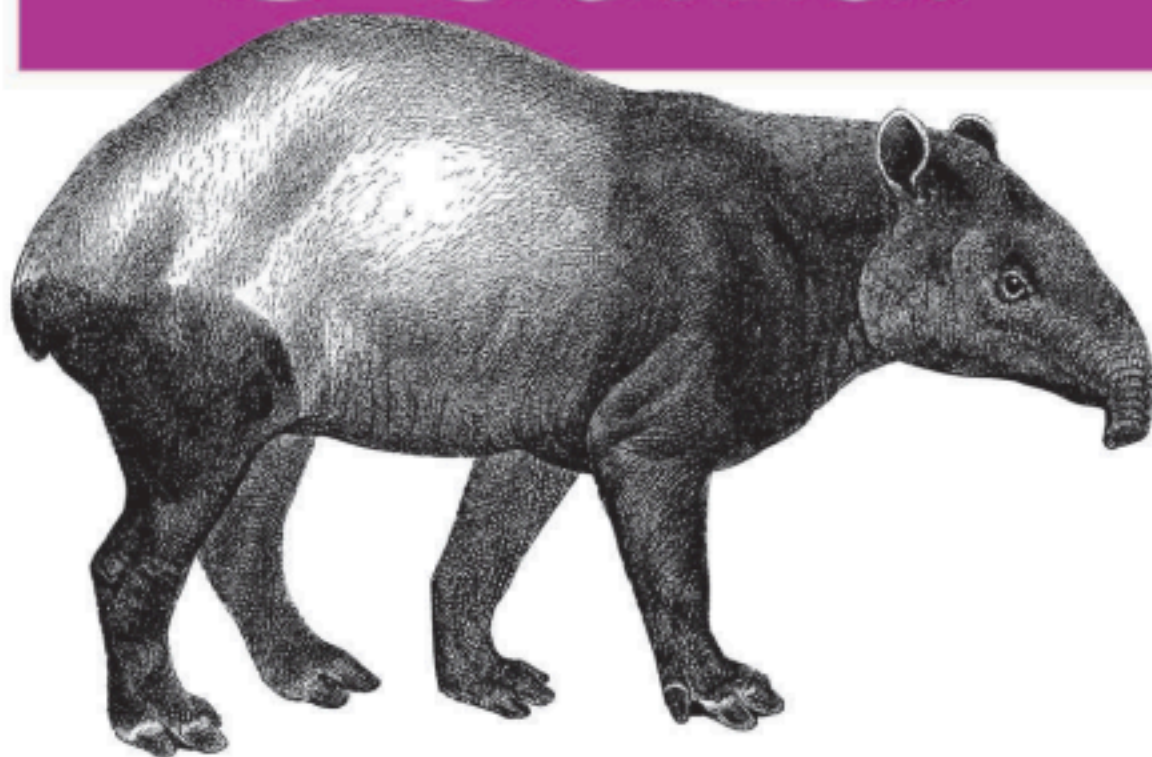


Scalability = Functional Programming + Objects

Programming

Scala



O'REILLY®

Dean Wampler & Alex Payne

Programming Scala

Learn how to be more productive with Scala, a new multi-paradigm language for the Java Virtual Machine (JVM) that integrates features of both object-oriented and functional programming. With this book, you'll discover why Scala is ideal for highly scalable, component-based applications that support concurrency and distribution.

Programming Scala clearly explains the advantages of Scala as a JVM language. You'll learn how to leverage the wealth of Java class libraries to meet the practical needs of enterprise and Internet projects more easily. Packed with code examples, this book provides useful information on Scala's command-line tools, third-party tools, libraries, and available language-aware plugins for editors and IDEs.

- Learn how Scala's succinct and flexible code helps you program faster
- Discover the notable improvements Scala offers over Java's object model
- Get a concise overview of functional programming, and learn how Scala's support for it offers a better approach to concurrency
- Know how to use mixin composition with traits, pattern matching, concurrency with Actors, and other essential features
- Take advantage of Scala's built-in support for XML
- Learn how to develop domain-specific languages
- Understand the basics of designing test-driven Scala applications

"Dean and Alex provide in-depth and insightful coverage of Scala. The first few chapters give an intense, get-up-to-speed overview of the entire language. Subsequent chapters dive into topics deeply enough to give you a firm understanding of what Scala programming is all about. This book is a great complement to the book I wrote with Martin Odersky and Lex Spoon for anyone who wants to use Scala in real-world projects."

—Bill Venners
coauthor of *Programming in Scala* (Artima)

Dean Wampler is a consultant, trainer, and mentor with Object Mentor, Inc. He specializes in Scala, Java, and Ruby.

Alex Payne is Platform Lead at Twitter, Inc. He studies and speaks about the design and implementation of programming languages and APIs.



Experience with one or more object-oriented languages, such as Java, C#, or Ruby, is recommended.

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EHOLHYHG LQ PH 7R \$QQ ZKR ZDV

DOZD\ WKHUH IRU PH ¢'HDQ

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DQG WKH DFFRPSDQLQJ

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ZKLFK WR DWWHPSW WR SURGXFH
LW
7R .ULVWHQ IRU KHU XQHQGLQJ
SDWLHQFH ORYH DQG NLQGQHVV
ø\$OH[

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1. Zero to Sixty: Introducing Scala

..... **1** :K\ 6FDOD" ,I <RX \$UH D -DYD
 3URJUDPPHU§ ,I <RX \$UH D 5XE\ 3\WKRQ HWF 3URJUDPPHU§
 ,QWURGXFLLQJ 6FDOD 7KH 6HGXFWRQV RI 6FDOD ,QVWDOOLQJ
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2. Type Less, Do More

..... **23** ,Q 7KLV &KDSWHU 6HPLFRORQV
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 ,QIHUULQJ 7SH ,QIRUPDWLRQ /LWHUDOV ,QWHJU
 /LWHUDOV)ORDWLQJ 3RLQW /LWHUDOV %RROHDQ
 /LWHUDOV &KDUDFWHU /LWHUDOV 6WULQJ /LWHUDOV
 6\PERO /LWHUDOV

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 ,PSRUWV DUH 5HODWLYH \$EVWUDFW 7SHV \$QG
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 :KDW\ 1H[W

3. Rounding Out the Essentials

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 (QXPHUDWLRQV 5HFDS DQG :KDW\ 1H[W

4. Traits

..... **75** ,QWURGXFLLQJ 7UDLWV 7UDLWV \$V
 0L[LQV

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6WDFNDEOH 7UDLWV &RQVWUXFWLQJ 7UDLWV &ODVV RU
7UDLW" 5HFDS DQG :KDW\|V 1H\|W

5. Basic Object-Oriented Programming in Scala

. **89** &ODVV DQG 2EMHFW %DVLFV
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&DOOLQJ 3DUHQW &ODVV &RQVWUXFWRUV 1HVWHG
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6. Advanced Object-Oriented Programming In Scala

. **111** 2YHUULGLQJ 0HPEHUV RI &ODVVHV
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2YHUULGLQJ \$EVWUDFW DQG &RQFUHWH)LHOGV LQ &ODVVHV
2YHUULGLQJ \$EVWUDFW 7\SHV
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8QDSSO\6HT IRU &ROOHFWLRQV &RPSDQLRQ 2EMHFWV DQG
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IRU %LQDU\ 2SHUDWLRQV 7KH FRS\ 0HWKRG LQ 6FDOD
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2EMHFWV 7KH HTXDOV 0HWKRG 7KH DQG 0HWKRGV
7KH QH DQG HT 0HWKRGV \$UUD\ (TXDOLW DQG WKH
VDPH(OHPHQWV 0HWKRG

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5HFDS DQG :KDW\|V 1H\|W

7. The Scala Object System

. **145** 7KH 3UHGH 2EMHFW &ODVVHV
DQG 2EMHFWV :KHUH \$UH WKH 6WDWLWLFV"
3DFNDJH 2EMHFWV 6HDOHG &ODVV +LHUDUFKLHV 7KH
6FDOD 7\SH +LHUDUFK\ /LQHDULJDWLRQ RI DQ 2EMHFW\|V
+LHUDUFK\ 5HFDS DQG :KDW\|V 1H\|W

8. Functional Programming in Scala

. **165** :KDW ,V)XQFWLRQDO 3URJUDPPQLQJ"
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)XQFWLRQ 3DUDPHWHUV)LQDO 7KRXJKWV RQ ,PSOLFLWV
&DOO E\ 1DPH &DOO E\ 9DOXH /D\ 9DOV 5HFDS)XQFWLRQDO
&RPSRQHQW \$EVWUDFWLRQV

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9. Robust, Scalable Concurrency with Actors

. **193** 7KH 3UREOHPV RI 6KDUHG
6\QFKURQLJHG 6WDWH \$FWRUV
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(IIHFWLYH \$FWRUV 7UDGLWLRQDO &RQFXUUHQF\ LQ 6FDOD
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XWLO FRQFXUUHQW (YHVV 5HFDS DQG :KDW\ 1H[W

10. Herding XML in Scala

. **211** 5HDGLQJ ;0/ ([SORULQJ ;0/ /RRSLQJ
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DQG :KDW\ 1H[W

11. Domain-Specific Languages in Scala

. **217** ,QWHUQDO '6/ \$ 3D\UROO
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&RQYHUVLRQV DQG 8VHU 'HILQH 7\SHV \$\$SO\ 0HWKRGV
3D\UROO 5XOHV '6/ ,PSOHPHQWDWLRQ ,QWHUQDO '6/)LQDO
7KRXJKWV ([WHUQDO '6/ ZLWK 3DUVHU &RPELDWRUV \$ERXW
3DUVHU &RPELDWRUV \$ 3D\UROO ([WHUQDO '6/ \$ 6FDOD
,PSOHPHQWDWLRQ RI WKH ([WHUQDO '6/ *UDPPDU *HQHUDWLQJ
3D\FKHFNV ZLWK WKH ([WHUQDO '6/ ,QWHUQDO 9HUVXV
([WHUQDO '6/)LQDO 7KRXJKWV 5HFDS DQG :KDW\ 1H[W

12. The Scala Type System

. **247** 5HIOHFWLQJ RQ 7\SHV
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 9DULDQFH RI 0XWDEOH 7ASHV 9DULDQFH ,Q 6FDOD 9HUVXV
 -DYD ,PSOHPHQWDWLRQ 1RWHV 7ASH %RXQGV 8SSHU
 7ASH %RXQGV /RZHU 7ASH %RXQGV \$ &ORVHU /RRN DW
 /LVWV 9LHZV DQG 9LHZ %RXQGV 1RWKLQJ DQG 1XOO
 8QGHUVWDQGLQJ \$EVWUDFW 7ASHV 3DUDPHWHULJHG 7ASHV
 9HUVXV \$EVWUDFW 7ASHV 3DWK 'HSHQGHQW 7ASHV & WKLV
 & VXSHU SDWK [9DOXH 7ASHV 7ASH 'HVLJQDWURUV
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 :KDW!V 1H[W

13. Application Design

. **289** \$QQRWDWLRQV (QXPHUDWLRQV
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 &DNH 3DWWHUQ %HWWHU 'HVLJQ ZLWK 'HVLJQ %\
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14. Scala Tools, Libraries, and IDE Support

. **343** &RPPDQG /LQH 7RROV VFDODF
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 DQG -DYD 3ULPLWLYHV 6FDOD 1DPHV LQ -DYD &RGH -DYD
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7HUUDFRWWD +DGRRS 5HFDS DQG :KDW|V 1H[W

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,I WKHUH KDV EHHQ D FRPPRQ WKHPH WKURXJKRXW P\ FDUHHU DV
D SURJUDPPHU LW KDV EHHQ WKH TXHVW IRU EHWWHU
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OHDGLQJ UROH LQ PRLQJ 7ZLWWHU\ V FRGH DQG
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FRPSDQLHV WR VXFFHVVIXOO\ GHSOR\ 6FDOD LQ SURGXFWLRQ

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\RXU 6FDOD FRGH KRZ WR XVH 6FDOD ZLWK SRSXODU IUDPHZRUNV
VXFK DV 6SULQJ +DGRRS DQG 7HUUDFRWW DQG PXFK PXFK

PRUH (QMR\ WKH ULGH , VXUH GLG

¢-RQDV %RQpU
,QGSHQGHQW &RQVXOWDQW 6FDODEOH 6ROXWLRQV \$%
\$XJXVW

Preface

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WHDFKHV \RX KRZ WR EH SURGXFWLYH ZLWK 6FDOD TXLFNO\ DQG
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/HDUQ PRUH DW [KWWS SURJUDPPLQJVFDOD FRP](#) RU DW WKH ERRN\ V [FDWDORJ SDJH](#)

Welcome to Programming Scala

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SDUWLFXODU ODQJXDJH RU RQH LV LQVWLWXWLRQDO\ JHG E\ D
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:LQGRZV SURJUDPPHUV XVH & DQG 1(7 ODQJXDJHV ORVW
HPEHGGHG V\WHPV GHYHORSHUV XVH & DQG &

6RPHWLPHV SRSXODULW\ GHULYHG IURP WHFKQLFDO PHULW JLYHV
ZD\ WR IDVKLRQ DQG IDQDWL FLVP & -DYD DQG 5XE\ KDYH
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SURJUDPPHUV

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WKH QHHGV RI LWV HUD -DYD ZDV LQLWLDOO\ VHHQ DV D
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DSSOLFDWLRQV 6PDOOWDON FDSWXUHG WKH HVVHQFH RI
REMHFW RULHQWHG SURJUDPPLQJ
223 DV WKDW PRGHO RI SUR JUDPPLQJ HQWHUHG WKH
PDLQVWUHP

7RGD\ FRQFXUUHQF\ KHWHURJHQHLW DOZDV RQ VHU\LFHV DQG
HYHU VKULQNLOJ GHYHORS PHQW VFKHGXOHV DUH GULYLQJ
LQWHUHVW LQ IXQFWLRQDO SURJUDPPLQJ
)3 ,W DSSHUV WKDW WKH GRPLQDQFH RI REMHFW RULHQWHG
SURJUDPPLQJ PD\ EH RYHU 0L[LQJ SDUDGLJPV LV EHFRP LQJ
SRSXODU HYHQ QHFHVVDU\

:H JUDYLWDWHG WR 6FDOD IURP RWKHU ODQJXDJHV EHFDXVH
6FDOD HPERGLHV PDQ\ RI WKH RS WLPDO TXDOLWLHV ZH ZDQW LQ
D JHQHUDO SXUSRVH SURJUDPPLQJ ODQJXDJH IRU WKH NLQGV RI
DSSOLFDWLRQV ZH EXLOG WRGD\ UHOLDEOH KLJK SHUIRUPDQFH
KLJKO\ FRQFXUUHQW ,QWHUQHW DQG HQWHUSULVH

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6FDOD LV D PXOWL SDUDGLJP ODQJXDJH VXSSRUWLQJ ERWK
 REMHFW RULHQWHG DQG IXQFWLRQDO SURJUDPPPLQJ DSSURDFKHV
 6FDOD LV VFDODEOH VXLWDEOH IRU HYHU\WKLQJ IURP VKRUW
 VFULSWV XS WR ODUJH VFDOH FRPSRQHQW EDVHG
 DSSOLFDWLRQV 6FDOD LV VRSKLVWLFDFWHG LQFRUSRUDWLQJ
 VWDWH RI WKH DUW LGHDV IURP WKH KDOOV RI FRPSXWHU
 VFLHQFH GHSDUWPHQWV ZRUOGZLGH <HW 6FDOD LV SUDFWLFDO
 ,WV FUHDWRU ODUWLQ 2GHUV\ SDUWLFLSDWHG LQ WKH
 GHYHORSHPHQW RI -DJD IRU \HDUV DQG XQGHUVWDQGV WKH QHHG
 RI SURIHVVLRQDO GHYHORSHU

%RWK RI XV ZHUH VHGXFHG E\ 6FDOD E\ LWV FRQFLVH HOHJDQW
 DQG H[SUHVVLYH \QWD\ DQG E\ WKH EUHDGWK RI WRROV LW SXW
 DW RXU GLVSRVDO ,Q WKLV ERRN ZH VWULYH WR GHPRQVUDWH
 ZK\ DOO WKHVVH TXDOLWLHV PDNH 6FDOD D FRPSHOOLQJ DQG
 LQGLVSHQVDEOH SURJUDPPPLQJ ODQJXDJH

,I \RX DUH DQ H[SHULHQFHG GHYHORSHU ZKR ZDQWV D IDWV
 WKRURXJK LQWURGXFWRQ WR 6FDOD WKLV ERRN LV IRU \RX <RX
 PD\ EH HYDOXDWLQJ 6FDOD DV D UHSODFHPHQW IRU RU
 FRPSOHPHQW WR \RXU FXUUHQW ODQJXDJHV OD\EH \RX KDYH
 DOUHDG\ GHFLGHG WR XVH 6FDOD DQG \RX QHHG WR OHDUQ LWV
 IHDWXUHV DQG KRZ WR XVH LW ZHO (LWKHU ZD\ ZH KRSH WR
 LOOXPLQDWH WKLV SRZHU\XO ODQJXDJH IRU \RX LQ DQ
 DFFHVVLEOH ZD\

:H DVVXPH WKDW \RX DUH ZHO YHUVHG LQ REMHFW RULHQWHG
 SURJUDPPPLQJ EXW ZH GRQ\W DVVXPH WKDW \RX KDYH SULRU
 H[SRVUH WR IXQFWLRQDO SURJUDPPPLQJ :H DVVXPH WKDW \RX
 DUH H[SHULHQFHG LQ RQH RU PRUH RWKHU SURJUDPPPLQJ
 ODQJXDJHV :H GUDZ SDUDOOHOV WR IHDWXUHV LQ -DJD & 5XE\
 DQG RWKHU ODQJXDJHV ,I \RX NQRZ DQ\ RI WKHVVH ODQJXDJHV
 ZH\OO SRLQW RXW VLPLODU IHDWXUHV LQ 6FDOD DV ZHO DV PDQ\
 IHDWXUHV WKDW DUH QHZ

:KWKHU \RX FRPH IURP DQ REMHFW RULHQWHG RU IXQFWLRQDO
 SURJUDPPPLQJ EDFNJURXQG \RX ZLOO VHH KRZ 6FDOD HOHJDQW\
 FRPELQHV ERWK SDUDGLJPV GHPRQVUDWLQJ WKHLU FRP
 SOHPHQWDU\ QDWXUH %DVHG RQ PDQ\ H[DPSOHV \RX ZLOO
 XQGHUVWDQG KRZ DQG ZKHQ WR DSSO\ 223 DQG)3 WHFKQLTXHV
 WR PDQ\ GLIIHUHQW GHVLJQ SUREOHPV

,Q WKH HQG ZH KRSH WKDW \RX WRR ZLOO EH VHGXFHG E\ 6FDOD
 (YHQ LI 6FDOD GRHV QRW HQG XS EHFRPLQJ \RXU GD\ WR GD\
 ODQJXDJH ZH KRSH \RX ZLOO JDLQ LQVLJKWV WKDW \RX FDQ DSSO\
 UHJDUGOHVV RI ZKLFK ODQJXDJH \RX DUH XVLQJ

Conventions Used in This Book

7KH IROORZLQJ W\SRJUDSKLFDO FRQYHQWLRQV DUH XVHG LQ WKLV ERRN

,WDOLF

,QGLFDWHV QHZ WHUPV 85/V HPDLO DGGUHV\HV ILOH QDPHV
DQG ILOH H[WHQVLRQV 0DQ\ LWDOLFLJHG WHUPV DUH GHILQHG
LQ WKH *ORVVDU\ RQ SDJH

Constant width

8VHG IRU SURJUDP OLVWLQJV DV ZHOO DV ZLWKLQ
SDUDJUDSKV WR UHIHU WR SURJUDP HOHPHQWV VXFK DV
YDULDEOH RU IXQFWLRQ QDPHV GDWDEDVHV GDWD WSHV
HQYLURQPHQW YDULDEOHV VWDWHPHQWV DQG NHZRUGV

Constant width bold

6KRZV FRPPDQGV RU RWKHU WH[W WKDW VKRXOG EH WSHG OLWHUDOO\ E\
WKH XVHU

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Constant width italic

6KRZV WH[W WKDW VKRXOG EH UHSODFHG ZLWK XVHU
VXSSOLHG YDOXHV RU E\ YDOXHV GHWHU PLQHG E\ FRQWH[W



7KLW LFRQ VLJQLILHV D WLS VXJJHVWLRQ RU JHQHUDO QRWH



7KLW LFRQ LQGLFDWHV D ZDUQLQJ RU FDXWLRQ

Using Code Examples

7KLW ERRN LV KHUH WR KHOS \RX JHW \RXU MRE GRQH ,Q JHQHUDO
\RX PD\ XVH WKH FRGH LQ WKLV ERRN LQ \RXU SURJUDPV DQG
GRFXPHQDWLRQ <RX GR QRW QHHG WR FRQWDFW XV IRU
SHUPLVVLRQ XQOHVV \RX\UH UHSURGXFHQJ D VLJQLILFDQW
SRUWLRQ RI WKH FRGH)RU H[DPSOH ZULWLQJ D SURJUDP WKDW
XVHV VHYHUDO FKXQNV RI FRGH IURP WKLV ERRN GRHV QRW
UHTXLUH SHUPLVVLRQ 6HOOLQJ RU GLVWULEXWLQJ D &' 520 RI
H[DPSOHV IURP 2;5HLOO\ ERRNV GRHV UHTXLUH SHUPLVVLRQ
\$QVZHULQJ D TXHVWLRQ E\ FLWLQJ WKLV ERRN DQG TXRWLQJ
H[DPSOH FRGH GRHV QRW UHTXLUH SHUPLVVLRQ ,QFRUSRUDWLQJ
D VLJQLILFDQW DPRXQW RI H[DPSOH FRGH IURP WKLV ERRN LQWR
\RXU SURGXFW\ GRFXPHQDWLRQ GRHV UHTXLUH SHUPLVVLRQ

:H DSSUHFLDWH EXW GR QRW UHTXLUH DWWULEXWLRQ \$Q
DWWULEXWLRQ XVXDOO\ LQFOXGHV WKH WLWOH DXWKRU
SXEOLVKHU DQG ,6%1)RU H[DPSOH £3URJUDPPLQJ 6FDOD E\ 'HDQ
:DPSOHU DQG \$OH[3D\QH &RSULJKW 'HDQ :DPSOHU DQG \$OH[
3D\QH

α

,I \RX IHHO \RXU XVH RI FRGH H[DPSOHV IDOOV RXWVLGH IDLU XVH
RU WKH SHUPLVVLRQ JLYHQ DERYH IHHO IUHH WR FRQWDFW XV DW

Getting the Code Examples

```
<RX  FDQ  GRZQORDG  WKH  FRGH  H[DPSOHV  IURP  KWWS
H[DPSOHV RUHLOO\ FRP 8Q]LS WKH ILOHV WR D
FRQYHQLHQW ORFDWLRQ 6HH WKH 5($'0( W[W ILOH LQ WKH
GLVWULEXWLRQ IRU LQVWUXFWLRQV RQ EXLOGQLQJ DQG XVLQJ WKH
H[DPSOHV
```

```
6RPH RI WKH H[DPSOH ILOHV FDQ EH UXQ DV VFULSWV XVLQJ WKH
scala FRPPDQG 2WKHUV PXVW EH FRPSLOHG LQWR FODVV ILOHV
6RPH ILOHV FRQWDLQ GHOLEHUDWH HUURUV DQG ZRQ'W FRPSLOH
:H KDYH DGRSWHG D ILOH QDPLQJ FRQYHQWLRQ WR LQGLFDWH
HDFK RI WKHVV FDVHV DOWKRXXJ DV \RX OHDUQ 6FDOD LW
VKRXOG EHFRPH REYLRXV IURP WKH FRQWHQWV RI WKH ILOHV LQ
PRVW FDVHV
```

VFULSW VFDOD

```
)LOHV WKDW HQG LQ VFULSW VFDOD FDQ EH UXQ RQ D
FRPPDQG OLQH XVLQJ scala H J scala foo-script.scala <RX FDQ
DOVR VWDUW scala LQ WKH LQWHUSUHWHU PRGH
ZKHQ \RX
```

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```
GRQ'W VSHFL\ D VFULSW ILOH DQG ORDG DQ\ VFULSW ILOH LQ
WKH LQWHUSUHWHU XVLQJ WKH :load filename FRPPDQG
```

ZRQW FRPSLOH VFDOD

```
)LOHV WKDW HQG LQ ZRQW FRPSLOH VFDOD FRQWDLQ
GHOLEHUDWH HUURUV WKDW ZLOO FDXVH WKHP WR IDLO WR
FRPSLOH :H XVH WKLV QDPLQJ FRQYHQWLRQ DORQJ ZLWK RQH
RU PRUH HP EHGGHG FRPPHQWV DERXW WKH HUURUV VR LW
ZLOO EH FOHDU WKDW WKH\ DUH LQYDOLG $OVR WKHVV ILOHV
DUH VNLSSHG E\ WKH EXLOG SURFHVV IRU WKH H[DPSOHV
```

VDNH VFDOD

```
)LOHV QDPHG VDNH VFDOD DUH XVHG E\ RXU EXLOG WRRO
FDOOHG sake 7KH 5($'0( W[W ILOH GHVFULEHV WKLV WRRO
```

VFDOD

```
$OO RWKHU 6FDOD ILOHV PXVW EH FRPSLOHG XVLQJ scalac ,Q
WKH GLVWULEXWLRQ WKH\ DUH XVHG HLWKHU E\ RWKHU
FRPSLOHG RU VFULSW ILOHV VXFK DV WHVWV QRW DOO RI
ZKLFK DUH OLVWHG LQ WKLV ERRN
```

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```
6DIDUL %RRNV 2QOLQH LV DQ RQ GHPDQG GLJLWDO
OLEUDU\ WKDW OHWV \RX HDVLO\ VHDUFK RYHU
```

WHFKQRORJ\ DQG FUHDWLYH UHIHUUHQFH ERRNV DQG
YLGHRV WR ILQG WKH DQVZHUV \RX QHHG TXLFNO\

:LWK D VXEVFULSWLRQ \RX FDQ UHDG DQ\ SDJH DQG ZDWFK DQ\
YLGHR IURP RXU OLEUDU\ RQOLQH 5HDG ERRNV RQ \RXU FHOO
SKRQH DQG PRELOH GHYLFHV \$FFHVV QHZ WLWOHV EHIRUH WKH\
DUH DYDLODEOH IRU SULQW DQG JHW H[FOXVLYH DFFHVV WR
PDQXVFULSWV LQ GHYHORSPHQW DQG SRVW IHGEDFN IRU WKH
DXWKRUV &RS\ DQG SDVWH FRGH VDPSOHV RUJDQLJH \RXU
IDYRULWHV GRZQ ORDG FKDSWHUV ERRNPDUN NH\ VHFWRQV
FUHDWH QRWHV SULQW RXW SDJHV DQG EHQHILW IURP WRQV RI
RWKHU WLPH VDYLOJ IHDWXUHV

2;5HLOO\ 0HGLD KDV XSORDGHG WKLW ERRN WR WKH 6DIDUL %RRNV
2QOLQH VHU\LFH 7R KDYH IXOO GLJLWDO DFFHVV WR WKLW ERRN
DQG RWKHUV RQ VLPLODU WRSLFV IURP 2;5HLOO\ DQG RWKHU SXE
OLVKHUV VLJQ XS IRU IUHH DW [KWWS P\ VPIDULERRNV RQOLQH FRP](#)

How to Contact Us

3OHDVH DGGUHV FRPPHQWV DQG TXHVVLRQV FRQFHUQLQJ WKLW ERRN WR
WKH SXEOLVKHU

2;5HLOO\ 0HGLD ,QF
*UDYHQVWLQ +LJKZD\ 1RUWK
6HEDVWRSRO &\$

LQ WKH 8QLWHG 6WDWHV RU &DQGD

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LQWHUQDWLRQDO RU ORFDO

ID[

:H KDYH D ZHE SDJH IRU WKLW ERRN ZKHUH ZH OLVW HUUDWD
H[VDPSOHV DQG DQ\ DGGLWLRQDO LQIRUPDWLRQ <RX FDQ DFFHVV
WKLW SDJH DW

[KWWS RUHLOO\ FRP FDWDORJ](#)

7R FRPPHQW RU DVN WHFKQLFDO TXHVVLRQV DERXW

WKLW ERRN VHQG HPDLO WR

[ERRN TXHVVLRQV#RUHLOO\ FRP](#)

)RU PRUH LQIRUPDWLRQ DERXW RXU ERRNV FRQIHUHQFHV
5HVRXUFH &HQWHUV DQG WKH 2;5HLOO\ 1HWZRUN VHH RXU
ZHEVLWH DW

Acknowledgments

\$V ZH GHYHORSHG WKL V ERRN PDQ\ SHRSOH UH DG HDUO\ GUDI WV
DQG VXJJHVW HG QXP HURXV LPSURYHPHQWV WR WKH WH[W IRU
ZKLFK ZH DUH HWHUQDOO\ JUDWHIXO :H DUH HVSHFLDOO\
JUDWHIXO WR 6WHYH -HQVHQ 5DPQLYDV /DGGDG 0DUFHO 0ROLQD
%LOO 9HQQUV DQG -RQDV %RQpU IRU WKHLU H[WHQVLYH
IHHGEDFN

0XFK RI WKH IHHGEDFN ZH UHFHLYHG FDPH WKURXJK WKH 6DIDUL
5RXJK &XWV UHOHDVHV DQG WKH RQOLQH HGLWLRQ DYDLODEOH
DW [KWWS SURJUDPPLQJVFDOD FRP](#) :H DUH JUDWHIXO IRU WKH
IHHGEDFN SURYLGHG E\

LQ QR SDUWLFXODU RUGHU ,XOLDQ 'UDJR\ 1LNRODM /LQGEHUJ
0DWW +HOOLJH 'DYL G 9GUD 5LFN\ &ODUNVRQ \$OH[&UXLVH -RVK
&URQHPH\HU 7\OHU -HQQLQJV \$ODQ 6XS\QXN 7RQ\ +LOOHUVRQ
5RJHU 9DXJKQ \$UEL 6RRNDJLDQ %UXFH /HLGO 'DQLHO 6REUDO
(GHU \$QGUHV \$YLOD 0DUHN .XELFD +HQLN +XWWXQHQ %KDVNDU
0DGGDOD *HG %\UQH 'HUHN 0DKDU *HRIIUH\ :LVHPDQ 3HWHU
5DZVWKRUQH *HRIIUH\ :LVHPDQ -RH %RZEHHU \$OH[DQGHU
%DWWLVWL 5RE 'LFNHQV 7LP 0DF(DFKHUQ -DVRQ +DUULV
6WHYHQ *UDG %RE)ROOHN \$ULHO 2UWL] 3DUWK 0DOZDQNDU
5HLG +RFKVWHGOHU -DVRQ =DXJJ -RQ +DQVRQ 0DULR
*OHLFKPDQQ 'DYL G *DWHV =HI +PHO 0LFKDHO <HH 0DULXV .UHLV
0DUWLQ 6 VVNUDXW -DYLHU 9HJDV 7RELDV +DXWK)UDQFHVFR
%RFLFFKL R 6WHSKHQ 'XQFDQ -U 3DWULN 'XGLWV -DQ
1LHKXVPDQQ %LOO %XUGLFN 'DYL G +ROEURR 6KDORP 'HLWFK
-HVSHU 1RUGHQEHUJ (V /DLQH *OHE)UDQ 6LPRQ \$QGHUVRQ
3DWULN 'XGLWV &KULV /HZLV -XOLDQ +RZDUWK 'LUN .XJHPFJDN
+HQUL *HUULWV -RKQ +HLQWJ 6WXDUW 5RHEXFN DQG -XQJKR .LP
0DQ\ RWKHU UH DG HUV IRU ZKRP ZH RQO\ KDYH XVHUQDPHV DOVR
SURYLGHG IHHGEDFN :H ZLVK WR WKDQN =DFN -RVK* HZLOOLJHU
DEFRDWHV EUDG WHWR SMFM PNOHLQW GDQGR\ RQ \$UHN UXH
DFDQJLDQR YNHOPDQ EU\DOO -HII PED[WHU SME N[HQ KLSHU
WUDFNHU FWUDQ 5DP 5 FRG\ 1RODQ -RVKXD \$MD\ -RH DQG
DQRQPRXV FRQWULEXWU V :H DSRORJLJH LI ZH KDYH
RYHUORRNHG DQ\ RQH

2XU HGLWRU 0LNH /RXNLGHV NQRZV KRZ WR SXVK DQG SURG
JHQWOH +H\ V EHHQ D JUHDW KHOS WKURXJKRXW WKL V FUD\,
SURFHVV 0DQ\ RWKHU SHRSOH DW 2\5HLOO\ ZHUH DOZDV WKHUH
WR DQVZHU RXU TXHVWL RQV DQG KHOS XV PRYH IRUZDUG

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:H WKDQN -RQDV %RQpU IRU ZULWLQJ WKH [RUHZRUG](#) IRU WKH
ERRN -RQDV LV D ORQJWLPH IULHQG DQG FROODERUDWRU IURP
WKH DVSHFW RULHQWHG SURJUDPPLQJ
\$23 FRPPXQLW\)RU \HDUV KH KDV GRQH SLRQHHULQJ ZRUN LQ
WKH -DYD FRPPXQLW\ 1RZ KH LV DSSO\LQJ KLV HQHUJLHV WR

SURPRWLQJ 6FDOD DQG JURZLQJ WKDW FRPPXQLW\

%LOO 9HQQHUV JUDFLRXVO\ SURYLGHG WKH TXRWH RQ WKH EDFN
FRYHU 7KH ILUVW SXEOLVKHG ERRN RQ 6FDOD 3URJUDPPLQJ LQ
6FDOD

\$UWLPD WKDW KH FRZURWH ZLWK 0DUWLQ 2GHUVN\ DQG /H
6SRRQ LV LQGLVSHQVDEOH IRU WKH 6FDOD GHYHORSHU %LOO
KDV DOVR FUHDWHG WKH ZRQGHUIXO 6FDOD7HVV OLEUD\

:H KDYH OHDUQHG D ORW IURP IHOORZ GHYHORSHUV DURXQG WKH
ZRUOG %HVLGHV -RQDV DQG %LOO 'HEDVLVK *KRVK -DPHV ,U
'DQLHO 6SLHZDN 'DYL 3ROODFN 3DXO 6QLYHO\ 2OD %LQL 'DQLHO
6REUDO -RVK 6XHUHWK 5REH\ 3RLQWHU 1DWKDQ +DPEOHQ -RUJH
2UWLJ DQG RWKHUV KDYH LOOXPLQDWHG GDUN FRUQHUV ZLWK
WKHLU EORJ HQWULHV IRUXP GLVFXVVRQV DQG SHUVRQDO
FRQYHUVDWLRQV

'HDQ WKDQNV KLV FROOHDJXH DW 2EMHFW 0HQWRU DQG
VHYHUDO GHYHORSHUV DW FOLHQW VLWHV IRU PDQ\
VWLPXODWLQJ GLVFXVVRQV RQ ODQJXDJHV VRIWZDUH GHVLJQ
DQG WKH SUDJPDWLF LVVXH IDFLQJ GHYHORSHUV LQ LQGXVWU\
7KH PHPEHUV RI WKH &KLFDJR \$UHD 6FDOD (QWKXVLDVWV
&\$6(JURXS KDYH DOVR EHHQ D VRXUFH RI YDOXDEOH IHHGEDFN
DQG LQVSLUDWRQ

\$OH\ WKDQNV KLV FROOHDJXH DW 7ZLWWHU IRU WKHLU
HQFRXUDJHPHQW DQG VXSHUE ZRUN LQ GHPRQVWUDWLQJ
6FDOD\V HIIHFWLYHQHVV DV D ODQJXDJH +H DOVR WKDQNV WKH
%D\ \$UHD 6FDOD (QWKXVLDVWV
%\$6(IRU WKHLU PRWLYDWRQ DQG FRPPXQLW\

0RVW RI DOO ZH WKDQN 0DUWLQ 2GHUVN\ DQG KLV WHDP IRU FUHDWLQJ 6FDOD

Zero to Sixty: Introducing Scala

Why Scala?

7RGD\|V HQWHUSULVH DQG ,QWHUQHW DSSOLFDWLRQV PXVW
 EDODQFH D QXPEHU RI FRQFHUQV 7KH\ PXVW EH LPSONPHQWHG
 TXLFNO\ DQG UHOLDEO\ 1HZ IHDWXUHV PXVW EH DGGHG LQ VKRUW
 LQ FUHPHQWDO F\FOHV %H\RQG VLPSO\ SURYLGGLQJ EXVLQHV
 ORJLF DSSOLFDWLRQV PXVW VXSSRUW VHFUXH DFFHVV
 SHUVLVWHQFH RI GDWD WUDQVDFWLRQDO EHKDYLRU DQG RWKHU
 DGYDQFHG IHDWXUHV \$\$SOLFDWLRQV PXVW EH KLJKO\
 DYDLODEOH DQG VFDODEOH UHTXLULQJ GHVLJQV WKDW
 VXSSRUW FRQFXUUHQF\ DQG GLVWULEXWLRQ \$\$SOLFDWLRQV DUH
 QHWZRUNHG DQG SURYLG LQWHUIDFHV IRU ERWK SHRSOH DQG
 RWKHU DSSOLFDWLRQV WR XVH

7R PHHW WKHVH FKDOOHQJHV PDQ\ GHYHORSHUV DUH ORRNQJ
 IRU QHZ ODQJXDJHV DQG WRROV 9HQHUDEOH VWDQGE\V OLNH
 -DYG & DQG & DUH QR ORQJHU RSWLPDO IRU GHYHORSQJ WKH
 QH\W JHQHUDWLRQ RI DSSOLFDWLRQV

If You Are a Java Programmer...

-DYG ZDV RIILFLDOO\ LQWURGXFHG E\ 6XQ 0LFURV\WHPV LQ 0D\ RI
 DW WKH DGYHQW RI ZLGHVSUHDG LQWHUHVW LQ WKH ,QWHUQHW
 -DYG ZDV LPPHGLDWHO\ KDLOHG DV DQ LGHDO ODQJXDJH IRU
 ZULWLQJ EURZVHU EDVHG DSSOHVV ZKHUH D VHFUXH SRUWDEOH
 DQG GHYHORSHU IULHQGO\ DSSOLFDWLRQ ODQJXDJH ZDV QHHGHG
 7KH UHLJQLQJ ODQJXDJH RI WKH GD\ & ZDV QRW VXLWDEOH IRU
 WKLV GRPDLQ

7RGD\ -DYG LV PRUH RIWHQ XVHG IRU VHUYHU VLGH
 DSSOLFDWLRQV ,W LV RQH RI WKH PRVW SRSXODU ODQJXDJHV LQ
 XVH IRU WKH GHYHORSPHQW RI ZHE DQG HQWHUSULVH
 DSSOLFDWLRQV

+RZHYHU -DYG ZDV D FKLOG RI LWV WLP 1RZ LW VKRZV LWV DJH
 ,Q -DYG SURYLGHG D \QWD\ VLPLDU HQRXJK WR & WR
 HQWLFH & GHYHORSHUV ZKLOH DYRLGLQJ PDQ\ RI WKDW
 ODQJXDJH\ GHILFLHQFLHV DQG £VKDUS HGJHV -DYG DGRSWHG
 WKH PRVW XVHIXO LGHDV IRU WKH GHYHORSPHQW SUREOHPV RI
 LWV HUD VXFK DV REMHFW RULHQWHG SURJUDPPLQJ

223 ZKLOH GLVFDUGLQJ PRUH WURXEOHVVRPH WHFKQLTXHV VXFK
DV PDQXDO PHPRU\ PDQDJHPPHQW 7KHVH GHVLJQ FKRLFHV VWUXFN
DQ H[FHOOHQW EDODQFH WKDW PLQLPL]HG FRPSOH[LW\ DQG
PD[LPL]HG GHYHORSHU SURGXFWLYLW\ ZKLOH WUDGLQJ RII
SHUIRUPDQFH FRPSDUHG WR QDWLYHO\ FRPSLOHG

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FRGH :KLOH -DYD KDV HYROYHG VLQFH LWV ELUWK PDQ\ SHRSOH
EHOLHYH LW KDV JURZQ WRR FRPSOH[ZLWKRXW DGHTXDWHO\
DGGUHVVLQJ VRPH QHZHU GHYHORSPPHQW FKDOOHQJHV

'HYHORSHUV ZDQW ODQJXDJHV WKDW DUH PRUH VXFFLQFW DQG
IOH[LEOH WR LPSURYH WKHLU SUR GXFWLYLW\ 7KLV LV RQH
UHDVRQ ZK\ VR FDOOHG VFULSWLQJ ODQJXDJHV OLNH 5XE\ DQG
3\WKRQ KDYH EHFRPH PRUH SRSXODU UHFHQWO\

7KH QHYHU HQGLQJ QHHG WR VFDOH LV GULYLQJ DUFKLWHFWXUHV
WRZDUG SHUYDVLVH FRQFXUUHQF\ +RZHYHU -DYD\V FRQFXUUHQF\
PRGHO ZKLK LV EDVHG RQ \QFKURQL]HG DFFHVV WR VKDUHG
PXWDEOH VWDWH UHVXOWV LQ FRPSOH[DQG HUURU SURQH
SURJUDPV

:KLOH WKH -DYD ODQJXDJH LV VKRZLQJ LWV DJH WKH -DYD
9LUWXDO ODFKLQH

-90 RQ ZKLK LV UXQV FRQWLQXHV WR VKLQH 7KH
RSWLPL]DWLRQV SHUIRUPHG E\ WRGD\ \V -90 DUH H[WUDRU GLQDU\
DOORZLQJ EWH FRGH WR RXWSHUIRUP QDWLYHO\ FRPSLOHG FRGH
LQ PDQ\ FDVHV 7RGD\ PDQ\ GHYHORSHUV EHOLHYH WKDW XLQJ
WKH -90 ZLWK QHZ ODQJXDJHV LV WKH SDWK IRUZDUG 6XQ LV
HPEUDFLQJ WKLV WUHQG E\ HPSOR\LQJ PDQ\ RI WKH OHDG
GHYHORSHUV RI -5XE\ DQG -WKRQ ZKLK DUH -90 SRUWV RI 5XE\
DQG 3\WKRQ UHVSHFWLYHO\

7KH DSSHDO RI 6FDOD IRU WKH -DYD GHYHORSHU LV WKDW LW
JLYHV IRX D QHZHU PRUH PRGHUQ ODQJXDJH ZKLOH OHYHUDJLQJ
WKH -90\ \V DPD[LQJ SHUIRUPDQFH DQG WKH ZHDOWK RI -DYD
OLEUDULHV WKDW KDYH EHHQ GHYHORSHG IRU RYHU D GHFDGH

If You Are a Ruby, Python, etc. Programmer...

'QDPLFDOO\ WSHG ODQJXDJHV OLNH 5XE\ 3\WKRQ *URRY\
-DYD6FULSW DQG 6PDOOWDON RIIHU YHU\ KLJK SURGXFWLYLW\
GXH WR WKHLU IOH[LEOLW\ SRZHUIXO PHWDSURJUDPPLQJ DQG
HOHJDQFH

Statically Typed Versus Dynamically Typed Languages

2QH RI WKH IXQGDPHQWDO ODQJXDJH GHVLJQ FKRLFHV LV VWDWLF YHUVXV G\QDPLF
WLSLQJ

7KH ZRUG £WLSLQJ LV XVHG LQ PDQ\ FRQWH[WV LQ VRIWZDUH 7KH
IROORZLQJ LV D £SODXVLEOH GHILQLWRQ WKDW LV XVHIXO IRU
RXU SXUSRVH

\$ WSH VVWHP LV D WUDFWDEOH \QWDFWLF PHWKRQ IRU

SUHVVHUYLQJ WKH DEVHQFH RI FHUWDLQ SURJUDP EHKDYLRUV
E\ FODVVLIQLQJ SKUDVHV DFFRUGLQJ WR WKH NLQGV RI YDOXH
WKH\ FRPSXWH

ø%HQMDPLQ & 3LHUFH 7ISHV DQG 3URJUDPPLQJ/DQJXDJHV
0,7 3UHV

1RWH WKH HPSKDVLV RQ KRZ D WSH VVWHP DOORZV UHDVRQLQJ
DERXW ZKDW D VVWHP H\ FOXGHV IURP KDSSHQLQJ 7KDW\ V
JHQHUDOO\ HDVLHU WKDQ WULQJ WR GHWHUPLQH WKH VHW RI
DOO DOORZHG SRVVLELOLWLHV \$ WSH VVWHP LV XVHG WR FDWFK
YDULRXV HUURUV OLNH XQVXSSRUWHG RSHUDWLRQV RQ
SDUWLFXODU GDWD VWUXFWXUH DWWHPSWLQJ WR FRPELQH
GDWD LQ DQ XQGHLQH ZD\
H J WULQJ WR DGG DQ LQWHJU WR D VWULQJ EUHDNLQJ
DEVWUDFWLRQV HWF

,QIRUPDOO\ LQ VWDWLF WSLQJ D YDULDEOH LV ERXQG WR D
SDUWLFXODU WSH IRU LWV OLIHWLPH ,WV WSH FDQ\W EH
FKDQJHG DQG LW FDQ RQO\ UHIHUEFH WSH FRPSDWLEOH
LQVWDQFHV 7KDW LV LI D YDULDEOH UHIHUV WR D YDOXH RI WSH
A \RX FDQ\W DVVLJQ D YDOXH RI D GLIHUEFH WSH B WR LW
XQOHVV B LV D VXEWSH RI A IRU VRPH UHDVRQDEOH GHILQLWLRQ
RI \VXEWSH ¶

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,Q G\QDPLF WSLQJ WKH WSH LV ERXQG WR WKH YDOXH QRW WKH
YDULDEOH 6R D YDULDEOH PLJKW UHIHU WR D YDOXH RI WSH A
WKHQ EH UHVVLIJQH ODWHU WR D YDOXH RI DQ XQUHODWHG
WSH X

7KH WHUP G\QDPLFDOO\ WSHG LV XVHG EHFDXVH WKH WSH RI D
YDULDEOH LV HYDOXDWHG ZKHQ LW LV XVHG GXULQJ UXQWLPH
ZKLOH LQ D VWDWLFDOO\ WSHG ODQJXDJH WKH WSH LV
HYDOXDWHG DW SDUVH WLP

7KLW PD\ VHHP OLNH D VPDOO GLVWLQFWLRQ EXW LW KDV D
SHUYDVLYH LPSDFW RQ WKH SKLORVRSK\ GHVLJQ DQG
LPSOHPHQDWLRQ RI D ODQJXDJH :H\OO H\SORUH VRPH RI WKHVH
LPSOLFDFWLRQV DV ZH JR WKURXJK WKH ERRN

6FDOD DQG -DYD DUH VWDWLFDOO\ WSHG ODQJXDJHV ZKHUHDV
5XE\ 3\WKRQ *URRY\ -DYD 6FULSW DQG 6PDOOWDON DUH
G\QDPLFDOO\ WSHG ODQJXDJHV

)RU VLPSOLFLW\ ZH ZLOO RIWHQ XVH WKH WHUPV VWDWLF
ODQJXDJH DQG G\QDPLF ODQJXDJH DV VKRUWKDQGV IRU
VWDWLFDOO\ WSHG ODQJXDJH DQG G\QDPLFDOO\ WSHG
ODQJXDJH UHVSFWLYHO\

\$Q RUWKRRQDO GHVLJQ FRQVLGHUDWLRQ LV VWURQJ YHUVXV
ZHDN WSLQJ ,Q VWURQJ WSLQJ HYHU\ YDULDEOH
IRU VWDWLF WSLQJ RU YDOXH
IRU G\QDPLF WSLQJ PXVW KDYH DQ XQDELIJRXV WSH ,Q ZHDN
WSLQJ D VSHFLILF WSH LV QRW UHTXLUHG :KLOH PRVW
ODQJXDJHV DOORZ VRPH PL\WXUH RI VWURQJ YHUVXV ZHDN WSLQJ
6FDOD -DYD DQG 5XE\ DUH SUHGRPLQDQWO\ VWURQJO\ WSHG
ODQJXDJHV 6RPH ODQJXDJHV OLNH & DQG 3HUO DUH PRUH
ZHDNO\ WSHG

'HVSLWH WKHLU SURGXFWLYLW DGYDQWDJHV G\QDPLF ODQJXDJHV
PD\ QRW EH WKH EHVW FKRLFHV IRU DOO DSSOLFDWLRQV
SDUWLFXODUO\ IRU YHU\ ODUJH FRGH EDVHV DQG KLJK
SHUIRUPDQFH DSSOL FDWLRQV 7KHUH LV D ORQJVWDQGLQJ
VSLULWHG GHEDWH LQ WKH SURJUDPPPLQJ FRPPXQLW\ DERXW WKH
UHODWLYH PHULWV RI G\QDPLF YHUVXV VWDWLF W\SLQJ ODQ\ RI
WKH SRLQWV RI FRPSDULVRQ DUH VRPHZKDW VXEMHFWLYH :H
ZRQ\W JR WKURXJK DOO WKH DUJXPHQWV KHUH EXW ZH ZLOO
RIIHU D IHZ WKRXJKWV IRU FRQVLGHUDWLRQ

2SWLPLJLQJ WKH SHUIRUPDQFH RI D G\QDPLF ODQJXDJH LV PRUH
FKDOOHQJLQJ WKDQ IRU D VWDWLF ODQJXDJH ,Q D VWDWLF
ODQJXDJH RSWLPLJHUV FDQ H[SORLW WKH W\SH LQIRUPDWLRQ WR
PDNH GHFLVLRQV ,Q D G\QDPLF ODQJXDJH IHZHU VXFK FOXHV
DUH DYDLODEOH IRU WKH RSWLPLJHU PDNLQJ RSWLPLJDWLRQ
FKRLFHV KDUGHU :KLOH UHFHQW DGYDQFHPHQWV LQ
RSWLPLJDWLRQV IRU G\QDPLF ODQJXDJHV DUH SURPLVLQJ WKH\
ODJ EHKLQG WKH VWDWH RI WKH DUW IRU VWDWLF ODQ JXDJHV
6R LI \RX UHTXLUH YHU\ KLJK SHUIRUPDQFH VWDWLF ODQJXDJHV
DUH SUREDEO\ D VDIHU FKRLFH

6WDWLF ODQJXDJHV FDQ DOVR EHQHILW WKH GHYHORSHPHQW
SURFHVV ,QWHJUDWHG GHYHORSHPHQW HQYLURQPHQW
,\ IHDXUHV OLNH *DXWRFRPSOHWLRQ*
VRPHWLPHV FDOOHG *FRGH VHQVH* DUH HDVLHU WR LPSOHPHQW
IRU VWDWLF ODQJXDJHV DJDLQ EHFDXVH RI WKH H[WUD W\SH
LQIRUPDWLRQ DYDLODEOH 7KH PRUH H[SOLFLW W\SH
LQIRUPDWLRQ LQ VWDWLF FRGH SURPRWHV EHWWHU £VHOI
GRFXPHQDWLRQ ▫ ZKLFK FDQ EH LPSRUWDQW IRU
FRPPXQLFDWLQJ LQWHQW DPRQJ GHYHORSHUV HVSHFLDOO\ DV D
SURMHFW JURZV

:KHQ XVLQJ D VWDWLF ODQJXDJH \RX KDYH WR WKLQN DERXW
DSSURSULDWH W\SH FKRLFHV PRUH RIWHQ ZKLFK IRUFHV \RX WR
ZHLJK GHVLJQ FKRLFHV PRUH FDUHIXOO\ :KLOH WKLW PD\ VORZ
GRZQ GDLO\ GHVLJQ GHFLVLRQV WKLQNQJ WKURXJK WKH W\SHV
LQ WKH DSSOLFDWLRQ FDQ UHVXOW LQ D PRUH FRKHUHQW
GHVLJQ RYHU WLP

Why Scala? | 3

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\$QRWKHU VPDOO EHQHILW RI VWDWLF ODQJXDJHV LV WKH H[WUD
FKHFNQJ WKH FRPSLOHU SHUIRUPV :H WKLQN WKLW DGYDQWDJH
LV RIWHQ RYHUVROG DV W\SH PLVPDWFK HUURUV DUH D VPDOO
IUDFWLRQ RI WKH UXQWLPH HUURUV \RX W\SLFDOO\ VHH 7KH
FRPSLOHU FDQ\W ILQG *ORJLF* HUURUV ZKLFK DUH IDU PRUH
VLJQLILFDQW 2QO\ D FRPSUHKHQVLYH DXWRPDWHG WHVW VXLWH
FDQ ILQG *ORJLF* HUURUV)RU G\QDPLFDOO\ W\SHG ODQJXDJHV WKH
WHVWV PXVW FRYHU SRVVLEOH W\SH HUURUV WRR ,I \RX DUH
FRPLQJ IURP D G\QDPLFDOO\ W\SHG ODQJXDJH \RX PD\ ILQG WKDW
\RXU WHVW VXLWHV DUH D OLWWOH VPDOOHU DV D UHVXOW EXW
QRW *WKDW* PXFK VPDOOHU

0DQ\ GHYHORSHUV ZKR ILQG VWDWLF ODQJXDJHV WRR YHUERVH
 RIWHQ EODPH VWDWLF WSLQJ IRU WKH YHUERVLW ZKHQ WKH
 UHDO SUREOHP LV D ODFN RI WSH LQIHUHQFH ,Q WSH LQIHUHQFH
 WKH FRPSLOHU LQIHUV WKH WSHV RI YDOXH EDVHG RQ WKH
 FRQWH[W]RU H[DPSOH WKH FRPSLOHU ZLOO UHFRJQLJH WKDW $x =$
 $1 + 3$ PHDQV WKDW x PXVW EH DQ LQWHUJHU 7\SH LQIHUHQFH
 UHGXFHV YHUERVLW VLJQLILFDQWO\ PDNLQJ WKH FRGH IHHO
 PRUH OLNH FRGH ZULWWHQ LQ D GQDPLF ODQJXDJH

:H KDYH ZRUNHG ZLWK ERWK VWDWLF DQG GQDPLF ODQJXDJHV
 DW YDULRXV WLPHV :H ILQG ERWK NLQGV RI ODQJXDJHV
 FRPSHOOLQJ IRU GLIIHUHQW UHDVRQV :H EHOLHYH WKH PRGHUQ
 VRIW ZDUH GHYHORSHU PXVW PDVWHU D UDQJH RI ODQJXDJHV
 DQG WRROV 6RPHWLPHV D GQDPLF ODQJXDJH ZLOO EH WKH
 ULJKW WRRO IRU WKH MRE \$W RWKHU WLPHV D VWDWLF
 ODQJXDJH OLNH 6FDOD LV MXVW ZKDWRX QHHG

Introducing Scala

6FDOD LV D ODQJXDJH WKDW DGGUHVHVH WKH PDMRU QHHGV RI
 WKH PRGHUQ GHYHORSHU ,W LV D VWDWLFDOO\ WSHG PLJHG
 SDUDGLJP -90 ODQJXDJH ZLWK D VXFFLQFW HOHJDQW DQG
 IOH[LEOH VQWD[D VRSKLVWLFDWHG WSH VVWHP DQG LGLRPV
 WKDW SURPRWH VFDODELOLW IURP VPDOO LQWHUSUHWHG
 VFULSWV WR ODUJH VRSKLVWLFDWHG DSSOLFDWLRQV 7KDW\ V D
 PRXWKIXO VR OHW\ V ORRN DW HDFK RI WKRVH LGHDV LQ PRUH
 GHWDLO

6WDWLFDOO\ WSHG

\$V ZH GHVFULEHG LQ WKH SUHYLRXV VHFWRU D VWDWLFDOO\
 WSHG ODQJXDJH ELQGV WKH WSH WR D YDULDEOH IRU WKH
 OLIHWLPH RI WKDW YDULDEOH ,Q FRQWUDVW GQDPLFDOO\
 WSHG ODQJXDJHV ELQGV WKH WSH WR WKH DFWXDO YDOXH
 UHIHUHQFHG E\ D YDULDEOH PHDQLQJ WKDW WKH WSH RI D
 YDULDEOH FDQ FKDQJH DORQJ ZLWK WKH YDOXH LW
 UHIHUHQFHV

2I WKH VHW RI QHZHU -90 ODQJXDJHV 6FDOD LV RQH RI WKH IHZ
 WKDW LV VWDWLFDOO\ WSHG DQG LW LV WKH EHVW NQRZQ
 DPRQJ WKHP

OLJHG SDUDGLJP REMHFW RULHQWHG SURJUDPPHQJ

6FDOD IXOO\ VXSSRUWV REMHFW RULHQWHG SURJUDPPHQJ
 223 6FDOD LPSURYHV XSRQ -DYD\ V VXSSRUW IRU 223 ZLWK
 WKH DGGLWLRQ RI WUDLWV D FOHDQ ZD\ RI LPSOHPHQWLQJ
 FODVVHV XVLQJ PLJLQ FRPSRVLWLRQ 6FDOD\ V WUDLWV ZRUN
 PXFK OLNH 5XE\ V PRGXOHV ,I RX\UH D -DYD SURJUDPPHU
 WKLQN RI WUDLWV DV XQL\LQJ LQWHUIDFHV ZLWK WKHLU
 LPSOHPHQWDWLRQV

,Q 6FDOD HYHUWKLQJ LV UHDOO\ DQ REMHFW 6FDOD GRHV
 QRW KDYH SULPLWLYH WSHV OLNH -DYD ,QVWHDG DOO
 QXPHULF WSHV DUH WUXH REMHFWV +RZHYHU IRU
 RSWLPDO

SHUIRUPDQFH 6FDOD XVHV WKH XQGHUO\LQJ SULPLWLYHV
 WSHV RI WKH UXQWLPH ZKHQHYHU SRVLEOH \$OVR 6FDOD
 GRHV QRW VXSSRUW £VWDWLF² RU FODVV OHYHO PHPEHUV RI
 WSHV VLQFH WKH\ DUH QRW DVVRFLDWHG ZLWK DQ DFWXDO
 LQVWDQFH ,QVWHDG 6FDOD VXSSRUWV D VLQJOHWRRQ
 REMHFW FRQVWUXFW WR VXSSRUW WKRVH FDVHV ZKHUH
 H\DFWO\ RQH LQVWDQFH RI D WSH LV QHHGHG

OL[HG SDUDGLJP&IXQFWLRQDO SURJUDPPLQJ

6FDOD IXOO\ VXSSRUWV IXQFWLRQDO SURJUDPPLQJ

)3)3 LV D SURJUDPPLQJ SDUDGLJP WKDW LV ROGHU WKDQ 223
 EXW LW KDV EHHQ VKHOWHUHG LQ WKH LYRU\ WRZHUV RI
 DFDGHPLD XQWLO UHFHQWO\ ,QWHUHVW LQ)3 LV LQFUHDVLQJ
 EHFDXVH RI WKH ZDV LW VLPSOLILHV FHUWDLQ GHVLJQ
 SUREOHPV HVSHFLDOO\ FRQFXUUHQF\ £3XUH² IXQFWLRQDO
 ODQJXDJHV GRQ\W DOORZ IRU DQ\ PXWDEOH VWDWH WKHUH\
 DYRLGLQJ WKH QHHG IRU \QFKURQLJDWLQJ RQ VKDUHG
 DFFHVV WR PXWDEOH VWDWH ,QVWHDG SURJUDPV ZULWWHQ
 LQ SXUH IXQFWLRQDO ODQJXDJHV FRPPXQLFDWH E\ SDVLQJ
 PHVVDJHV EHWZHHQ FRQFXUUHQW DXWRQRPRXV SURFHVVHV
 6FDOD VXSSRUWV WKLV PRGHO ZLWK LWV \$FWRUV OLEUDU\
 EXW LW DOORZV IRU ERWK PXWDEOH DQG LPPXWDEOH
 YDULDEOHV

)XQFWLRQV DUH £ILUVW FODVV² FLWLJHQV LQ)3 PHDQLQJ WKH\
 FDQ EH DVVLJQHG WR YDULDEOHV SDVVHG WR RWKHU
 IXQFWLRQV HWF MXVW OLNH RWKHU YDOXHV 7KLV IHDWXUH
 SURPRWHV FRP SRVLWLQJ RI DGYDQFHG EHKDYLRU XVLQJ
 SULPLWLYH RSHUDWLQJ %HFDXVH 6FDOD DGKHUV WR WKH
 GLFWXP WKDW *HYHU\WKLQJ LV DQ REMHFW* IXQFWLRQV DUH
 WKHPVHOYHV REMHFWV LQ 6FDOD 6FDOD DOVR RIIHUV
FORVXUHV D IHDWXUH WKDW G\QDPLF ODQJXDJHV OLNH
 3\WKRRQ DQG 5XE\ KDYH DGRSWHG IURP WKH IXQFWLRQDO
 SURJUDPPLQJ ZRUOG DQG RQH VDGO\ DEVHQW IURP UHFHQW
 YHUVLRQV RI -DYD &ORVXUHV DUH IXQFWLRQV WKDW
 UHIHUUHQFH YDULDEOHV IURP WKH VFRSH HQFORVLQJ WKH
 IXQFWLRQ GHILQLWLQJ 7KDW LV WKH YDULDEOHV DUHQ\W
 SDVVHG LQ DV DUJXPHQWV RU GHILQH DV ORFDO
 YDULDEOHV ZLWKLQ WKH IXQFWLRQ \$ FORVUH £FORVHV
 DURXQG² WKHVH UHIHUUHQFHV VR WKH IXQFWLRQ LQYRFDWLQJ
 FDQ VDIHO\ UHIHU WR WKH YDULDEOHV HYHQ ZKHQ WKH
 YDULDEOHV KDYH JRQH RXW RI VFRSH &ORVXUHV DUH VXFK D
 SRZHUIXO DEVWUDFWLRQ WKDW REMHFW \VWHPV DQG
 IXQGDPHQWDO FRQWURO VWUXFWXUHV DUH RIWHQ
 LPSOHPHQWHG XVLQJ WKHP

\$ -90 DQG 1(7 ODQJXDJH

:KLOH 6FDOD LV SULPDULO\ NQRZQ DV D -90 ODQJXDJH PHDQLQJ
 WKDW 6FDOD JHQHUDWHV -90 E\WH FRGH D 1(7 YHUVLRQ RI
 6FDOD WKDW JHQHUDWHV &RPPRQ/DQJXDJH 5XQWLPH
 &/5 E\WH FRGH LV DOVR XQGHU GHYHORSPHQW :KHQ ZH
 UHIHU WR WKH XQGHUO\LQJ £UXQWLPH ² ZH ZLOO XVXDOO\
 GLVFXVV WKH -90 EXW PRVW RI ZKDW ZH ZLOO VD\ DSSOLHV

HTXDOO\ WR ERWK UXQWLPHV :KHQ ZH GLVFXVV -90 VSHFLILF
GHWDLOV WKH\ JHQHUDOLJH WR WKH 1(7 YHUVLRQ H[FHSW
ZKHUH QRWHG

7KH 6FDOD FRPSLOHU XVHV FOHYHU WHFKQLTXHV WR PDS
6FDOD H[WHQVLRQV WR YDOLG EWH FRGH LGLRPV)URP 6FDOD
\RX FDQ HDVLO\ LQYRNH EWH FRGH WKDW RULJLQDWHG DV
-DYD VRXUFH

IRU WKH -90 RU & VRXUFH

IRU 1(7 &RQYHUVHO\ \RX FDQ LQYRNH 6FDOD FRGH IURP -DYD
& HWF 5XQQLQJ RQ WKH -90 DQG &5 DOORZV WKH 6FDOD
GHYHORSHU WR OHYHUDJH DYDLODEOH OLEUDULHV DQG WR
LQWHURSHUDWH ZLWK RWKHU ODQJXDJHV KRVWHG RQ WKRVH
UXQWLPHV

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\$ VXFFLQFW HOHJDQW DQG IOH[LEOH \IQWD[

-DYD \IQWD[FDQ EH YHUERVH 6FDOD XVHV D QXPEHU RI
WHFKQLTXHV WR PLQLPLJH XQ QHFHVVDU\ \IQWD[PDNLQJ
6FDOD FRGH DV VXFFLQFW DV FRGH LQ PRVW G\QDPLFDOO\
WSHG ODQJXDJHV 7\SH LQIHUHQFH PLQLPLJHV WKH QHHG IRU
H[SOLFLW WSH LQIRUPDWLRQ LQ PDQ\ FRQWH[VW
'HFODUDWLRQV RI WSHV DQG IXQFWLRQV DUH YHU\ FRQFLVH

6FDOD DOORZV IXQFWLRQ QDPHV WR LQFOXGH QRQ
DOSKDQXPHULF FKDUFWHU\ &RPELQHG ZLWK VRPH
\QWDFWLF VXJDU WKLV IHDWXUH SHUPLWV WKH XVHU WR
GHILQH PHWKRGV WKDW ORRN DQG EHKDYH OLNH RSHUDWRUV
\$V D UHVXOW OLEUDULHV RXWVLGH WKH FRUH RI WKH
ODQJXDJH FDQ IHHO £QDWLYH WR XVHUV

\$ VRSKLVWLFDWHG WSH \VWHP

6FDOD H[WHQGV WKH WSH \VWHP RI -DYD ZLWK PRUH IOH[LEOH
JHQHULFV DQG D QXPEHU RI PRUH DGYDQFHG WSLQJ
FRQVWUXFWV 7KH WSH \VWHP FDQ EH LQWLPLGDWLQJ DW
ILUVW EXW PRVW RI WKH WLPH \RX ZRQ\W QHHG WR ZRUU\
DERXW WKH DGYDQFHG FRQVWUXFWV 7\SH LQIHUHQFH KHOSV
E\ DXWRPDWLFDOO\ LQIHUULQJ WSH VLJQDWXUHV VR WKDW
WKH XVHU GRHVQ\W KDYH WR SURYLGH WULYLD OWSH
LQIRUPDWLRQ PDQXDOO\ :KHQ \RX QHHG WKHP WKRXJK WKH
DGYDQFHG WSH IHDWXUHV SURYLGH \RX ZLWK JUHDWHU
IOH[LEOLW\ IRU VROYLQJ GHVLJQ SUREOHPV LQ D WSH VDIH ZD\

6FDODEOH&DUFKLWHFWXUHV

6FDOD LV GHVLJQHG WR VFDOH IURP VPDOO LQWHUSUHHWG
VFULSWV WR ODUJH GLVWULEXWHG DS SOLFDWLRQV 6FDOD
SURYLGHV IRXU ODQJXDJH PHFKDQLVPV WKDW SURPRWH
VFODEOH FRP SRVLWLRQ RI \VWHPV H[SOLFLW VHOI WSHV
DEVWUDFW WSH PHPEHUV DQG JHQHULFV QHVVHG FODVVHV
DQG PL[LQ FRPSRVLWLRQ XVLQJ WUDLWV

1R RWKHU ODQJXDJH SURYLGHV DOO WKHVH PHFKDQLVPV
7RJHWKHU WKH\ DOORZ DSSOLFD WLRQV WR EH

FRQVWUXFWHG IURP UHXVDEOH £FRPSRQHQWV LQ D WSH
VDIH DQG VXFFLQFW PDQQHU \$V ZH ZLOO VHH PDQ\ FRPPRQ
GHVLJQ SDWWHUQV DQG DUFKLWHFWXUDO WHFKQL TXHV OLNH
GHSHQGHHQ\ LQMHFWLRQ DUH HDV\ WR LPSOHPHQW LQ 6FDOD
ZLWKRXW WKH ERLOHU SODWH FRGH RU OHQJWK\ ;0/
FRQILJXUDWLRQ ILOHV WKDW FDQ PDNH -DYG GHYHORSPHQW
WHGLRXV

6FDODEOH&SHUIRUPDQFH

%HFDXVH 6FDOD FRGH UXQV RQ WKH -90 DQG WKH &/5 LW
EHQHILWV IURP DOO WKH SHUIRUP DQFH RSWLPLJDWLRQV
SURLGHG E\ WKRVL UXQWLPHV DQG DOO WKH WKLUG SDUW
WRROR WKDW VXSSRUW SHUIRUPDQFH DQG VFDODELOLW\ VXF
DV SURILOHUV GLVWULEXWHG FDFKH OLEUDULHV FOXVWHULQJ
PHFKDQLVPV HWF ,I\RX WUXVW -DYG\ V DQG & \ V SHUIRUPDQFH
I\RX FDQ WUXVW 6FDOD\ V SHUIRUPDQFH 2I FRXUVH VRPH
SDUWLFXODU FRQVWUXFWV LQ WKH ODQJXDJH DQG VRPH
SDUWV RI WKH OLEUDU\ PD\ SHUIRUP VLJQLILFDQWO\ EHWWHU
RU ZRUVH WKDQ DOWHUQDWLYH RSWLRQV LQ RWKHU
ODQJXDJHV \$V DOZDV\ I\RX VKRXOG SURILOH I\RXU FRGH DQG
RSWLPLJH LW ZKHQ QHFHVVDU\

,W PLJKW DSSHU WKDW 223 DQG)3 DUH LQFRPSDWLEOH ,Q IDFW
D GHVLJQ SKLORVRSK\ RI 6FDOD LV WKDW 223 DQG)3 DUH PRUH
V\QHUJLVWLF WKDQ RSSRVHG 7KH IHDWXUHV RI RQH DSSURDFK
FDQ HQKDQFH WKH RWKHU

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,Q)3 IXQFWLRQV KDYL QR VLGH HHHFWV DQG YDULDEOHV DUH
LPPXWDEOH ZKLOH LQ 223 PX WDEOH VWDWH DQG VLGH HHHFWV
DUH FRPPRQ HYHQ HQFRXUDJHG 6FDOD OHWV I\RX FKRRVL WKH
DSSURDFK WKDW EHVW ILWV I\RXU GHVLJQ SUREOHPV)XQFWLRQDO
SURJUDPPQLJ LV HVSHFLDOO\ XVHIXO IRU FRQFXUUHQF\ VLQFH LW
HOLPLQDWHV WKH QHHG WR V\QFKURQLJH DFFHVV WR PXWDEOH
VWDWH +RZHYHU £SXUH LQ)3 FDQ EH UHVWULFWLYH 6RPH GHVLJQ
SUREOHPV DUH HDVLHU WR VROYH ZLWK PXWDEOH REMHFWV

7KH QDPH 6FDOD LV D FRQWUDFWLRQ RI WKH ZRUGV VFDODEOH
ODQJXDJH :KLOH WKLW VXJJHVWV WKDW WKH SURQXQFLDWLRQ
VKRXOG EH VFDOH DK WKH FUHDWRUV RI 6FDOD DFWXDOO\
SURQRXQFH LW VFDK ODK OLNH WKH ,WDOLDQ ZRUG IRU £VWDLUV
7KH WZR £DV DUH SURQRXQFHG WKH VDPH

6FDOD ZDV VWDUWHG E\ 0DUWLQ 2GHUVN\ LQ 0DUWLQ LV D
SURIHVVURU LQ WKH 6FKRRO RI &RPSXWHU DQG &RPPXQLFDWLRQ
6FLHQFHV DW WKH (FROH 3RO\WHFKQLTXH)pGpUDOH GH /DX VDQQH
(3)/ +H VSHQW KLV JUDGXDWH \HUV ZRUNLQJ LQ WKH JURXS
KHDGHG E\ 1LNODXV :LUWK RI 3DVDO IDPH 0DUWLQ ZRUNHG RQ
3LJJD DQ HDUO\ IXQFWLRQDO ODQJXDJH RQ WKH -90 +H ODWHU
ZRUNHG RQ *- D SURWRW\SH RI ZKDW ODWHU EHFDPH *HQHULFV LQ
-DYG ZLWK 3KLOLS :DGOHU RI +DVNHOO IDPH 0DUWLQ ZDV KLUHG E\
6XQ 0LFURV\WHPV WR SURGXFH WKH UHIHUUHQFH

LPSOHPHQWDWLRQ RI javac WKH -DYD FRPSLOHU WKDW VKLSV ZLWK
WKH -DYD 'HYHORSHU .LW
-'. WRGD\

0DUWLQ 2GHUVN\|V EDFNJURXQG DQG H[SHULHQFH DUH HYLGHQW
LQ WKH ODQJXDJH \$V \RX OHDUQ 6FDOD \RX FRPH WR
XQGHUVWDQG WKDW LW LV WKH SURGXFW RI FDUHIXOO\
FRQVLGHUHGH GHVLJQ GHFLVLRQV H[SORLWLQJ WKH VWDWH RI
WKH DUW LQ W\SH WKHRU\ 223 DQG)3 0DUWLQ\|V H[SH ULHQFH
ZLWK WKH -90 LV HYLGHQW LQ 6FDOD\|V HOHJDQW LQWHJUDWLRQ
ZLWK WKDW SODWIRUP 7KH \QWKHVLV LW FUHDWHV EHWZHHQ 223
DQG)3 LV DQ H[FHOHQW £EHVW RI ERWK ZRUOGV VROXWLRQ

The Seductions of Scala

7RGD\ RXU LQGXFVWU\ LV IRUWXQDWH WR KDYH D ZLGH YDULHW\ RI
ODQJXDJH RSWLRQV 7KH SRZHU IOH[LELOLW\ DQG HOHJDQFH RI
G\QDPLFDOO\ W\SHG ODQJXDJHV KDYH PDGH WKHP YHU\ SRSXODU
DJDLQ <HW WKH ZHDOWK RI -DYD DQG 1(7 OLEUDULHV DQG WKH
SHUIRUPDQFH RI WKH -90 DQG &/5 PHHW PDQ\ SUDFWLFDO QHHGV
IRU HQWHUSULVH DQG ,QWHUQHW SURMHFWV

6FDOD LV FRPSHOOLQJ EHFDXVH LW IHHOV OLNH D G\QDPLFDOO\
W\SHG VFULSWLQJ ODQJXDJH GXH WR LWV VXFFLQFW \QWD\ DQG
W\SH LQIHUHQFH <HW 6FDOD JLYHV \RX DOO WKH EHQHILWV RI
VWDWLF WSLQJ D PRGHUQ REMHFW PRGHO IXQFWLRQDO
SURJUDPPLQJ DQG DQ DGYDQFHG W\SH \VWHP 7KHVH WRROV
OHW \RX EXLOG VFDODEOH PRGXODU DSSOLFDWLRQV WKDW FDQ
UHXVH OHJDF\ -DYD DQG 1(7 \$3,V DQG OHYHUDJH WKH
SHUIRUPDQFH RI WKH -90 DQG &/5

6FDOD LV D ODQJXDJH IRU *SURIHVVLRQDO* GHYHORSHUV
&RPSDUHG WR ODQJXDJHV OLNH -DYD DQG 5XE\ 6FDOD LV D PRUH
GLIILFXOW ODQJXDJH WR PDVWHU EHFDXVH LW UHTXLUHV
FRPSHWHQF\ ZLWK 223)3 DQG VWDWLF WSLQJ WR XVH LW PRVW
HIIHFWLYHO\ ,W LV WHPSWLQJ WR SUHIHU WKH UHODWLYH
VLPSOLFLW\ RI G\QDPLFDOO\ W\SHG ODQJXDJHV <HW WKL
VLPSOLFLW\ FDQ EH GHFHSWLYH ,Q D G\QDPLFDOO\ W\SHG
ODQJXDJH LW LV RIWHQ QHFHVVDU\ WR XVH PHWDSURJUDPPLQJ
IHDWXUHV WR LPSOHPHQW DGYDQFHG GHVLJQV :KLOH
PHWDSURJUDPPLQJ LV SRZHUIXO XVLQJ LW ZHOO WDNHV
H[SHULHQFH DQG WKH UHVXOWLQJ FRGH WHQGV WR EH KDUG WR
XQGHUVWDQG PDLQWDLQ DQG GHEXJ

Why Scala? | 7

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,Q 6FDOD PDQ\ RI WKH VDPH GHVLJQ JRDOV FDQ EH DFKLHYHG LQ
D W\SH VDIH PDQQHU E\ H[SORLWLQJ LWV W\SH \VWHP DQG PL[LQ
FRPSRVLWLRQ WKURXJK *WUDLWV*

:H IHHO WKDW WKH H[WUD HIIRUW UHTXLUHG GD\ WR GD\ WR XVH
6FDOD ZLOO SURPRWH PRUH FDUHIXO UHIOHFWLRQ DERXW \RXU
GHVLJQV 2YHU WLPD WKLV GLVFLSOLQH ZLOO \LHOG PRUH
FRKHUHQW PRG XODU DQG PDLQWDLQDEOH DSSOLFDWLRQV

)RUWXQDWHO\ \RX GRQ\W QHHG DOO RI WKH VRSKLVWLFD WLRQ RI
6FDOD DOO RI WKH WLPH 0XFK RI \RXU FRGH ZLOO KDYH WKH
VLPSOLFLW\ DQG FODULW\ RI FRGH ZULWWHQ LQ \RXU IDYRULWH
G\QDPLFDOO\ WSHG ODQJXDJH

\$Q DOWHUQDWLYH VWUDWHJ\ LV WR FRPELQH VHYHUDO VLPSOHU
ODQJXDJHV H J -DYD IRU REMHFW RULHQWHG FRGH DQG (UODQJ
IRU IXQFWLRQDO FRQFXUUHQW FRGH 6XFK D GHFRPSRVLWLRQ
FDQ ZRUN EXW RQO\ LI \RXU \VWHP GHFRPSRVHV FOHDQO\ LQWR
VXFK GLVFUHHW SDUWV DQG \RXU WHDP FDQ PDQDJH D
KHWHURJHQHRXV HQYLURQPHQW 6FDOD LV DWWUDFWLYH IRU
VLWXDWLRQV LQ ZKLFK D VLQJOH DOO LQ RQH ODQJXDJH LV
SUHIHUHG 7KDW VDLG 6FDOD FRGH FDQ KDSSLO\ FR H\LVW ZLWK
RWKHU ODQJXDJHV HVSHFLDOO\ RQ WKH -90 RU 1(7

Installing Scala

7R JHW XS DQG UXQQQLQJ DV TXLFNO\ DV SRVVLEOH WKLV VHFWRUQ
GHVFULEHV KRZ WR LQVWDOO WKH FRPPDQG OLQH WRROV IRU
6FDOD ZKLFK DUH DOO \RX QHHG WR ZRUN ZLWK WKH H\DP SOHV LQ
WKH ERRN)RU GHWDLOV RQ XVLRQ 6FDOD LQ YDULRXV HGLWRUV
DQG '(V VHH [£,QWHJUDWLRQ ZLWK '\(V RQ SDJH](#) 7KH H\DP SOHV
XVHG LQ WKLV ERRN ZHUH ZULWWHQ DQG FRPSLOHG XVLRQ 6FDOD
YHUVLRQ ILQDO WKH ODWHVW UHOHDVH DW WKH WLPH RI
WKLV ZULWLQJ DQG £QLJKWO\ EXLOGV RI 6FDOD YHUVLRQ
ZKLFK PD\ EH ILQDOLJHG E\ WKH WLPH \RX UH DG WKLV

9HUVLRQ LQWURGXFHV PDQ\ QHZ IHDWXUHV ZKLFK ZH ZLOO
KLJKOLJKW
WKURXJKRXW WKH ERRN

:H ZLOO ZRUN ZLWK WKH -90 YHUVLRQ RI 6FDOD LQ WKLV ERRN
)LUVW \RX PXVW KDYH -DYD RU JUHDWHU LQVWDOOHG
RU JUHDWHU LV UHFRPPHQGHG ,I \RX QHHG WR LQVWDOO -DYD
JR WR [KWWS](#) [ZZZ MDYD FRP HQ GRZQORDG PDQXDO MVS](#) DQG
IROORZ WKH LQVWUXFWLRQV WR LQVWDOO -DYD RQ \RXU PDFKLQH
7KH RIILFLDO 6FDOD ZHEVLWH LV [KWWS](#) [ZZZ VFDOD ODQJ RUJ](#)
7R LQVWDOO 6FDOD JR WR WKH [GRZQORDGV SDJH](#) 'RZQORDG
WKH LQVWDOOHU IRU \RXU HQYLURQPHQW DQG IROORZ WKH
LQVWUXF WLRQV RQ WKH GRZQORDGV SDJH

7KH HDVLHVW FURVV SODWIRUP LQVWDOOHU LV WKH ,J3DFN
LQVWDOOHU 'RZQORDG WKH 6FDOD -\$5 ILOH HLWKHU VFDOD
ILQDO LQVWDOOHU MDU RU VFDOD 1 LQVWDOOHU MDU ZKHUH N
LV WKH ODWHVW UHOHDVH RI WKH YHUVLRQ *R WR WKH
GRZQORDG GLUHFWRU\ LQ D WHUPLQDO ZLQGRZ DQG LQVWDOO
6FDOD ZLWK WKH java FRPPDQG \$VVXPLQJ \RX GRZQORDGHG
VFDOD ILQDO LQVWDOOHU MDU UXQ WKH IROORZLQJ FRPPDQG
ZKLFK ZLOO JXLGH \RX WKURXJK WKH SURFHVV

java -jar scala-2.8.0.final-installer.jar



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2Q 0DF 26 ; WKH HDVLHVW URXWH WR D ZRUNLQJ 6FDOD LQVWDOODWLRQ
LV YLD

0DF3RUWV)ROORZ WKH LQVWDOODWLRQ LQVWUXFWLRQV DW [KWWS](#)
[ZZZ PDFSRUWV](#)

[RUJ](#) WKHQ sudo port install scala <RX\OO EH XS DQG UXQQLQJ LQ D IHZ
PLQXWHV

7KURXJKRXW WKLV ERRN ZH ZLOO XVH WKH \PERO *scala-home* WR
UHIHU WR WKH £URRW GLUHFWRU\ RI \RXU 6FDOD
LQVWDOODWLRQ

2Q 8QL[/LQX[DQG 0DF 26 ; \VWHPV \RX ZLOO QHHG WR UXQ WKLV
FRPPDQG DV WKH root XVHU RU XVLQJ WKH sudo FRPPDQG LI \RX ZDQW WR
LQVWDOO 6FDOD XQGHU D \VWHP GLUHFWRU\ H J *scala-home* *XVU ORFDO*
VFDOD ILQDO

\$V DQ DOWHUQDWLYH \RX FDQ GRZQORDG DQG H[SDQG WKH
FRPSUHVVHG 7\$5 ILOH

H J *VFDOD ILQDO WJ* RU =,3 ILOH

VFDOD ILQDO jLS 2Q 8QL[OLNH \VWHPV H[SDQG WKH
FRPSUHVVHG ILOH LQWR D ORFDWLRQ RI \RXU FKRRVLQJ
\$IWHUZDUG DGG WKH *scala-home* *ELQ* VXEGLUHFWRU\ LQ WKH QHZ
GLUHFWRU\ WR \RXU PATH)RU H[DPSOH LI \RX LQVWDOOHG
LQWR *XVU ORFDO VFDOD ILQDO* WKHQ DGG *XVU ORFDO VFDOD*
ILQDO ELQ WR \RXU PATH

7R WHVW \RXU LQVWDOODWLRQ UXQ WKH IROORZLQJ

FRPPDQG RQ WKH FRPPDQG OLQH *scala -version*

:H\OO OHDUQ PRUH DERXW WKH *scala* FRPPDQG OLQH WRRO ODWHU
<RX VKRXOG JHW VRPHWKLQJ OLNH WKH IROORZLQJ RXWSXW

Scala code runner version 2.8.0.final -- Copyright 2002-2009, LAMP/EPFL

2I FRXUVH WKH YHUVLRQ QXPEHU \RX VHH ZLOO EH GLIIHUHQW LI
\RX LQVWDOOHG D GLIIHUHQW UH OHDVH)URP QRZ RQ ZKHQ ZH
VKRZ FRPPDQG RXWSXW WKDW FRQWDLQV WKH YHUVLRQ QXPEHU
ZH\OO VKRZ LW DV version 2.8.0.final

&RQJUDWXODWLRQV \RX KDYH LQVWDOOHG 6FDOD ,I \RX JHW DQ
HUURU PHVVDJH DORQJ WKH OLQHV RI *scala: command not found* PDNH
VXUH \RXU HQYLURQPHQW\V PATH LV VHW SURSHU\ WR LQFOXGH
WKH FRUHHFW *ELQ* GLUHFWRU\

6FDOD YHUVLRQV ; DQG HDUOLHU DUH FRPSDWLEOH ZLWK -. DQG
ODWHU

6FDOD YHUVLRQ GURSV FRPSDWLEOLW 1RWH WKDW 6FDOD XVHV PDQ\
-. FODVVHV DV LWV RZQ IRU H[DPSOH WKH String FODVV 2Q 1(7 6FDOD
XVHV WKH FRUHHVSRQGLQJ 1(7 FODVVHV

<RX FDQ DOVR ILQG GRZQORDGV IRU WKH \$3, GRFXPHQWDWLRQ
DQG WKH VRXUFHV IRU 6FDOD LWVHOI RQ WKH VDPH GRZQORDGV
SDJH

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\$V \RX H[SORUH 6FDOD \RX ZLOO ILQG RWKHU XVHIXO UHVRXUFHV
WKDW DUH DYDLODEOH RQ [KWWWS](#) [VFDOD ODQJ RUJ](#) <RX ZLOO
ILQG OLQNV IRU GHYHORSPHQW VXSSRUW WRROV DQG OLEUDULHV
WXWRULDOV WKH ODQJXDJH VSHFLILFDWLRQ >[6FDOD6SHF](#) @
DQG DFDGHPLF SDSHUV WKDW GHVFULEH IHDWXUHV RI WKH
ODQJXDJH

7KH GRFXPHQWDWLRQ IRU WKH 6FDOD WRROV DQG \$3,V DUH
HVSHFLDOO\ XVHIXO <RX FDQ EURZVH WKH \$3, DW [KWWWS](#) [ZZZ](#)
[VFDOD ODQJ RUJ](#) [GRFX](#) [ILOHV](#) [DSL](#) [LQGH](#) [KWPO](#) 7KL
GRFXPHQWDWLRQ ZDV JHQHUDWHG XVLQJ WKH scaladoc WRRO
DQDORJRJV WR -DYD\ javadoc WRRO 6HH [£7KH VFDODGRF &RPPDQG](#)
[/LQH 7RRO](#) RQ [SDJH](#) IRU PRUH LQIRUPDWLRQ

<RX FDQ DOVR GRZQORDG D FRPSUHVHG ILOH RI WKH \$3,
GRFXPHQWDWLRQ IRU ORFDO EURZVLQJ XVLQJ WKH DSSURSULDWH
OLQN RQ WKH [GRZQORDGV SDJH](#) RU \RX FDQ LQVWDOO LW ZLWK
WKH sbaz SDFNDJH WRRO DV IROORZV

sbaz install scala-devel-docs

sbaz LV LQVWDOOHG LQ WKH VDPH [ELQ](#) GLUHFWRU\ DV WKH scala
DQG scalac FRPPDQG OLQH WRROV 7KH LQVWDOOHG
GRFXPHQWDWLRQ DOVR LQFOXGHV GHWDLOV RQ WKH VFDOD
WRRO FKDLQ
LQFOXGLQJ sbaz DQG FRGH H[DPSOHV)RU PRUH LQIRUPDWLRQ RQ
WKH 6FDOD FRPPDQG OLQH WRROV DQG RWKHU UHVRXUFHV VHH
[&KDSWHU](#)

A Taste of Scala

,W\ V WLPH WR ZKHW \RXU DSSHWLWH ZLWK VRPH UHDO 6FDOD
FRGH ,Q WKH IROORZLQJ H[DPSOHV ZH\OO GHVFULEH MXVW
HQRXJK RI WKH GHWDLOV VR \RX XQGHUVWDQG ZKDW\ V JRLQJ RQ
7KH JRDO LV WR JLYH \RX D VHQVH RI ZKDW SURJUDPPLQJ LQ 6FDOD
LV OLNH :H\OO H[SORUH WKH GHWDLOV RI WKH IHDWXUHV LQ
VXEVTXHQQW FKDSWHUV

)RU RXU ILUVW H[DPSOH \RX FRXOG UXQ LW RQH RI WZR ZD\ LQWHUFWLYHO\ RU DV D
£VFULSW

/HW\ V VWDUW ZLWK WKH LQWHUFWLYH PRGH 6WDUW WKH VFDOD
LQWHUSUHWHU E\ \SLQJ scala DQG WKH UHWXUQ NH\ RQ \RXU

FRPPDQG OLQH <RX!OO VHH WKH IROORZLQJ RXWSXW
6RPH RI WKH YHUVLRQ QXPEHUV PD\ YDU\

Welcome to Scala version 2.8.0.final (Java ...).
Type in expressions to have them evaluated.
Type :help for more information.

scala>

7KH ODVW OLQH LV WKH SURPSW WKDW LV ZDLWLQJ IRU \RXU
LQSW 7KH LQWHUDFWLYH PRGH RI WKH scala FRPPDQG LV YHU\
FRQYHQLHQW IRU H[SHULPHQWDWLRQ
VHH £7KH VFDOD &RPPDQG /LQH 7RRO= RQ SDJH IRU PRUH
GHWDLOV \$Q LQWHUDFWLYH LQWHUSUHWHU OLNH WKLV LV
FDOOHG D 5(3/ 5HDG (YDOXDWH 3ULQW /RRS

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7SH LQ WKH IROORZLQJ WZR OLQHV RI FRGH

```
val book = "Programming Scala"  
println(book)
```

7KH DFWXDO LQSW DQG RXWSXW VKRXOG ORRN OLNH WKH IROORZLQJ

```
scala> val book = "Programming Scala"  
book: java.lang.String = Programming Scala
```

```
scala> println(book)  
Programming Scala
```

scala>

7KH ILUVW OLQH XVHV WKH val NH\ZRUG WR GHFODUH D UHDG RQO\
YDULDEOH QDPHG book 1RWH WKDW WKH RXWSXW UHWXUQH G IURP
WKH LQWHUSUHWHU VKRZV \RX WKH WSH DQG YDOXH RI book 7KL
FDQ EH YHU\ KDGQ\ IRU XQGHUVWDQGLQJ FRPSOH\
GHFODUDWLRQV 7KH VHFRQG OLQH SULQVV WKH YDOXH RI book
ZKLFK LV £3URJUDPPLQJ 6FDOD=

([SHULPHQLQJ ZLWK WKH scala FRPPDQG LQ WKH LQWHUDFWLYH PRGH
5(3/
LV D JUHDW ZD\ WR OHDUQ WKH GHWDLOV RI 6FDOD

0DQ\ RI WKH H[DPSOHV LQ WKLV ERRN FDQ EH H[FXWHG LQ WKH
LQWHUSUHWHU OLNH WKLV +RZHYHU LW\V RIWHQ PRUH
FRQYHQLHQW WR XVH WKH VHFRQG RSWLRQ ZH PHQWLHQHG
ZULWLQJ 6FDOD VFULSWV LQ D WH[W HGLWRU RU ,(DQG H[FXWLQJ
WKHP ZLWK WKH VDPH scala FRPPDQG :H!OO GR WKDW IRU PRVW RI
WKH UHPDLQLQJ H[DPSOHV LQ WKLV FKDSWHU

,Q \RXU WH[W HGLWRU RI FKRLFH VDYH WKH 6FDOD FRGH LQ WKH
IROORZLQJ H[DPSOH WR D ILOH QDPHG XSSHU VFULSW VFDOD LQ D
GLUHFWRU\ RI \RXU FKRRVLQJ

```
// code-examples/IntroducingScala/upper1-script.scala
```

```
class Upper {  
  def upper(strings: String*): Seq[String] = {  
    strings.map((s:String) => s.toUpperCase())  
  }  
}  
  
val up = new Upper  
Console.println(up.upper("A", "First", "Scala", "Program"))
```

7KL V 6FDOD VFULSW FRQYHUWV VWULQJV WR XSSHUFDVH

%\ WKH ZD\ WKDW\ V D FRPPHQW RQ WKH ILUVW OLQH
ZLWK WKH QDPH RI WKH VRXUFH ILOH IRU WKH FRGH H[DPSOH
6FDOD IROORZV WKH VDPH FRPPHQW FRQYHQWLRQV DV -DYD & &
HWF \$ // comment JRHV WR WKH HQG RI D OLQH ZKLOH D /* comment */
FDQ FURVV OLQH ERXQGDULHV

A Taste of Scala | 11

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7R UXQ WKL V VFULSW JR WR D FRPPDQG ZLQGRZ FKDQJH WR WKH
VDPH GLUHFWRU\ DQG UXQ WKH IROORZLQJ FRPPDQG

```
scala upper1-script.scala
```

7KH ILOH LV LQWHUSUHHWG PHDQLQJ LW LV FRPSLOHG DQG
H[HFXXHG LQ RQH VVHS <RX VKRXOG JHW WKH IROORZLQJ
RXWSXW

```
Array(A, FIRST, SCALA, PROGRAM)
```

Interpreting Versus Compiling and Running Scala Code

7R VXPPDULJH LI \RX WSH scala RQ WKH FRPPDQG OLQH ZLWKRXW
D ILOH DUJXPHQW WKH LQWHUSUHHWU UXQV LQ LQWHUDFWLYH
PRGH <RX WSH LQ GHILQLWRQV DQG VWDWHPHQWV WKDW DUH
HYDOXDWHG RQ WKH IO\ ,I \RX JLYH WKH FRPPDQG D VFDOD
VRXUFH ILOH DUJXPHQW LW FRPSLOHV DQG UXQV WKH ILOH DV D
VFULSW DV LQ RXU scala upper1-script.scala H[DPSOH)LQDOO\ \RX
FDQ FRPSLOH 6FDOD ILOHV VHSODUDWHO\ DQG H[HFXXH WKH class
ILOH DV ORQJ DV LW KDV D main PHWKRJ MXVW DV \RX ZRXOG
QRUPDOO\ GR ZLWK WKH java FRPPDQG
:H\OO VKRZ DQ H[DPSOH VKRUWO\

7KHUH DUH VRPH VXEWOHWLHV \RX\OO QHHG WR XQGHUVWDQG
DERXW WKH OLPLDWRQV RI XVLRQ WKH LQWHUSUHHWU PRGHV
YHUVXV VHSODUDWH FRPSODWRQ DQG H[HFXXH VVHSV :H
GLVFXVV WKHVH VXEWOHWLHV LQ £&RPPDQG /LQH 7RROV RQ
SDJH

:KHQYHU ZH UHIHU WR H[HFXXH D VFULSW ZH PHDQ UXQQLQJ
D 6FDOD VRXUFH ILOH ZLWK WKH scala FRPPDQG

,Q WKH FXUUHQW H[DPSOH WKH upper PHWKRG LQ WKH Upper FODVV
QR SXQ LQWHQGHHG FRQYHUWV WKH LQSWV VWULQJV WR
XSSHUFDVH DQG UHWXUQV WKHP LQ DQ DUUD\ 7KH ODVW OLQH LQ
WKH H[DPSOH FRQYHUWV IRXU VWULQJV DQG SULQWV WKH
UHVXOWLQJ Array

/HW\ V H[DPLQH WKH FRGH LQ GHWDLO VR ZH FDQ EHJLQ WR OHDUQ
6FDOD \VQWD\ 7KHUH DUH D ORW RI GHWDLOV LQ MXVW VL\ OLQHV
RI FRGH :H\OO H[SODLQ WKH JHQHUDO LGHDV KHUH \$OO WKH
LGHDV XVHG LQ WKLV H[DPSOH ZLOO EH H[SODLQHG PRUH
WKRURXJKO\ LQ ODWHU VHFWRUQV RI WKH ERRN

,Q WKH H[DPSOH WKH Upper FODVV EHJLQV ZLWK WKH class NHZRUG
7KH FODVV ERG\ LV LQVLGH WKH RXWHUPRVW FXUO\ EUDFHV
{...}

7KH upper PHWKRG GHILQLWLRQ EHJLQV RQ WKH VHFRRG OLQH
ZLWK WKH def NHZRUG IROORZHG E\ WKH PHWKRG QDPH DQG DQ
DUJXPHQW OLVW WKH UHWXUQ WSH RI WKH PHWKRG DQ HTXDOV
VLJQ
= DQG WKHQ WKH PHWKRG ERG\

7KH DUJXPHQW OLVW LQ SDUHQWKHVV LV DFWXDOO\ D *YDULDEOH*
OHQJWK DUJXPHQW OLVW RI StringV LQGLFDWHG E\ WKH String* WSH
IROORZLQJ WKH FRORQ 7KDW LV \RX FDQ SDVV LQ DV PDQ\ FRPPD
VHSDUDWHG VWULQJV DV \RX ZDQW
LQFOXGLQJ DQ HPSW\ OLVW 7KHVH VWULQJV DUH VWRUHG LQ D
SDUDPHWHU QDPHG strings ,QVLGH WKH PHWKRG strings LV
DFWXDOO\ DQ Array

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:KHQ H[SOLFLW WSH LQIRUPDWLRQ IRU YDULDEOHV LV ZULWWHQ LQ WKH
FRGH WKHVH
WSH DQQRWDWLRQV IROORZ WKH FRORQ DIWHU WKH LWHP QDPH
L H 3DVDO OLNH
\VQWD\ :K\ GRHVQ\W 6FDOD IROORZ -DYD FRQYHQWLRQV" 5HFDOD WKDW WSH
LQIRUPDWLRQ LV RIWHQ *LQIHUHG* LQ 6FDOD
XQOLNH -DYD PHDQLQJ ZH GRQ\W
DOZDV VKRZ WSH DQQRWDWLRQV H[SOLFWO\ &RPSDUHG WR -DYD\W
type item FRQYHQWLRQ WKH item: type FRQYHQWLRQ LV HDVLHU IRU WKH FRP
SLOHU WR DQDO\JH XQDPXLJRXVO\ ZKHQ \RX RPLW WKH FRORQ DQG WKH WSH
DQQRWDWLRQ DQG MXVW ZULWH item

7KH PHWKRG UHWXUQ WSH DSSHUV DIWHU WKH DUJXPHQW OLVW
,Q WKLV FDVH WKH UHWXUQ WSH LV Seq[String] ZKHUH Seq
£VHTXHQFH LV D SDUWLFXODU NLQG RI FROOHFWLRQ ,W LV D
SDUDPH WHULJHG WSH
OLNH D *JHQHULF* WSH LQ -DYD SDUDPHWHULJHG KHUH ZLWK String
1RWH WKDW 6FDOD XVHV VTXDUH EUDFNHWV
[...] IRU SDUDPHWHULJHG WSHV ZKHUHDV -DYD XVHV DQJOH



6FDOD DOORZV DQJOH EUDFNHWV WR EH XVHG LQ PHWKRQ QDPHV H J
 QDPLQJ D
 £OHVV WKDQ PHWKRQ < LV FRPPRQ 6R WR DYRLG DPELJXLWLHV 6FDOD XVHV
 VTXDUH EUDFNHWV LQVWHDG IRU SDUDPHWHULJHG WSHV 7KH\ FDQ\W EH XVHG LQ
 PHWKRQ QDPHV \$OORZLQJ < DQG > LQ PHWKRQ QDPHV LV ZK\ 6FDOD GRHVQ\W
 IROORZ -DYD\ V FRQYHQWLRQ IRU DQJOH EUDFNHWV

7KH ERG\ RI WKH upper PHWKRQ FRPHV DIWHU WKH HTXDOV VLJQ
 = :K\ DQ HTXDOV VLJQ" :K\ QRW MXVW FXUO\ EUDFHV
 {...} OLNH LQ -DYD" %HFDXVH VHPLFRORQV IXQFWLRQ UHWXUQ
 WSHV PHWKRQ DUJXPHQWV OLVWV DQG HYHQ WKH FXUO\ EUDFHV
 DUH VRPHWLPHV RPLWWHG XVLQJ DQ HTXDOV VLJQ SUHYHQWV
 VHYHUDO SRVVLEOH SDUVLQJ DPELJXLWLHV 8VLQJ DQ HTXDOV
 VLJQ DOVR UHPLQGV XV WKDW HYHQ IXQFWLRQV DUH YDOXH V LQ
 6FDOD ZKLFK LV FRQVLVWHQW ZLWK 6FDOD\ V VXSSRUW RI
 IXQFWLRQDO SURJUDPPLQJ GHVFULEHG LQ PRUH GHWDLO LQ
[&KDSWHU](#)

7KH PHWKRQ ERG\ FDOOV WKH map PHWKRQ RQ WKH strings DUUD\
 ZKLFK WDNHV D IXQFWLRQ OLWHUDO DV DQ DUJXPHQW)XQFWLRQ
 OLWHUDOV DUH £DQRQ\PRXV PHWKRQV 7KH\ DUH VLPLODU WR
 ODPEGDV FORVXUHV EORFNV RU SURFV LQ RWKHU ODQJXDJHV ,Q
 -DYD \RX ZRXOG KDYH WR XVH DQ DQRQ\PRXV LQQHU FODVV KHUH
 WKDW LPSOHPHQWV D PHWKRQ GHILQH E\ DQ LQWHUIDFH HWF

,Q WKLV FDVH ZH SDVVHG LQ WKH IROORZLQJ IXQFWLRQ OLWHUDO

(s:String) => s.toUpperCase()

,W WDNHV DQ DUJXPHQW OLVW ZLWK D VLQJOH String DUJXPHQW
 QDPHG s 7KH ERG\ RI WKH IXQFWLRQ OLWHUDO LV DIWHU WKH
 £DUURZ PHWKRQ => ,W FDOOV toUpperCase() RQ s 7KH UHVXOW RI WKLV FDOO
 LV UHWXUQH E\ WKH IXQFWLRQ OLWHUDO ,Q 6FDOD WKH ODVW
 H[SUHVVLRQ LQ D IXQFWLRQ LV WKH UHWXUQ YDOXH DOWKRXJK
 \RX FDQ KDYH return VWDWHPHQWV HOVHZKHUH WRR 7KH return NH\
 ZRUG LV RSWLRQDO KHUH DQG LV UDUHO\ XVHG H[FHSW ZKHQ
 UHWXUQLQJ RXW RI WKH PLGGOH RI D EORFN
 H J LQ DQ if VWDWHPHQW



7KH YDOXH RI WKH ODVW H[SUHVVLRQ LV WKH GHIDXOW UHWXUQ YDOXH
 RI D IXQFWLRQ

1R return LV UHTXLUHG

6R map SDVVHV HDFK String LQ strings WR WKH IXQFWLRQ OLWHUDO
 DQG EXLOGV XS D QHZ FRO OHFWLRQ ZLWK WKH UHVXOWV

UHWXUQHGEI\WKHIXQFWLWRQOLWHUDO

7R H[HUFLVH WKH FRGH ZH FUHDWH D QHZ Upper LQVWDQFH DQG
DVVLJQ LW WR D YDULDEOH QDPHG up \$V LQ -DYD & DQG VLPLODU
ODQJXDJHV WKH \QWD[new Upper FUHDWHV D QHZ LQVWDQFH 7KH
up YDULDEOH LV GHFODUHG DV D UHDO RQ\ £YDOXH² XVLQJ WKH
val NHZRUG

```
)LQDOO\ ZH FDOO WKH upper PHWKRQ RQ D OLVW RI VWULQJV DQG  
SULQW RXW WKH UHVXOW ZLWK Console.println(...) ZKLFK LV  
HTXLYDOHQW WR -DYD\`V System.out.println(...)
```

:H FDQ DFWXDOO\ VLPSOLI\ RXU VFULSW HYHQ IXUWKHU &RQVLGHU
WKLW VLPSOLILHG YHUVLRQ RI WKH VFULSW

```
// code-examples/IntroducingScala/upper2-script.scala
```

```
object Upper {  
  def upper(strings: String*) = strings.map(_._toUpperCase())  
}
```

```
println(Upper.upper("A", "First", "Scala", "Program"))
```

7KLW FRGH GRHV H[DFWO\ WKH VDPH WKLQJ EXW ZLWK D WKLUG IHZHU
FKDUDFWHUV

2Q WKH ILUVW OLQH Upper LV QRZ GHFODUHG DV DQ object ZKLFK LV
D VLQJOHWRRQ :H DUH GH FODULQJ D FODVV EXW WKH 6FDOD
UXQWLPH ZLOO RQ\ HYHU FUHDWH RQH LQVWDQFH RI Upper
<RX FDQ\W ZULWH new Upper IRU H[DPSOH 6FDOD XVHV objects IRU
VLWXDWLRQV ZKHUH RWKHU ODQ JXDJHV ZRXOG XVH £FODVV
OHYHO² PHPEHUV OLNH staticV LQ -DYD :H GRQ\W UHDOO\ QHHG
PRUH WKDQ RQH LQVWDQFH KHUH VR D VLQJOHWRRQ LV ILQH



```
:K\ GRHVQ\W 6FDOD VXSSRUW staticV" 6LQFH HYHU\WKLQJ LV DQ  
REMHFW LQ
```

```
6FDOD WKH object FRQVWUXFW NHHSV WKLW SROLF\ FRQLVWHQW -DYD\`V static  
PHWKRQV DQG ILHOGV DUH QRW WLHG WR DQ DFWXDO LQVWDQFH
```

1RWH WKDW WKLW FRGH LV IXOO\ WKUHDG VDIH :H GRQ\W
GHFODUH DQ\ YDULDEOHV WKDW PLJKW FDXVH WKUHDG VDIHW\
LVVXH 7KH \$, PHWKRQV ZH XVH DUH DOVR WKUHDG VDIH
7KHUHIRUH ZH GRQ\W QHHG PXOWLSOH LQVWDQFHV \$
VLQJOHWRRQ object ZRUNV ILQH

7KH LPSOHPHQWDWLRQ RI upper RQ WKH VHFRQG OLQH LV DOVR
VLPSOHU 6FDOD FDQ XVDOO\ LQIHU WKH UHWXUQ WSH RI WKH
PHWKRQ

EXW QRW WKH WSHV RI WKH PHWKRQ DUJXPHQW VR ZH GURS
WKH H[SOLFLW GHFODUDWLRQ \$OVR EHFDXVH WKUH LV RQ\
RQH H[SUHVVLRQ LQ WKH PHWKRQ ERG\ ZH GURS WKH EUDFHV
DQG SXW WKH HQWLUH PHWKRQ GHILQLWRQ RQ RQH OLQH 7KH
HTXDOV VLQJ EHIRUH WKH PHWKRQ ERG\ WHOOV WKH FRPSLOHU
DV ZHOO DV WKH KXPQ UHGHU ZKHUH WKH PHWKRQ ERG\
EHJLQV

```
:H KDYH DOVR H[SORLWHG D VKRUWKDQG IRU WKH IXQFWLRQ
OLWHUDO 3UHYLRXVO\ ZH ZURWH LW DV IROORZV
```

```
(s:String) => s.toUpperCase()
```

```
:H FDQ VKRUWHQ LW WR WKH IROORZLQJ H[SUHVVLQJ
```

```
_.toUpperCase()
```

```
%HFDXVH map WDNHV RQH DUJXPHQW D IXQFWLRQ ZH FDQ XVH
WKH £SODFHKROGHU LQGLFDWRU _ LQVWHDG RI D QDPHG
SDUDPHWHU 7KDW LV WKH _ DFWV OLNH DQ DQRQPRXV
YDULDEOH WR ZLFK HDFK VWULQJ ZLOO EH DVVLJQH EHIRUH
toUpperCase LV FDOOHG 1RWH WKDW WKH String WSH LV LQIHUHG
IRU XV WRR $V ZH ZLOO VHH 6FDOD XVHV _ DV D £ZLOGFDUG LQ
VHYUDO FRQWH[WV
```

```
<RX FDQ DOVR XVH WKLW VKRUWKDQG \QWD[ LQ VRPH PRUH
FRPSOH[ IXQFWLRQ OLWHUDOV DV ZH ZLOO VHH LQ &KDSWHU
```

```
2Q WKH ODVW OLQH XVLQJ DQ object UDWKHU WKDQ D class
VLPSOLILHV WKH FRGH ,QVWHDG RI FUHDWLQJ DQ LQVWDQH ZLWK
new Upper ZH FDQ MXVW FDOO WKH upper PHWKRQ RQ WKH Upper
REMHFW GLUHFWO\
QRWH KRZ WKLW ORRN OLNH WKH \QWD[ \RX ZRXOG XVH ZKHQ
FDOOLQJ VWDWLW PHWKRQV LQ D -DYD FODV
```

```
)LQDOO\ 6FDOD DXWRPDWLFDOO\ LPSRUWV PDQ\ PHWKRQV IRU , 2
OLNH println VR ZH GRQ'W QHHG WR FDOO Console.println() :H FDQ
MXVW XVH println E\ LWVHOI
```

```
6HH £7KH 3UHGHI 2EMHFW RQ SDJH IRU GHWDLOV RQ WKH WSHV
DQG PHWKRQV WKDW DUH DXWRPDWLFDOO\ LP SRUWHG RU
GHILQH
```

```
/HW\ GR RQH ODVW UHIDFWRLQJ &RQYHUW WKH VFULSW LQWR
```

```
D FRPSLOHG FRPPDQG OLQH WRRO //
```

```
code-examples/IntroducingScala/upper3.scala
```

```
object Upper {
  def main(args: Array[String]) = {
    args.map(_.toUpperCase()).foreach(printf("%s ", _))
    println("")
  }
}
```

```
1RZ WKH upper PHWKRQ KDV EHHQ UHQDPHG main %HFDXVH Upper LV
DQ object WKLW main PHWKRQ ZRUNV H[DFWO\ OLNH D static main
PHWKRQ LQ D -DYD FODV ,W LV WKH HQWU\ SRLQW WR WKH Upper
DSSOLFDWLRQ
```

```
,Q 6FDOD main PXVW EH D PHWKRQ LQ DQ object
```

```
,Q -DYD main PXVW EH D
```

```
static PHWKRQ LQ D class 7KH FRPPDQG OLQH DUJXPHQWV IRU WKH DSSOL
FDWLRQ DUH SDVVHG WR main LQ DQ DUUD\ RI VWULQJV H J args: Array[String]
```

```
7KH ILUVW OLQH LQVLGH WKH main PHWKRQ XVHV WKH VDPH
VKRUWKDQG QRWDWLRQ IRU map WKDW ZH MXVW H[DPLQH
```



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```
7KH FDOO WR map UHWXUQV D QHZ FROOHFWLRQ :H LWHUDWH
WKURXJK LW ZLWK foreach :H XVH D _ SODFHKROGHU VKRUWFXW
DJDQLQ LQ DQRWKHU IXQFWLRQ OLWHUDO WKDW ZH SDVV WR foreach
,Q WKLW FDVH HDFK VWULQJ LQ WKH FROOHFWLRQ LV SDVVHG DV
DQ DUJXPHQW WR printf
```

```
...foreach(printf("%s ",_))
```

```
7R EH FOHDU WKHVH WZR XVHV RI _ DUH FRPSOHWHO\
LQGHSHQGHHQW RI HDFK RWKHU 0HWKRG FKDLQLQJ DQG IXQFWLRQ
OLWHUDO VKRUWKDQGV DV LQ WKLW H[DPSOH FDQ WDNH VRPH
JHWWLQJ XVHG WR EXW RQFH \RX DUH FRPIRUWDEOH ZLWK WKHP
WKH\ \LHOG YHU\ UHGDDEOH FRGH ZLWK PLQLPDO XVH RI
WHPSRUDU\ YDULDEOHV
```

```
7KH ODVW OLQH LQ main DGGV D ILQDO OLQH IHHG WR WKH RXWSXW
7KLW WLPH \RX PXVW ILUVW FRPSLOH WKH FRGH WR D -90
```

```
FODVV ILOH XVLQJ scalac scalac upper3.scala
```

```
<RX VKRXOG QRZ KDYH D ILOH QDPHG 8SSHU FODVV MXVW DV LI \RX
KDG MXVW FRPSLOHG D -DYD FODVV
```



```
<RX PD\ KDYH QRWLFHG WKDW WKH FRPSLOHU GLG QRW FRPSODLQ ZKHQ
WKH
ILOH ZDV QDPHG XSSHU VFDOD DQG WKH object ZDV QDPHG Upper 8QOLNH
-DYD WKH ILOH QDPH GRHVQ\W KDYH WR PDWFK WKH QDPH RI WKH WASH ZLWK
public VFRSH
:H\OO H[SORUH WKH YLVLELOLW\ UXOHV LQ £9LVLELOLW\
5XOHV RQ SDJH ,Q IDFW XQOLNH -DYD \RX FDQ KDYH DV PDQ\ SXEOLF
WASHV LQ D VLQJOH ILOH DV \RX ZDQW \XUWKHUPRUH WKH GLUHFWRU\ ORFDWLRQ
RI
D ILOH GRHVQ\W KDYH WR PDWFK WKH SDFNDJH GHFODUDWLRQ +RZHYHU \RX FDQ
FHUWDLQO\ IROORZ WKH -DYD FRQYHQWLRQV LI \RX ZDQW WR
```

```
1RZ \RX FDQ H[FXWH WKLW FRPPDQG IRU DQ\ OLVW RI
```

```
VWULQJV +HUH LV DQ H[DPSOH scala -cp . Upper Hello World!
```

```
7KH -cp . RSWLRQ DGGV WKH FXUUHQW GLUHFWRU\ WR WKH VHDFK
£FODVV SDWK R <RX VKRXOG JHW WKH IROORZLQJ RXWSXW
```

```
HELLO WORLD!
```

```
7KHUHIRUH ZH KDYH PHW WKH UHTXLUHPHQW WKDW D
SURJUDPPLQJ ODQJXDJH ERRN PXVW VWDUW ZLWK D £KHOOR
ZRUOG R SURJUDP
```

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```
7KHUH DUH PDQ\ UHIVRQV WR EH VHGXFHG E\ 6FDOD 2QH UHIVRQ
```

LV WKH \$FWRUV \$3, LQFOXGHG LQ WKH 6FDOD OLEUDU\ ZKLFK LV
EDVHG RQ WKH UREXVW \$FWRUV FRQFXUUHQF\ PRGHO EXLOW
LQWR (UODQJ
VHH >+DOOHU @ +HUH LV DQ H[DPSOH WR ZKHW \RXU DSSHWLWH
,Q WKH \$FWRU PRGHO RI FRQFXUUHQF\
>\$JKD @ LQGHSHQGHW VRIWZDUH HQWLWLHV FDOOHG \$FWRUV
VKDUH QR VWDWH LQIRUPDWLRQ ZLWK HDFK RWKHU ,QVWHDG
WKH\ FRPPXQLFDWH E\

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H[FKDQJLQJ PHVVDJHV %\ HOLPLQDWLQJ WKH QHHG WR
\QFKURQLJH DFFHVV WR VKDUHG PXWDEOH VWDWH LW LV IDU
HDVLHU WR ZULWH UREXVW FRQFXUUHQW DSSOLFDWLRQV
,Q WKLV H[DPSOH LQVWDQFHV LQ D JHRPHWULF Shape KLHUDUFK\
DUH VHQW WR DQ \$FWRU IRU GUDZ LQJ RQ D GLVSOD\ ,PDJLQH D
VFHQDULR ZKHUH D UHQGHULQJ £IDUP± JHQHUDWHV VFHQHV LQ DQ
DQLPDWLRQ \$V WKH UHQGHULQJ RI D VFHQH LV FRPSOHWHG WKH
VKDSH £SULPLWLYHV± WKDW DUH SDUW RI WKH VFHQH DUH VHQW
WR DQ \$FWRU IRU D GLVSOD\ VXEVVWHP

7R EHJLQ ZH GHILQH D Shape FODVV KLHUDUFK\

[// code-examples/IntroducingScala/shapes.scala](#)

```
package shapes {
  class Point(val x: Double, val y: Double) {
    override def toString() = "Point(" + x + "," + y + ")"
  }

  abstract class Shape() {
    def draw(): Unit
  }

  class Circle(val center: Point, val radius: Double) extends Shape { def draw()
= println("Circle.draw: " + this)
    override def toString() = "Circle(" + center + "," + radius + ")" }

  class Rectangle(val lowerLeft: Point, val height: Double, val width: Double) extends
Shape {
    def draw() = println("Rectangle.draw: " + this)
    override def toString() =
      "Rectangle(" + lowerLeft + "," + height + "," + width + ")"
  }

  class Triangle(val point1: Point, val point2: Point, val point3: Point) extends
Shape {
    def draw() = println("Triangle.draw: " + this)
    override def toString() =
      "Triangle(" + point1 + "," + point2 + "," + point3 + ")"
  }
}
```

7KH Shape FODVV KLHUDUFK\ LV GHILQHG LQ D shapes SDFNDJH <RX
FDQ GHFODUH WKH SDFNDJH XVLQJ -DYD \QWD[EXW 6FDOD DOVR
VXSSRUWV D \QWD[VLPLODU WR & \V £QDPHVSDFH± \QWD[
ZKHUH WKH HQWLUH GHFODUDWLRQ LV VFRSHG XVLQJ FXUO\
EUDFHV DV XVHG KHUH 7KH -DYD VWOH SDFNDJH GHFODUDWLRQ

VQWD[LV IDU PRUH FRPPRQ\ XVHG KRZHYHU EHLQJ ERWK
FRPSDFW DQG UHGDDEOH

7KH Point FODVV UHSUHVHQWV D WZR GLPHQVLRQDO SRLQW RQ D
SODQH 1RWH WKH DUJXPHQW OLVW DIWHU WKH FODVV QDPH
7KRVH DUH FRQVWUXFWRU SDUDPHWHUV ,Q 6FDOD WKH ZKROH
FODVV ERG\ LV WKH FRQVWUXFWRU VR\RX OLVW WKH DUJXPHQWV
IRU WKH *SULPDU* FRQVWUXFWRU DIWHU WKH FODVV QDPH DQG
EHIRUH WKH FODVV ERG\
:H\OO VHH KRZ WR GHILQH DX[LOLDU\ FRQVWUXFWRUV LQ
[£&RQVWUXFWRUV LQ 6FDOD RQ SDJH](#) %HFDXVH ZH SXW WKH val
NH\ZRUG EHIRUH HDFK SDUDPHWHU GHFODUDWLRQ WKH\ DUH
DXWRPDWLFDOO\ FRQYHUWHG WR UHGD RQO\ ILHOGV ZLWK WKH

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VDPH QDPHV ZLWK SXEOLF UHGDHU PHWKRGV RI WKH VDPH QDPH
7KDW LV ZKHQ\RX LQVWDQJ WLDWH D Point LQVWDQFH H J point \RX
FDQ UHGD WKH ILHOGV XVLQJ point.x DQG point.y ,I \RX ZDQW
PXWDEOH ILHOGV WKHQ XVH WKH NH\ZRUG var :H\OO H[SORUH
YDULDEOH GHFODUDWLRQV DQG WKH val DQG var NH\ZRUGV LQ
[£9DULDEOH 'HFODUDWLRQV RQ SDJH](#)

7KH ERG\ RI Point GHILQHV RQH PHWKRG DQ *RYHUULGH* RI WKH
IDPLOLDU toString PHWKRG LQ -DYD
OLNH ToString LQ & 1RWH WKDW 6FDOD OLNH & UHTXLUHV WKH
override NH\ZRUG ZKHQHYHU \RX RYHUULGH D FRQFUHWH PHWKRG
8QOLNH & \RX GRQ\W KDYH WR XVH D virtual NH\ZRUG RQ WKH
RULJLQDO FRQFUHWH PHWKRG ,Q IDFW WKHUH LV QR virtual
NH\ZRUG LQ 6FDOD \$V EHIRUH ZH RPLW WKH FXUO\ EUDFHV
{...} DURXQG WKH ERG\ RI toString VLQFH LW KDV RQO\ RQH
H[SUHVVLRQ

Shape LV DQ DEVWUDFW FODVV \$EVWUDFW FODVVHV LQ 6FDOD DUH
VLPLODU WR WKRWH LQ -DYD DQG & :H FDQ\W LQVWDQJWLDWH
LQVWDQFHV RI DEVWUDFW FODVVHV HYHQ ZKHQ DOO WKHLU
ILHOG DQG PHWKRG PHPEHUV DUH FRQFUHWH

,Q WKLW FDVH 6KDSH GHFODUHV DQ *DEVWUDFW* draw PHWKRG :H
NQRZ LW LV DEVWUDFW EHFDXVH LW KDV QR ERG\ 1R abstract
NH\ZRUG LV UHTXLUHG RQ WKH PHWKRG \$EVWUDFW PHWKRGV LQ
6FDOD DUH MXVW OLNH DEVWUDFW PHWKRGV LQ -DYD DQG &
6HH [£2YHUULGLQJ 0HEHUV RI &ODVVHV DQG 7UDLWV RQ SDJH](#)
IRU PRUH GHWDLOV

7KH draw PHWKRG UHWXUQV Unit ZKLFK LV D WASH WKDW LV URXJKO\
HTXLYDOHQW WR void LQ & GHULYHG ODQJXDJHV OLNH -DYD HWF
6HH [£7KH 6FDOD 7ASH +LHUDUFK RQ SDJH](#) IRU PRUH GHWDLOV

Circle LV GHFODUHG DV D FRQFUHWH VXEFOFVV RI Shape ,W GHILQHV
WKH draw PHWKRG WR VLP SO\ SULQW D PHVVDJH WR WKH FRQVROH
Circle DOVR RYHUULGHV toString

Rectangle LV DOVR D FRQFUHWH VXEFOFVV RI Shape WKDW
GHILQHV draw DQG RYHUULGHV toString)RU VLPSOLFLW ZH DVVXPH
LW LV QRW URWDWHG UHODWLYH WR WKH [DQG \D]HV +HQFH DOO

ZH QHHG LV RQH SRLQW WKH ORZHU OHIWKDQG SRLQW ZLOO GR
DQG WKH KHLJKW DQG ZLGWK RI WKH UHFWDQJOH

Triangle IROORZV WKH VDPH SDWWHUQ ,W WDNHV WKUHH Points DV LWV FRQVWUXFWRU
DUJXPHQVV

%RWK draw PHWKRGV LQ Circle Rectangle DQG Triangle XVH this \$V LQ
-DYD DQG & this LV KRZ DQ LQVWDQFH UHIHUV WR LWVHOI ,Q WKLV
FRQWH[W ZKHUH this LV WKH ULJKWKDQG VLGH RI D 6WULQJ
FRQFDWHQDWLRQ H[SUHVVLRQ
XVLQJ WKH SOXV VLQJ this.toString LV LQYRNHG LPSOLFLWO\



2I FRXUVH LQ D UHDO DSSOLFDWLRQ \RX ZRXOG QRW LPSOHPHQW
GUDZLQJ LQ

£GRPDQ PRGHO FODVVHV OLNH WKLV VLQFH WKH LPSOHPHQWDWLRQV ZRXOG
GH

SHQG RQ GHWDLOV OLNH WKH RSHUDWLQJ \VWHP SODWIRUP JUDSKLFV \$3, HWF
:H ZLOO VHH D EHWWHU GHVLJQ DSSURDFK ZKHQ ZH GLVFXVV WUDLWLQ
&KDSWHU

1RZ WKDW ZH KDYH GHILQH RXU VKDSHV WSHV OHW\U UHWXUQ
WR \$FWRUV :H GHILQH DQ \$FWRU WKDW UHFHLYHV £PHVVDJHV
WKDW DUH VKDSHV WR GUDZ

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[// code-examples/IntroducingScala/shapes-actor.scala](#)

```
package shapes {  
  import scala.actors._  
  import scala.actors.Actor._  
  
  object ShapeDrawingActor extends Actor {  
    def act() {  
      loop {  
        receive {  
          case s: Shape => s.draw()  
          case "exit" => println("exiting..."); exit  
          case x: Any => println("Error: Unknown message! " + x)  
        }  
      }  
    }  
  }  
}
```

7KH \$FWRU LV GHFODUHG WR EH SDUW RI WKH shapes SDFNDJH
1H[W ZH KDYH WZR LPSRUW VWDWHPHQWV

7KH ILUVW LPSRUW VWDWHPHQW LPSRUWV DOO WKH WSHV LQ
WKH scala.actors SDFNDJH ,Q 6FDOD WKH XQGHUVFRUH _ LV XVHG
WKH ZD\ WKH VWDU * LV XVHG LQ -DYD



%HFDXVH * LV D YDOLG FKDUFDWHU IRU D IXQFWLRQ QDPH LW FDQ\W EH
XVHG DV WKH
import ZLOGFDUG ,QVWHDG _ LV UHVHUYHG IRU WKLV SXUSRVH

\$OO WKH PHWKRGV DQG SXEOLF ILHOGV IURP Actor DUH LPSRUWHG
E\ WKH VHFRQG import 7KHVH DUH QRW static LPSRUWV IURP WKH Actor
WASH DV WKH\ ZRXOG EH LQ -DYG 5DWKHU WKH\ DUH LPSRUWHG
IURP DQ object WKDW LV DOVR QDPHG Actor 7KH class DQG object FDQ
KDYH WKH VDPH QDPH DV ZH ZLOO VHH LQ [£&RPSDQLRQ 2EMHFVV](#)
[RQ SDJH](#)

2XU \$FWRU FODVV GHILQLWLRQ ShapeDrawingActor LV DQ object WKDW
H\WHQGV Actor
WKH WASH QRW WKH object 7KH act PHWKRG LV RYHUULGGHQ WR GR
WKH XQLTXH ZRUN RI WKH \$FWRU %HFDXVH act LV DQ DEVWUDFW
PHWKRG ZH GRQ\W QHHG WR H\SOLFLWO\ RYHUULGH LW ZLWK WKH
override NH\ZRUG 2XU \$FWRU ORRSV LQGHILQLWHO\ ZDLWLQJ IRU
LQFRPLQJ PHVVDJHV

'XULQJ HDFK SDVV LQ WKH ORRS WKH receive PHWKRG LV FDOOHG
,W EORFNV XQWLO D QHZ PHVVDJH DUULYHV :K\ LV WKH FRGH
DIWHU receive HQFORVHG LQ FXUO\ EUDFHV {...} DQG QRW
SDUHQWKHVVH (...)" :H ZLOO OHDUQ ODWHU WKDW WKHUH DUH
FDVHV ZKHU WKLV VXEVLWXWLRQ LV DOORZHG DQG LV TXLWH
XVHIXO

VHH [&KDSWHU](#))RU QRZ ZKD\RX QHHG WR NQRZ LV WKDW WKH
H\SUHVVLQV LQVLGH WKH EUDFHV FRQVLWXWH D VLQJOH
IXQFWLRQ OLWHUDO WKDW LV SDVVHG WR receive 7KLXQF WLRQ
OLWHUDO GRHV D *SDWWHUQ PDWFK* RQ WKH PHVVDJH LQVWDQFH
WR GHFLGH KRZ WR KDQGOH WKH PHVVDJH %HFDXVH RI WKH case
FODXVHV LW ORRNV OLNH D WSLFDO VZLWFK VWDWHPHQW LQ
-DYG IRU H\DPSON DQG WKH EH KDYL RU LV YHU\ VLPLODU

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7KH ILUVW case GRHV D WASH FRPSDULVRQ ZLWK WKH PHVVDJH
7KHUH LV QR H\SOLFLW YDULDEOH IRU WKH PHVVDJH LQVWDQFH LQ
WKH FRGH LW LV LQIHUUHG ,I WKH PHVVDJH LV RI WASH Shape WKH
ILUVW case PDWFKHV 7KH PHVVDJH LQVWDQFH LV FDVW WR D Shape
DQG DVVLJQH WR WKH YDULDEOH s DQG WKHQ WKH draw PHWKRG
LV FDOOHG RQ LW

,I WKH PHVVDJH LV QRW D Shape WKH VHFRQG case LV WULHG ,I
WKH PHVVDJH LV WKH VWULQJ "exit" WKH \$FWRU SULQVV D
PHVVDJH DQG WHUPLQDWHV H\HFXWLRQ \$FWRUV VKRXOG
XVXDOO\ KDYH D ZD\ WR H\LW JUDFHIXOO\

7KH ODVW case FODXVH KDQGOHV DQ\ RWKHU PHVVDJH LQVWDQFH
WKHUHE\ IXQFWLRQLQJ DV WKH *GHIDXOW* FDVH 7KH \$FWRU
UHSRUWV DQ HUURU DQG WKHQ GURSV WKH PHVVDJH Any LV WKH
SDUHQW RI DOO WASHV LQ WKH 6FDOD WASH KLHUDUFK\ OLNH Object
LV WKH URRW WASH LQ -DYG DQG RWKHU ODQJXDJHV +HQFH WKLV
FDVH FODXVH ZLOO PDWFK DQ\ PHVVDJH RI DQ\ WASH 3DWWHUQ
PDWFKLQJ LV HDJHU ZH KDYH WR SXW WKLV FDVH FODXVH DW WKH

HQG VR LW GRHVQIW FRQVXP WKH PHVVDJHV ZH DUH H[SHFWLQJ
 5HFDOO WKDW ZH GHFODUHG draw DV DQ DEWUDFW PHWKRG
 LQ Shape DQG ZH LPSOHPHQWHG draw LQ WKH FRQFUHWH
 VXEFODVVHV +HQFH WKH FRGH LQ WKH ILUVW case VWDWHPHQW
 LQYRNHV D SROI\PRUSKLF RSHUDWLRQ

Pattern Matching Versus Polymorphism

3DWWHUQ PDWFKLQJ SODV D FHQWUDO UROH LQ IXQFWLRQDO
 SURJUDPPLQJ MXVW DV SROI\PRUSKLV SODV D FHQWUDO UROH
 LQ REMHFW RULHQWHG SURJUDPPLQJ)XQFWLRQDO SDWWHUQ
 PDWFKLQJ LV PXFK PRUH LPSRUWDQW DQG VRSKLVWLFDWHG
 WKDQ WKH FRUHVSRQGLQJ switch/case VWDWH PHQWV IRXQG LQ
 PRVW LPSHUDWLYH ODQJXDJHV OLNH -DYD :H ZLOO H[DPLQH
 6FDODV VXSSRUW IRU SDWWHUQ PDWFKLQJ LQ PRUH GHWDLO
 LQ &KDSWHU ,Q RXU H[DPSOH KHUH ZH FDQ EHJLQ WR VHH
 WKDW MRLQLQJ IXQFWLRQDO VW\OH SDWWHUQ PDWFKLQJ ZLWK
 REMHFW RULHQWHG SROI\PRUSKLF GLVSDWFKLQJ LV D SRZHUOXO
 FRPELQDWLRQ WKDW LV D EHQHILW RI PL[HG SDUDGLJP
 ODQJXDJHV OLNH 6FDOD

)LQDOO\ KHUH LV D VFULSW WKDW XVHV WKH ShapeDrawingActor \$FWRU

[// code-examples/IntroducingScala/shapes-actor-script.scala](#)

```
import shapes._
```

```
ShapeDrawingActor.start()
```

```
ShapeDrawingActor ! new Circle(new Point(0.0,0.0), 1.0)
ShapeDrawingActor ! new Rectangle(new Point(0.0,0.0), 2, 5)
ShapeDrawingActor ! new Triangle(new Point(0.0,0.0),
  new Point(1.0,0.0),
  new Point(0.0,1.0))
ShapeDrawingActor ! 3.14159
```

```
ShapeDrawingActor ! "exit"
```

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7KH VKDSHV LQ WKH shapes SDFNDJH DUH LPSRUWHG

7KH ShapeDrawingActor \$FWRU LV VWDUWHG %\ GHIDXOW LW UXQV LQ
 LWV RZQ WKUHDG

WKHUH DUH DOWHUQDWLYHV ZH ZLOO GLVFXVV LQ &KDSWHU
 ZDLWLQJ IRU PHVVDJHV

)LYH PHVVDJHV DUH VHQW WR WKH \$FWRU XVLQJ WKH \QWD[actor !
 message 7KH ILUVW PHVVDJH VHQQV D Circle LQVWDQFH 7KH \$FWRU
 £GUDZV WKH FLUFOH 7KH VHFRQG PHVVDJH VHQQV D Rectangle
 PHVVDJH 7KH \$FWRU £GUDZV WKH UHFWDQJOH 7KH WKLUG
 PHVVDJH GRHV WKH VDPH WKLQJ IRU D Triangle 7KH IRXUWK
 PHVVDJH VHQQV D Double WKDW LV DSSUR[LPDWHO\ HTXDO WR 3L
 7KLW LV DQ XQNQRZQ PHVVDJH IRU WKH \$FWRU VR LW MXVW
 SULQWV DQ HUURU PHVVDJH 7KH ILQDO PHVVDJH VHQQV DQ £H[LW
 VWULQJ ZKLFK FDXVHV WKH \$FWRU WR WHUPLQDWH

```

7R WU\ RXW WKH $FWRU H[DP SOH VWDUW E\ FRPSLOLQJ WKH
ILUVW WZR ILOHV <RX FDQ JHW WKH VRXUFHV IURP WKH
2;5HLOO\ GRZQORDG VLWH
VHH £*HWWLQJ WKH &RGH ([DP SOHV RQ SDJH [L IRU GHWDLOV
RU \RX FDQ FUHDWH WKHP \RXUVHOI

8VH WKH IROORZLQJ FRPPDQG WR FRPSLOH WKH ILOHV

```

```
scalac shapes.scala shapes-actor.scala
```

```

:KLOH WKH VRXUFH ILOH QDPHV DQG ORFDWLRQV GRQ!W KDYH WR
PDWFK WKH ILOH FRQWHQWV \RX ZLOO QRWLFH WKDW WKH
JHQHUDWHG FODVV ILOHV DUH ZULWWHQ WR D VKDSHV GLUHFWRU\
DQG WKHUH LV RQH FODVV ILOH IRU HDFK FODVV ZH GHILQH 7KH
FODVV ILOH QDPHV DQG ORFDWLRQV PXVW FRQIRUP WR WKH -90
UHTXLUHPHQWV

```

```
1RZ \RX FDQ UXQ WKH VFULSW WR VHH WKH $FWRU LQ DFWLRQ
```

```
scala -cp . shapes-actor-script.scala
```

```
<RX VKRXOG VHH WKH IROORZLQJ RXWSXW
```

```

Circle.draw: Circle(Point(0.0,0.0),1.0)
Rectangle.draw: Rectangle(Point(0.0,0.0),2.0,5.0)
Triangle.draw: Triangle(Point(0.0,0.0),Point(1.0,0.0),Point(0.0,1.0)) Error:
Unknown message! 3.14159
exiting...

```

```
)RU PRUH RQ $FWRUV VHH &KDSWHU
```

Recap and What's Next

```

:H PDGH WKH FDVH IRU 6FDOD DQG JRW \RX VWDUWHG ZLWK WZR
VDP SOH 6FDOD SURJUDPV RQH RI ZKLFK JDYH \RX D WDVWH RI
6FDOD!V $FWRUV OLEUDU\ IRU FRQFXUUHQF\ 1H[W ZH!OO GLYH
LQWR PRUH 6FDOD \QWD[ HPSKDV]LQJ YDULRXV NH\WURNH
HFRQRPLFDO ZD\ RI JHWWLQJ ORVV RI ZRUN GRQH

```

Recap and What's Next | 21

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CHAPTER 2 Type Less, Do More

In This Chapter

:H HQGHG WKH SUHYLRXV FKDSWHU ZLWK D IHZ £WHDVHU^α
H[DPSOHV RI 6FDOD FRGH 7KLV FKDSWHU GLVFXVVHV XVHV RI
6FDOD WKDW SURPRWH VXFFLQFW IOH[LEOH FRGH :H!OO GLVFXVV
RUJDQLJDWLRQ RI ILOHV DQG SDFNDJHV LPSRUWLQJ RWKHU WSHV
YDULDEOH GHFODUDWLRQV PLVFHOODQHRXV \!Q WD[
FRQYHQWLRQV DQG D IHZ RWKHU FRQFHSWV :H!OO HPSKDV\JH KRZ
WKH FRQFLVH \!QWD[RI 6FDOD KHOSV \RX ZRUN EHWWHU DQG
IDVWHU

6FDOD\!V \!QWD[LV HVSHFLDOO\ XVHIXO ZKHQ ZULWLQJ VFULSWV
6HSDUDWH FRPSLOH DQG UXQ VWHSV DUHQ\W UHTXLUHG IRU
VLPSOH SURJUDPV WKDW KDYH IHZ GHSHQGHQFLHV RQ OLEUDULHV
RXWVLGH RI ZKDW 6FDOD SURYLGHV <RX FRPSLOH DQG UXQ VXFK
SURJUDPV LQ RQH VKRW ZLWK WKH scala FRPPDQG ,I \RX\YH
GRZQORDGHG WKH H[DPSOH FRGH IRU WKLW ERRN PDQ\ RI WKH
VPDOOHU H[DPSOHV FDQ EH UXQ XVLQJ WKH scala FRPPDQG H J
scala *filename.scala* 6HH WKH 5(\$'0(W\W ILOHV LQ HDFK FKDSWHU\!V
FRGH H[DPSOHV IRU PRUH GHWDLOV 6HH DOVR **£&RP PDQG /LQH**
7RROV^α RQ SDJH IRU PRUH LQIRUPDWLRQ DERXW XVLQJ WKH scala
FRPPDQG

Semicolons

<RX PD\ KDYH DOUHDG\ QRWLFHG WKDW WKHUH ZHUH YHU\ IHZ
VHPLFRORQV LQ WKH FRGH H[DPSOHV LQ WKH SUHYLRXV FKDSWHU
<RX FDQ XVH VHPLFRORQV WR VHSUDUDWH VWDWHPHQWV DQG
H[SUHVV VLRQV DV LQ -DYD & 3+3 DQG VLPLODU ODQJXDJHV ,Q
PRVW FDVHV WKRXJK 6FDOD EHKDYHV OLNH PDQ\ VFULSWLQJ
ODQJXDJHV LQ WUHDWLQJ WKH HQG RI WKH OLQH DV WKH HQG RI D
VWDWHPHQW RU DQ H[SUHVVLHQ :KHQ D VWDWHPHQW RU
H[SUHVVLHQ LV WRR ORQJ IRU RQH OLQH 6FDOD FDQ XVXDOO\
LQIHU ZKHQ \RX DUH FRQLQXLQJ RQ WR WKH QH[W OLQH DV
VKRZQ LQ WKLW H[DPSOH

```
// code-examples/TypeLessDoMore/semicolon-example-script.scala
```

```
// Trailing equals sign indicates more code on next line
```

```
def equalsign = {  
  val reallySuperLongValueNameThatGoesOnForeverSoYouNeedANewLine =  
    "wow that was a long value name"
```

```
println(reallySuperLongValueNameThatGoesOnForeverSoYouNeedANewLine)
```

}

// Trailing opening curly brace indicates more code on next line

```
def equalsign2(s: String) = {
  println("equalsign2: " + s)
}
```

// Trailing comma, operator, etc. indicates more code on next line

```
def commas(s1: String,
s2: String) = {
  println("comma: " + s1 +
", " + s2)
}
```

```
:KHQ \RX ZDQW WR SXW PXOWLSOH VWDWHPHQWV RU
H[ SUHVVLQRQV RQ WKH VDPH OLQH \RX FDQ XVH VHPLFRORQV WR
VHSDUDWH WKHP :H XVHG WKLV WHFKQLTXH LQ WKH
ShapeDrawingActor H[DPSOH LQ £$ 7DVWH RI &RQFXUUHQF RQ SDJH
```

```
case "exit" => println("exiting..."); exit
```

```
7KLW FRGH FRXOG DOVR EH ZULWWHQ DV IROORZV
```

```
...
case "exit" =>
  println("exiting...")
  exit
...
```

```
<RX PLJKW ZRQGHU ZK\ \RX GRQ\W QHHG FXUO\ EUDFHV
{...} DURXQG WKH WZR VWDWHPHQWV DIWHU WKH case ... => OLQH
<RX FDQ SXW WKHP LQ LI \RX ZDQW EXW WKH FRPSLOHU NQRZV
ZKHQ \RX\YH UHDFKHG WKH HQG RI WKH £EORFN ZKHQ LW ILQGV
WKH QH[W case FODXVH RU WKH FXUO\ EUDFH
} WKDW HQGV WKH HQFORVLQJ EORFN IRU DOO WKH case FODXVHV
2PLWWLQJ RSWLRQDO VHPLFRORQV PHDQV IHZHU FKDUFWHU
WR WASH DQG IHZHU FKDUFWHUV WR FOXWWHU \RXU FRGH
%UHDNLQJ VHSDUDWH VWDWHPHQWV RQWR WKHLU RZQ OLQHV
LQFUHDVHV \RXU FRGH\V UHGDGDELWL
```

Variable Declarations

```
6FDOD DOORZV \RX WR GHFLGH ZKHWKHU D YDULDEOH LV
LPPXWDEOH
UHDG RQO\ RU QRW
UHDG ZULWH ZKHQ \RX GHFODUH LW $Q LPPXWDEOH
£YDULDEOH LV GHFODUHG ZLWK WKH NHZRUG val
WKLQN YDOXH REMHFW
```

```
val array: Array[String] = new Array(5)
```

```
7R EH PRUH SUHFLVH WKH array UHIHUUHQFH FDQQRW EH FKDQJHG
WR SRLQW WR D GLIIHUHQW Array EXW WKH DUUD\ LWVHOI FDQ EH
PRGLLHG DV VKRZQ LQ WKH IROORZLQJ scala VHVLRQ
```

```
scala> val array: Array[String] = new Array(5)
array: Array[String] = Array(null, null, null, null, null)
```

```
scala> array = new Array(2)
<console>:5: error: reassignment to val
array = new Array(2)
```

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^

scala> array(0) = "Hello"

scala> array

res3: Array[String] = Array(Hello, null, null, null, null)

scala>

\$Q LPPXWDEOH val PXVW EH LQLWLDOLJHG WKDW LV

GHILQHGH ZKHQ LW LV GHFODUHG \$ PXWDEOH YDULDEOH LV

GHFODUHG ZLWK WKH NHZRUG var

scala> var stockPrice: Double = 100.

stockPrice: Double = 100.0

scala> stockPrice = 10.

stockPrice: Double = 10.0

scala>

6FDOD DOVR UHTXLUHV \RX WR LQLWLDOLJH D var ZKHQ LW LV
 GHFODUHG <RX FDQ DVVLJQ D QHZ YDOXH WR D var DV RIWHQ DV
 \RX ZDQW \$JDLQ WR EH SUHFLVH WKH stockPrice UHIHUUHQFH FDQ EH
 FKDQJHG WR SRLQW WR D GLIIHUHQW Double REMHFW
 H J 10. ,Q WKLW FDVH WKH REMHFW WKDW stockPrice UHIHUV WR
 FDQ\W EH FKDQJHG EHFDXVH Doubles LQ 6FDOD DUH LPPXWDEOH

7KHUH DUH D IHZ H[FHSWLRQV WR WKH UXOH WKDW \RX PXVW
 LQLWLDOLJH val DQG varV ZKHQ WKH\ DUH GHFODUHG %RWK
 NHZRUGV FDQ EH XVHG ZLWK FRQVWUXFWRU SDUDPHWHUV :KHQ
 XVHG DV FRQVWUXFWRU SDUDPHWHUV WKH PXWDEOH RU
 LPPXWDEOH YDULDEOHV VSHFLILHG ZLOO EH LQLWLDOLJHG ZKHQ
 DQ REMHFW LV LQVWDQWLWDWHG %RWK NHZRUGV FDQ EH XVHG
 WR GHFODUH DEVWUDFW

XQ LQLWLDOLJHG YDULDEOHV LQ DEVWUDFW WSHV \$OVR
 GHULYHG WSHV FDQ RYHUULGH val GHFODUHG LQVLGH SDUHQW
 WSHV :H\OO GLVFXVV WKHVH H[FHSWLRQV LQ &KDSWHU

6FDOD HQFRXUDJHV \RX WR XVH LPPXWDEOH YDOXHV ZKHQHYHU
 SRVVLEOH \$V ZH ZLOO VHH WKLW SURPRWHV EHWWHU REMHFW
 RULHQWHG GHVLJQ DQG LV FRQVLVWHQW ZLWK WKH SULQFLSOHV
 RI £SXUH IXQFWLRQDO SURJUDPPHQJ ,W PD\ WDNH VRPH JHWWLQJ
 XVHG WR EXW \RX\OO ILQG D QHZIRXQG FRQILGHQFH LQ \RXU FRGH
 ZKHQ LW LV ZULWWHQ LQ DQ LPPXWDEOH VWOH



7KH var DQG val NHZRUGV RQ\ VSHFL\ ZKHWKHU WKH UHIHUUHQFH FDQ EH
 FKDQJHG WR UHIHU WR D GLIIHUHQW REMHFW
 var RU QRW
 val 7KH\ GRQ\W VSHF
 LI\ ZKHWKHU RU QRW WKH REMHFW WKH\ UHIHUUHQFH LV PXWDEOH

Method Declarations

```
,Q &KDSWHU ZH VDZ VHYHUDO H[DPSOHV RI KRZ WR GHILQH
PHWKRGV ZKLFK DUH IXQFWLRQV WKDW DUH PHPEHUV RI D FODVV
0HWKRG GHILQLWLRQV VWDUW ZLWK WKH def NH\ZRUG IROORZHG
E\ RSWLRQDO DUJXPHQW OLVWV D FRORQ FKDUDFWHU
: DQG WKH UHWXUQ WSH RI WKH PHWKRG DQ HTXDOV VLJQ
= DQG ILQDOO\ WKH PHWKRG ERG\ 0HWKRGV DUH LPSOLFLWO\
GHFODUHG £DEVWUDFW¤
```

Method Declarations | 25

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```
LI \RX OHDYH RII WKH HTXDOV VLJQ DQG PHWKRG ERG\ 7KH
HQFORVLQJ WSH LV WKHQ LWVHOI DEVWUDFW :H\OO GLVFXVV
DEVWUDFW WSHV LQ PRUH GHWDLO LQ &KDSWHU

:H VDLG £RSWLRQDO DUJXPHQW OLVWV ¤ PHDQLQJ PRUH WKDQ
RQH 6FOD 0HWV \RX GHILQH PRUH WKDQ RQH DUJXPHQW OLVW
IRU D PHWKRG 7KLV LV UHTXLUHG IRU FXUU\LQJ PHWKRGV ZKLFK
ZH\OO GLVFXVV LQ £&XUU\LQJ¤ RQ SDJH ,W LV DOVR YHU\ XVHIXO
IRU GHILQLQJ \RXU RZQ 'RPDLQ 6SHFLILF /DQJXDJHV
'6/V DV ZH\OO VHH LQ &KDSWHU 1RWH WKDW HDFK DUJXPHQW
OLVW LV VXUURXQG HG E\ SDUHQWKHVVH DQG WKH DUJXPHQW
DUH VHSUDUDWHG E\ FRPPDV

,I D PHWKRG ERG\ KDV PRUH WKDQ RQH H[SUHVVLRQ \RX PXVW
VXUURXQG LW ZLWK FXUO\ EUDFHV
{...} <RX FDQ RPLW WKH EUDFHV LI WKH PHWKRG ERG\ KDV MXVW
RQH H[SUHVVLRQ
```

Method Default and Named Arguments (Scala Version 2.8)

```
0DQ\ ODQJXDJHV OHW \RX GHILQH GHIDXOW YDOXHV IRU VRPH RU
DOO RI WKH DUJXPHQWV WR D PHWKRG &RQVLGHU WKH
IROORZLQJ VFULSW ZLWK D StringUtil REMHFW WKDW OHWV \RX MRLQ
D OLVW RI VWULQJV ZLWK D XVHU VSHFLILHG VHSUDUDWRU
```

```
// code-examples/TypeLessDoMore/string-util-v1-script.scala
// Version 1 of "StringUtil".
```

```
object StringUtil {
  def joiner(strings: List[String], separator: String): String =
    strings.mkString(separator)

  def joiner(strings: List[String]): String = joiner(strings, " ")
}
import StringUtil._ // Import the joiner methods.

println( joiner(List("Programming", "Scala")) )
```

```
7KHUH DUH DFWXDOO\ WZR £RYHUORDGHG¤ joiner PHWKRGV 7KH
VHFRQG RQH XVHV D VLQJOH VSDFH DV WKH £GHIDXOW¤
VHSUDUDWRU +DYLQJ WZR PHWKRGV VHPV D ELW ZDVWHIXO ,W
ZRXOG EH QLFH LI ZH FRXOG HOLPLQDWH WKH VHFRQG joiner
PHWKRG DQG GHFODUH WKDW WKH separator DUJXPHQW LQ WKH
ILUVW joiner KDV D GHIDXOW YDOXH ,Q IDFW LQ 6FOD YHUVLRQ
\RX FDQ QRZ GR WKL
```

```
// code-examples/TypeLessDoMore/string-util-v2-v28-script.scala
// Version 2 of "StringUtil" for Scala v2.8 only.
```

```
object StringUtil {
  def joiner(strings: List[String], separator: String = " "): String =
    strings.mkString(separator)
}
import StringUtil._ // Import the joiner methods.
```

```
println(joiner(List("Programming", "Scala")))
```

```
7KHUH LV DQRWKHU DOWHUQDWLYH IRU HDUOLHU YHUVLRQV RI
6FDOD <RX FDQ XVH LPSOLFLW DUJXPHQWV ZKLFK ZH ZLOO
GLVFXVV LQ £,PSOLFLW )XQFWLRQ 3DUDPHWHUV RQ SDJH
```

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```
6FDOD YHUVLRQ RIIHUV DQRWKHU HQKDQFHPHQW IRU PHWKRG
DUJXPHQW OLVWV QDPHG DUJX PHQWV :H FRXOG DFWXDOO\
ZULWH WKH ODVW OLQH RI WKH SUHYLRXV H[DPSOH LQ VHYHUDO
ZD\V $OO RI WKH IROORZLQJ println VWDWHPHQWV DUH
IXQFWLRQDOO\ HTXLYDOHQW
```

```
println(joiner(List("Programming", "Scala")))
println(joiner(strings = List("Programming", "Scala")))
println(joiner(List("Programming", "Scala"), " ")) // #1
println(joiner(List("Programming", "Scala"), separator = " ")) // #2
println(joiner(strings = List("Programming", "Scala"), separator = " "))
```

```
:K\ LV WKLW XVHIXO" )LUVW LI \RX FKRRVH JRRG QDPHV IRU WKH
PHWKRG DUJXPHQWV WKHQ \RX FDOOV WR WKRVH PHWKRGV
GRFXPHQW HDFK DUJXPHQW ZLWK D QDPH )RU H[DPSOH FRP SDUH
WKH WZR OLQHV ZLWK FRPPHQWV DQG ,Q WKH ILUVW OLQH LW
PD\ QRW EH REYLRXV ZKDW WKH VHFRQG " " DUJXPHQW LV IRU ,Q
WKH VHFRQG FDVH ZH VXS$OO\ WKH QDPH separator ZKLFK
VXJJHVWV WKH SXUSRVH RI WKH DUJXPHQW
```

```
7KH VHFRQG EHQHILW LV WKDW \RX FDQ VSHFL\ WKH SDUDPHWHUV
LQ DQ\ RUGHU ZKHQ \RX VSHFL\ WKHP E\ QDPH &RPELQHG ZLWK
GHIDXOW YDOXHV \RX FDQ ZULWH FRGH OLNH WKH IROORZLQJ
```

```
// code-examples/TypeLessDoMore/user-profile-v28-script.scala
// Scala v2.8 only.
```

```
object OptionalUserProfileInfo {
  val UnknownLocation = ""
  val UnknownAge = -1
  val UnknownWebSite = ""
}
```

```
class OptionalUserProfileInfo(
  location: String = OptionalUserProfileInfo.UnknownLocation,
  age: Int = OptionalUserProfileInfo.UnknownAge,
  webSite: String = OptionalUserProfileInfo.UnknownWebSite)
```

```
println( new OptionalUserProfileInfo )
println( new OptionalUserProfileInfo(age = 29) )
println( new OptionalUserProfileInfo(age = 29, location="Earth") )
```

```
OptionalUserProfileInfo UHSUHVHQWV DOO WKH £RSWLRQDO XVHU
```

SURILOH GDWD LQ IRXU QH[W :HE VRFLDO QHWZRUNLQJ VLWH ,W
GHILQHV GHIDXOW YDOXHV IRU DOO LWV ILHOGV 7KH VFULSW
FUHDWHV LQVWDQFHV ZLWK JHUR RU PRUH QDPHG SDUDPHWHUV
7KH RUGHU RI WKRVH SDUDPHWHUV LV DUELWUDU\

7KH H[DPSOHV ZH KDYH VKRZQ XVH FRQVWDQW YDOXHV DV WKH
GHIDXOWV 0RVW ODQJXDJHV ZLWK GHIDXOW DUJXPHQW YDOXHV
RQO\ DOORZ FRQVWDQWV RU RWKHU YDOXHV WKDW FDQ EH
GHWHUPLQHG DW SDUVH WLPH +RZHYHU LQ 6FDOD DQ\
H[SUHVVLRQ FDQ EH XVHG DV WKH GHIDXOW DV ORQJ DV LW FDQ
FRPSLOH ZKHUH XVHG)RU H[DPSOH DQ H[SUHVVLRQ FRXOG QRW
UHIHU WR DQ LQVWDQFH ILHOG WKDW ZLOO EH FRPSXWHG LQVLGH
WKH FODVV RU REMHFW ERG\ EXW LW FRXOG LQYRNH D PHWKRG
RQ D VLQJOHWRRQ REMHFW

\$ UHODWHG OLPLDWDLRQ LV WKDW D GHIDXOW H[SUHVVLRQ IRU
RQH SDUDPHWHU FDQ\W UHIHU WR DQRWKHU SDUDPHWHU LQ WKH
OLVW XQOHVV WKH SDUDPHWHU WKDW LV UHIHUHQFHG DSSHDUV
HDUOLHU LQ WKH OLVW DQG WKH SDUDPHWHUV DUH *FXUULHG* D
FRQFHSW ZH\OO GLVFXVV LQ [£&XUU\LQJ RQ SDJH](#)

Method Declarations | 27

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)LQDOO\ DQRWKHU FRQVWUDLQW RQ QDPHG SDUDPHWHUV LV
WKDW RQH \RX SURYLGH D QDPH IRU D SDUDPHWHU LQ D PHWKRG
LQYRFDWLRQ WKH UHVV RI WKH SDUDPHWHUV DSSHDULQJ DIWHU
LV PXVW DOVR EH QDPHG)RU H[DPSOH `new OptionalUserProfileInfo(age =`
`29, "Earth") ZRXOG QRW FRPSLOH EHFDXVH WKH VHFRQG DUJXPHQW`
`LV QRW LQYRNHG E\ QDPH`

:H\OO VHH DQRWKHU XVHIXO H[DPSOH RI QDPHG DQG GHIDXOW
DUJXPHQWV ZKHQ ZH GLVFXVV *FDVH FODVVHV* LQ [£&DVH &ODVVHV RQ SDJH](#)

Nesting Method Definitions

0HWKRG GHILQLWLRQV FDQ DOVR EH QHVWHG +HUH LV DQ
LPSOHPHQWDWLRQ RI D IDFWRULDO FDOFX ODWRU ZKHUH ZH XVH
D FRQYHQWLRQDO WHFKQLTXH RI FDOOLQJ D VHFRQG QHVWHG
PHWKRG WR GR WKH ZRUN

[// code-examples/TypeLessDoMore/factorial-script.scala](#)

```
def factorial(i: Int): Int = {  
  def fact(i: Int, accumulator: Int): Int = {  
    if (i <= 1)  
      accumulator  
    else  
      fact(i - 1, i * accumulator)  
  }  
  
  fact(i, 1)  
}  
  
println( factorial(0) )  
println( factorial(1) )  
println( factorial(2) )  
println( factorial(3) )
```

```
println( factorial(4) )
println( factorial(5) )
```

```
7KH VHFRRG PHWKRQ FDOOV LWVHOI UHFUVLYHO\ SDVVLQJ DQ
accumulator SDUDPHWHU ZKHUH WKH UHVXOW RI WKH FDOFXODWLRQ
LV £DFFXPXODWHG ¢ 1RWH WKDW ZH UHWXUQ WKH DFFXPXODWHG
YDOXH ZKHQ WKH FRXQWHU i UHDFKHV
:H\UH LJQRULQJ LQYDOLG QHJDWLYH LQWHJHUV 7KH IXQFWLRQ
DFWXDOO\ UHWXUQV IRU i < 0 $IWHU WKH GHILQLWLRQ RI WKH
QHVVHG PHWKRQ factorial FDOOV LW ZLWK WKH SDVVHG LQ YDOXH i
DQG WKH LQLWLDO DFFXPXODWRU YDOXH RI
```

```
/LNH D ORFDO YDULDEOH GHFODUDWLRQ LQ PDQ\ ODQJXDJHV D
QHVVHG PHWKRQ LV RQO\ YLVLEOH LQVLGH WKH HQFORVLQJ
PHWKRQ ,I \RX WU\ WR FDOO fact RXWVLGH RI factorial \RX ZLOO JHW
D FRPSLOHU HUURU
```

```
'LG \RX QRWLFH WKDW ZH XVH i DV D SDUDPHWHU QDPH WZLFH
ILUVW LQ WKH factorial PHWKRQ DQG DJDLQ LQ WKH QHVVHG fact
PHWKRQ" $V LQ PDQ\ ODQJXDJHV WKH XVH RI i DV D SDUDPHWHU
QDPH IRU fact £VKDGRZV¢ WKH RXWHU XVH RI i DV D SDUDPHWHU
QDPH IRU factorial 7KLV LV ILQH EHFDXVH ZH GRQ\W QHHG WKH
RXWHU YDOXH RI i LQVLGH fact :H RQO\ XVH LW WKH ILUVW WLPH ZH
FDOO fact DW WKH HQG RI factorial
```

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```
:KDW LI ZH QHHG WR XVH D YDULDEOH WKDW LV GHILQHG RXWVLGH
D QHVVHG IXQFWLRQ" &RQVLGHU WKLW FRQWULYHG H[DPSOH
```

[// code-examples/TypeLessDoMore/count-to-script.scala](http://code-examples/TypeLessDoMore/count-to-script.scala)

```
def countTo(n: Int): Unit = {
  def count(i: Int): Unit = {
    if (i <= n) {
      println(i)
      count(i + 1)
    }
  }
  count(1)
}

countTo(5)
```

```
1RWH WKDW WKH QHVVHG count PHWKRQ XVHV WKH n YDOXH WKDW
LV SDVVHG DV D SDUDPHWHU WR count 7KHUH LV QR QHHG WR
SDVV n DV DQ DUJXPHQW WR count %HFDXVH count LV QHVVHG
LQVLGH count n LV YLVLEOH WR LW
```

```
7KH GHFODUDWLRQ RI D ILHOG
PHPEHU YDULDEOH FDQ EH SUHIL[HG ZLWK NHZRUGV LQGLFDWLQJ
WKH YLVLEOLW\ MXVW DV LQ ODQJXDJHV OLNH -DYD DQG &
6LPLODUO\ WKH GHFODUDWLRQ RI D QRQ QHVVHG PHWKRQ FDQ
EH SUHIL[HG ZLWK WKH VDPH NHZRUGV :H ZLOO GLVFXVV WKH
YLVLEOLW\ UXOHV DQG NHZRUGV LQ £9LVLEOLW\ 5XOHV¢ RQ
SDJH
```

Inferring Type Information

6WDWLFDOO\ W\SHG ODQJXDJHV FDQ EH YHU\ YHUERVH &RQVLGHU WKLW W\SLFDO
GHFODUDWLRQ LQ -DJD

```
import java.util.Map;  
import java.util.HashMap;
```

```
...  
Map<Integer, String> intToStringMap = new HashMap<Integer, String>();
```

:H KDYH WR VSHFL\ WKH W\SH SDUDPHWHUV <Integer, String> WZLFH
6FDOD XVHV WKH WHUP W\SH DQQRWDWLRQV IRU H[SOLFLW W\SH
GHFODUDWLRQV OLNH HashMap<Integer, String>

6FDOD VXSSRUWV W\SH LQIHUHQFH

VHH IRU H[DPSOH >7\SH,QIHUHQFH@ DQG >3LHUFH @ 7KH
ODQJXDJH\ FRPSLOHU FDQ GLVFHUQ TXLWH D ELW RI W\SH
LQIRUPDWLRQ IURP WKH FRQWH[W ZLWK RXW H[SOLFLW W\SH
DQQRWDWLRQV +HUH\ WKH VDPH GHFODUDWLRQ UHZULWWHQ LQ
6FDOD ZLWK LQ IHUHHG W\SH LQIRUPDWLRQ

```
import java.util.Map;  
import java.util.HashMap;
```

```
...  
val intToStringMap: Map[Integer, String] = new HashMap
```

5HFDOO IURP &KDSWHU WKDW 6FDOD XVHV VTXDUH EUDFNHWV
[...] IRU JHQHULF W\SH SDUDPHWHUV :H VSHFL\ Map[Integer, String] RQ
WKH OHIWKDQG VLGH RI WKH HTXDOV VLJQ
:H DUH VWLFNLQJ ZLWK -DJD W\SHV IRU WKH H[DPSOH 2Q WKH
ULJKWKDQG VLGH ZH LQVWDQWLDWH WKH DFWXDO W\SH ZH ZDQW
D HashMap EXW ZH GRQ\W KDYH WR UHSHDW WKH W\SH
SDUDPHWHUV

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)RU FRPSOHWHQHVV VXSSRVH ZH GRQ\W DFWXDOO\ FDUH LI WKH
LQVWDQFH LV RI W\SH Map
WKH -DJD LQWHUIDFH W\SH ,W FDQ EH RI W\SH HashMap IRU DOO ZH
FDUH

```
import java.util.Map;  
import java.util.HashMap;
```

```
...  
val intToStringMap2 = new HashMap[Integer, String]
```

7KLW GHFODUDWLRQ UHTXLUHV QR W\SH DQQRWDWLRQV RQ WKH
OHIWKDQG VLGH EHFDXVH DOO RI WKH W\SH LQIRUPDWLRQ
QHGHG LV RQ WKH ULJKWKDQG VLGH 7KH FRPSLOHU
DXWRPDWLFDOO\ PDNHV intToStringMap2 D HashMap[Integer,String]

7\SH LQIHUHQFH LV XVHG IRU PHWKRGV WRR ,Q PRVW FDUHV WKH
UHWXUQ W\SH RI WKH PHWKRG FDQ EH LQIHUHHG VR WKH : DQG
UHWXUQ W\SH FDQ EH RPLWWHG +RZHYHU W\SH DQQRWDWLRQV
DUH UHTXLUHG IRU DOO PHWKRG SDUDPHWHUV

3XUH IXQFWLRQDO ODQJXDJHV OLNH +DVNHOO

VHH H J >2\6XOOLYDQ @ XVH W\SH LQIHUHQFH DOJRULWKPV

```

OLNH +LQGOH\ 0LOQHU
VHH >6SLHZDN @ IRU DQ HDVLO\ GLJHVWHG H[SODQDWLRQ &RGH
ZULWWHQ LQ WKHVH ODQJXDJHV UHTXLUH WSH DQQRWDWLRQV
OHVV RIWHQ WKDQ LQ 6FDOD EH FDXVH 6FDOD\V WSH LQIHUHQFH
DOJRULWKP KDV WR VXSSRUW REMHFW RULHQWHG WSLQJ DV
ZHOO DV IXQFWLRQDO WSLQJ 6R 6FDOD UHTXLUHV PRUH WSH
DQQRWDWLRQV WKDQ ODQJXDJHV OLNH +DVNHOO +HUH LV D
VXPPDU\ RI WKH UXOHV IRU ZKHQ H[SOLFLW WSH DQQRWDWLRQV
DUH UHTXLUHG LQ 6FDOD

```

When Explicit Type Annotations Are Required

```

,Q SUDFWLFDQ WHUPV \RX KDYH WR SURYLGH H[SOLFLW WSH
DQQRWDWLRQV IRU WKH IROORZLQJ VLWXDWLRQV

$ YDULDEOH GHFODUDWLRQ XQOHVV \RX DVVLJQ D YDOXH WR
WKH YDULDEOH
H J val name = "Programming Scala"

$OO PHWKRQ SDUDPHWHUV
H J def deposit(amount: Money)...

OHWKRQ UHWXUQ YDOXHV LQ WKH IROORZLQJ FDVHV

D :KHQ \RX H[SOLFLW\ FDOO return LQ D PHWKRQ
HYHQ DW WKH HQQ

E :KHQ D PHWKRQ LV UHFXUVLYH

F :KHQ D PHWKRQ LV RYHUORDGHG DQG RQH RI WKH
PHWKRGV FDOOV DQRWKHU WKH FDOOLQJ PHWKRQ
QHHGV D UHWXUQ WSH DQQRWDWLRQ

G :KHQ WKH LQIHUHQH UHWXUQ WSH ZRXOG EH PRUH
JHQHUDO WKDQ \RX LQWHQG H J Any

```



```

7KH Any WSH LV WKH URRW RI WKH 6FDOD WSH KLHUDUFK\
VHH £7KH 6FDOD 7\SH
+LHUDUFK\ RQ SDJH IRU PRUH GHWDLOV ,I D EORFN RI FRGH UHWXUQV
D
YDOXH RI WSH Any XQH[SHFWHG\ FKDQFHV DUH JRRG WKDW WKH WSH
LQIHUHQFHU
FRXOGQ\W ILJXUH RXW ZKDW WSH WR UHWXUQ VR LW FKRVL WKH PRVW JHQHULF
WSH

SRVVLEOH

```

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```

/HW\V ORRN DW H[DPSOHV ZKHUH H[SOLFLW GHFODUDWLRQV RI
PHWKRQ UHWXUQ WSHV DUH UHTXLUHG ,Q WKH IROORZLQJ
VFULSW WKH upCase PHWKRQ KDV D FRQGLWLRQDO UHWXUQ
VWDWPHPHQW IRU JHUR OHQJWK VWULQJV

```

```

// code-examples/TypeLessDoMore/method-nested-return-script.scala
// ERROR: Won't compile until you put a String return type on upCase.

```

```

def upCase(s: String) = {
  if (s.length == 0)

```

```

return s
else
s.toUpperCase()
}

println( upCase("") )
println( upCase("Hello") )

```

5XQQQLQJ WKLW VFULSW JLYHV \RX WKH IROORZLQJ HUURU

```

... 6: error: method upCase has return statement; needs result type
return s
^

```

<RX FDQ IL[WKLW HUURU E\ FKDQJLQJ WKH ILUVW OLQH RI WKH

```

PHWKRQ WR WKH IROORZLQJ def upCase(s: String): String = {

```

\$FWXDOO\ IRU WKLW SDUWLFXODU VFULSW DQ DOWHUQDWLYH IL[
LV WR UHPRYH WKH return NHZRUG IURP WKH OLQH ,W LV QRW
QHGHGHG IRU WKH FRGH WR ZRUN SURSHUO\ EXW LW
LOOXVWUDWHV RXU SRLQW

5HFXUVLYH PHWKRQV DOVR UHTXLUH DQ H[SOLFLW UHWXUQ WSH
5HFDQO RXU factorial PHWKRQ LQ £1HVWLQJ 0HWKRQ 'HILQLWLRQV
RQ SDJH /HW\U UHPRYH WKH : Int UHWXUQ WSH RQ WKH QHVWHG
fact PHWKRQ

```

// code-examples/TypeLessDoMore/method-recursive-return-script.scala //
ERROR: Won't compile until you put an Int return type on "fact".

```

```

def factorial(i: Int) = {
  def fact(i: Int, accumulator: Int) = {
    if (i <= 1)
      accumulator
    else
      fact(i - 1, i * accumulator)
  }

  fact(i, 1)
}

```

1RZ LW IDLOV WR FRPSLOH

```

... 9: error: recursive method fact needs result type
fact(i - 1, i * accumulator)
^

```

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2YHUORDGHG PHWKRQV FDQ VRPHWLPHV UHTXLUH DQ H[SOLFLW
UHWXUQ WSH :KHQ RQH VXFK PHWKRQ FDOOV DQRWKHU ZH KDYH
WR DGG D UHWXUQ WSH WR WKH RQH GRLQJ WKH FDOOLQJ DV LQ
WKLW H[DPSOH

```

// code-examples/TypeLessDoMore/method-overloaded-return-script.scala //
Version 1 of "StringUtil" (with a compilation error).
// ERROR: Won't compile: needs a String return type on the second "joiner".

```

```

object StringUtil {
  def joiner(strings: List[String], separator: String): String =
    strings.mkString(separator)
}

```

```
def joiner(strings: List[String]) = joiner(strings, " ")
}
import StringUtil._ // Import the joiner methods.
```

```
println( joiner(List("Programming", "Scala"))) )
```

7KH WZR joiner PHWKRGV FRQFDWHQDWH D List RI VWULQJW
 WRJHWKHU 7KH ILUVW PHWKRG DOVR WDNHV DQ DUJXPHQW IRU
 WKH VHSUDUDWRU VWULQJ 7KH VHFRQG PHWKRG FDOOV WKH
 ILUVW ZLWK D £GHIDXOW VHSUDUDWRU RI D VLQJOH VSDFH

,I\RX UXQ WKLV VFULSW \RX JHW WKH IROORZLQJ HUURU

... 9: error: overloaded method joiner needs result type

```
def joiner(strings: List[String]) = joiner(strings, "")
```

^

6LQFH WKH VHFRQG joiner PHWKRG FDOOV WKH ILUVW LW UHTXLUHV
 DQ H[SOLFLW String UHWXUQ WSH ,W VKRXOG ORRN OLNH WKLV

```
def joiner(strings: List[String]): String = joiner(strings, " ")
```

7KH ILQDO VFHQDULR FDQ EH VXEWOH ZKHQ D PRUH JHQHUDO
 UHWXUQ WSH LV LQIHUUHG WKDQ ZKDW \RX H[SHFWHG <RX
 XVXDOO\ VHH WKLV HUURU ZKHQ \RX DVVLQJ D YDOXH UHWXUQH
 IURP D IXQF WLRQ WR D YDULDEOH ZLWK D PRUH VSHFLILF
 WSH)RU H[DPSOH \RX ZHUH H[SHFWLQJ D String EXW WKH
 IXQFWLRQ LQIHUUHG DQ Any IRU WKH UHWXUQH REMHFW /HW\ VHH
 D FRQWULYHG H[DPSOH WKDQ UHIOHFWV D EXJ ZKHU WKLV
 VFHQDULR FDQ RFFXU

// code-examples/TypeLessDoMore/method-broad-inference-return-script.scala //

ERROR: Won't compile; needs a String return type on the second "joiner".

```
def makeList(strings: String*) = {
  if (strings.length == 0)
    List(0) // #1
  else
    strings.toList
}
```

```
val list: List[String] = makeList()
```

5XQQLQJ WKLV VFULSW UHWXUQV WKH IROORZLQJ HUURU

...11: error: type mismatch;

found : List[Any]

required: List[String]

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```
val list: List[String] = makeList()
```

^

:H LQWHQG HG IRU makeList WR UHWXUQ D List[String] EXW ZKHQ
 strings.length HTXDOV]HUR ZH UHWXUQH List(0) LQFRUHHFWO\
 £DVVXPLQJ WGDW WKLV H[SUHVLRQ LV WKH FRUHHFW ZD\ WR
 FUHDWH DQ HPSW OLVW ,Q IDFV ZH UHWXUQH D List[Int] ZLWK RQH
 HOHPHQW 0 :H VKRXOG KDYH UHWXUQH List() 6LQFH WKH else
 H[SUHVLRQ UHWXUQV D List[String] WKH UHVXOW RI strings.toList WKH
 LQIHUUHG UHWXUQ WSH IRU WKH PHWKRG LV WKH FORVHVW
 FRPPRQ VXSHUWSH RI List[Int] DQG List[String] ZKLFK LV List[Any] 1RWH

WKDW WKH FRPSLOD WLRQ HUURU GRHVQ\W RFFXU LQ WKH
IXQFWLRQ GHILQLWLRQ :H RQO\ VHH LW ZKHQ ZH DWWHPSW WR
DVVLJQ WKH YDOXH UHWXUQH IURP makeList WR D List[String]
YDULDEOH

,Q WKLW FDVH IL[LQJ WKH EXJ LV WKH VROXWLRQ \$OWHUQDWLYHO\
ZKHQ WKHUH LVQ\W D EXJ LW PD\ EH WKDW WKH FRPSLOHU MXVW
QHGGV WKH £KHOS α RI DQ H[SOLFLW UHWXUQ WSH GHFODUDWLRQ
,QYHV WLJDWH WKH PHWKRQ WKDW DSSHDUV WR UHWXUQ WKH
XQH[SHFWHG WSH ,Q RXU H[SHULHQFH \RX RIWHQ ILQG WKDW \RX
PRGLLHG WKDW PHWKRQ

RU DQRWKHU RQH LQ WKH FDOO SDWK LQ VXFK D ZD\ WKDW WKH
FRPSLOHU QRZ LQIHUV D PRUH JHQHUDO UHWXUQ WSH WKDQ
QHGHVVDU\ \$GG WKH H[SOLFLW UHWXUQ WSH LQ WKLW FDVH

\$QRWKHU ZD\ WR SUHYHQW WKHVH SUREOHPV LV WR DOZD\
GHFODUH UHWXUQ WSHV IRU PHWKRQV HVSHFLDOO\ ZKHQ
GHILQLQJ PHWKRQV IRU D SXEOLF \$3, /HW\V UHYLVLW RXU StringUtil
H[DPSOH DQG VHH ZK\ H[SOLFLW GHFODUDWLRQV DUH D JRRG LGHD
DGDSWHG IURP >6PLWK D@

+HUH LV RXU StringUtil £\$3, α DJDLQ ZLWK D QHZ PHWKRQ toCollection

```
// code-examples/TypeLessDoMore/string-util-v3.scala  
// Version 3 of "StringUtil" (for all versions of Scala).
```

```
object StringUtil {  
  def joiner(strings: List[String], separator: String): String =  
    strings.mkString(separator)  
  
  def joiner(strings: List[String]): String = strings.mkString(" ")  
  
  def toCollection(string: String) = string.split(' ')  
}
```

7KH toCollection PHWKRQ VSOLWV D VWULQJ RQ VSDFHV DQG
UHWXUQV DQ Array FRQWDLQLQJ WKH VXE VWULQJV 7KH UHWXUQ
WSH LV LQIHUUHG ZKLFK LV D SRWHQWLDO SUREOHP DV ZH ZLOO
VHH 7KH PHWKRQ LV VRPHZKDW FRQWULYHG EXW LW ZLOO
LOOXVWUDWH RXU SRLQW +HUH LV D FOLHQW RI StringUtil WKDW
XVHV WKLW PHWKRQ

```
// code-examples/TypeLessDoMore/string-util-client.scala
```

```
import StringUtil._  
  
object StringUtilClient {  
  def main(args: Array[String]) = {  
    args foreach { s => toCollection(s).foreach { x => println(x) } }  
  }  
}
```

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,I \RX FRPSLOH WKHVH ILOHV ZLWK scala \RX FDQ UXQ WKH FOLHQW DV IROORZV

```
$ scala -cp ... StringUtilClient "Programming Scala"  
Programming  
Scala
```



```

)RU WKH -cp ... FODVV SDWK DUJXPHQW XVH WKH GLUHFWRU\ ZKHUH scalac
    ZURWH WKH FODVV ILOHV ZKLFK GHIDXOWV WR WKH FXUUHQW
    GLUHFWRU\
    L H XVH
-cp . ,\RX XVHG WKH EXLOG SURFHVV LQ WKH GRZQORDGHG FRGH H[DPSOHV
    WKH FODVV ILOHV DUH ZULWWHQ WR WKH EXLOG GLUHFWRU\
    XVLQJ scalac -d

build ... ,Q WKLW FDVH XVH -cp build

```

```

(YHUWKLQJ LV ILQH DW WKLW SRLQW EXW QRZ LPDJLQH WKDW
WKH FRGH EDVH KDV JURZQ StringUtil DQG LWV FOLHQWV DUH QRZ
EXLOW VHSUDUDWHO\ DQG EXQGOHG LQWR GLIIHUHQW -$5V
,PDJLQH DOVR WKDW WKH PDLQWDLQHUV RI StringUtil GHFLGH WR
UHWXUQ D List LQVWHDG RI WKH GHIDXOW

```

```

object StringUtil {
    ...

    def toCollection(string: String) = string.split(' ').toList // changed! }

```

```

7KH RQO\ GLIIHUHQFH LV WKH ILQDO FDOO WR toList WKDW
FRQYHUWV WKH FRPSXWHG Array WR D List <RX UHFRPSLOH StringUtil
DQG UHGHSOR\ LWV -$5 7KHQ \RX UXQ WKH VDPH FOLHQW
ZLWKRXW UHFRPSLOLQJ LW ILUVW

```

```

$ scala -cp ... StringUtilClient "Programming Scala"
java.lang.NoSuchMethodError: StringUtil$.toCollection(...)
    at StringUtilClient$$anonfun$main$1.apply(string-util-client.scala:6)    at
StringUtilClient$$anonfun$main$1.apply(string-util-client.scala:6) ...

```

```

:KDW KDSSHQH" :KHQ WKH FOLHQW ZDV FRPSLOHG
StringUtil.toCollection UHWXUQH DQ Array 7KHQ toCollection ZDV FKDQJH
WR UHWXUQ List ,Q ERWK YHUVLRQV WKH PHWKRQ UHWXUQ YDOXH
ZDV LQIHUHG 7KHUHIRUH WKH FOLHQW VKRXOG KDYH EHHQ
UHFRPSLOHG WRR

```

```

+RZHYHU KDG DQ H[SOLFLW UHWXUQ WSH RI Seq EHHQ GHFODUH
ZKLFK LV D SDUHQW IRU ERWK Array DQG List WKHQ WKH
LPSOHPHQDWLRQ FKDQJH ZRXOG QRW KDYH IRUFHG D
UHFRPSLODWLRQ RI WKH FOLHQW

```



```

:KHQ GHYHORSLOJ $3,V WKDW DUH EXLOW VHSUDUDWHO\ IURP WKHLU
    FOLHQWV GH
FODUH PHWKRQ UHWXUQ WSHV H[SOLFLWO\ DQG XVH WKH PRVW JHQUDO
    UHWXUQ WSH

```

```

\RX FDQ 7KLV LV HVSHFLDOO\ LPSRUWDQW ZKHQ $3,V GHFODUH DEVWUDFW PHWK
    RGV
    VHH H J &KDSWHU

```

```

7KHUH LV DQRWKHU VFHQDULR WR ZDWFK IRU ZKHQ XVLQJ
GHFODUDWLRQV RI FROOHFWLRQV OLNH val map = Map() DV LQ WKH
IROORZLQJ H[DPSOH

```

```
val map = Map()

map.update("book", "Programming Scala")

... 3: error: type mismatch;
found   : java.lang.String("book")
required: Nothing
map.update("book", "Programming Scala")
^
```

```
:KDW KDSSHQHG" 7KH WSH SDUDPHWHUV RI WKH JHQHULF WSH
Map ZHUH LQIHUUHG DV [Nothing,Nothing] ZKHQ WKH PDS ZDV
FUHDWHG
```

```
:H\OO GLVFXVV Nothing LQ £7KH 6FDOD 7\SH +LHUDUFK\ RQ SDJH
EXW LWV QDPH LV VXJJHVWLYH :H DWWHPSWHG WR LQVHUW DQ
LQFRPSDWLEOH NH\ YDOXH SDLU RI WASHV String DQG String &DOO LW
D Map WR QRZKHU 7KH VROXWLRQ LV WR SDUDPHWHUL]H WKH
LQLWLDO PDS GHFODUDWLRQ H J val map = Map[String, String]() RU WR
VSHFL\ LQLWLDO YDOXHV VR WKDW WKH PDS SDUDPHWHUV DUH
LQIHUUHG H J val map = Map("Programming" -> "Scala")
```

```
]LQDOO\ WKHUH LV D VXEWOH EHKDYLRU ZLWK LQIHUUHG UHWXUQ
WASHV WKDW FDQ FDXVH XQH[SHFWHG DQG EDIOLQJ UHVXOWV
VHH >6FDOD7LSV@ &RQVLGHU WKH IROORZLQJ H[DPSOH scala
VHVLRQ
```

```
scala> def double(i: Int) { 2 * i }
double: (Int)Unit
```

```
scala> println(double(2))
()
```

```
:K\ GLG WKH VHFRQG FRPPDQG SULQW () LQVWHDG RI 4" /RRN
FUDHIXOO\ DW ZKW WKH scala LQWHUSUHHU VDLG WKH ILUVW
FRPPDQG UHWXUQH double (Int)Unit :H GHILQH D PHWKRQ QDPHG
double WKDW WDNHV DQ Int DUJXPHQW DQG UHWXUQV Unit 7KH
PHWKRQ GRHVQ\W UHWXUQ DQ Int DV ZH ZRXOG H[SHFW
```

```
7KH FDXVH RI WKLW XQH[SHFWHG EHKDYLRU LV D PLVVLQJ HTXDOV
VLJQ LQ WKH PHWKRQ GHILQLWLRQ +HUH LV WKH GHILQLWLRQ ZH
DFWXDOO\ LQWHQGHG
```

```
scala> def double(i: Int) = { 2 * i }
double: (Int)Int
```

```
scala> println(double(2))
4
```

```
1RWH WKH HTXDOV VLJQ EHIRUH WKH ERG\ RI double 1RZ WKH
RXWSXW DV\ ZH KDYH GHILQH double WR UHWXUQ DQ Int DQG WKH
VHFRQG FRPPDQG GRHV ZKW ZH H[SHFW LW WR GR
```

```
7KHUH LV D UHDVRQ IRU WKLW EHKDYLRU 6FDOD UHJDUGV D
PHWKRQ ZLWK WKH HTXDOV VLJQ EHIRUH WKH ERG\ DV D IXQFWLRQ
GHILQLWLRQ DQG D IXQFWLRQ DOZDV UHWXUQV D YDOXH LQ
IXQFWLRQDO SURJUDPPLQJ 2Q WKH RWKHU KDQG ZKHQ 6FDOD
VHHV D PHWKRQ ERG\ ZLWKRXW WKH OHDGLQJ HTXDOV VLJQ LW
DVVXPHV WKH SURJUDPPHU LQWHQGHG WKH PHWKRQ WR EH D
£SURFHGXUH GHILQLWLRQ PHDQW IRU SHUIRUPPLQJ VLGH HHHFWV
RQO\ ZLWK WKH UHWXUQ YDOXH Unit ,Q SUDFWLFH LW LV PRUH
OLNHO\ WKDW WKH SURJUDPPHU VLPSO\ IRUJRW WR LQVHUW WKH
```

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:KHQ WKH UHWXUQ WSH RI D PHWKRG LV LQIHUUHG DQG \RX GRQ\W XVH
DQ HTXDOV
VLJQ EHIRUH WKH RSHQLQJ SDUHQWKHVLV IRU WKH PHWKRG ER\
6FDOD LQIHUV D
Unit UHWXUQ WSH HYHQ ZKHQ WKH ODVW H[SUHVVLRQ LQ WKH PHWKRG LV D
YDOXH

RI DQRWKHU WSH

%\ WKH ZD\ ZKHUH GLG WKDW () FRPH IURP WKDW ZDV SULQWHG
EHIRUH ZH IL[HG WKH EXJ" ,W LV DFWXDOO\ WKH UHDO QDPH RI WKH
VLQJOHWRRQ LQVWDQFH RI WKH Unit WSH
7KLW QDPH LV D IXQFWLRQDO SURJUDPPHQJ FRQYHQWLRQ

Literals

2IWHQ D QHZ REMHFW LV LQLWLDOL]HG ZLWK D *OLWHUDO* YDOXH
VXFK DV `val book = "Programming Scala" /HW\` GLVFXVV WKH NLQGV RI
OLWHUDO YDOXHV VXSSRUWHG E\ 6FDOD +HUH ZH\OO OLPLW
RXUVHOYHV WR OH[LFDO \QWD[OLWHUDOV :H\OO FRYHU
OLWHUDO \QWD[IRU IXQFWLRQV
XVHG DV *YDOXHV* QRW PHPEHU PHWKRGV WXSOHV DQG
FHUWDLQ WSHV OLNH Lists DQG Maps DV ZH FRPH WR WKHP

Integer Literals

,QWHJHU OLWHUDOV FDQ EH H[SUHVVHG LQ GHFLPDO
KH[DGHFLPDO RU RFWDO 7KH GHWDLOV DUH VXP PDUL]HG LQ
[7DEOH](#)

7DEOH ,QWHJHU OLWHUDOV

Kind Format Examples Decimal 0 *or* a nonzero digit followed by zero
or more digits (0–9) 0, 1, 321 Hexadecimal 0x followed by one or more
hexadecimal digits (0–9, A–F, a–f) 0xFF, 0x1a3b Octal 0 followed by
one or more octal digits (0–7) 013, 077

)RU Long OLWHUDOV LW LV QHFHVVDU\ WR DSSHQG WKH L RU I
FKDUDFWHU DW WKH HQG RI WKH OLWHUDO 2WKHUZLVH DQ Int LV
XVHG 7KH YDOLG YDOXHV IRU DQ LQWHJHU OLWHUDO DUH
ERXQG HG E\ WKH WSH RI WKH YDULDEOH WR ZKLFK WKH YDOXH
ZLOO EH DVVLJQH [7DEOH](#) GHILQH WKH OLPLWV ZKLFK DUH
LQFOXVLYH

7DEOH 5DQJHV RI DOORZHG YDOXHV IRU LQWHJHU OLWHUDOV
ERXQGDULHV DUH LQFOXVLYH

Target type Minimum (inclusive) Maximum (inclusive)

Long -2^{63} $2^{63} - 1$

Int -2^{31} $2^{31} - 1$

Short -2^{15} $2^{15} - 1$

Char 0 $2^{16} - 1$

Byte -2^7 $2^7 - 1$

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```
$ FRPSLOH WLPH HUURU RFFXUV LI DQ LQWHJHU OLWHUDO QXPEHU
LV VSHFLILHG WKDW LV RXWVLGH WKHVH UDQJHV DV LQ WKH
IROORZLQJ H[DPSOHV
```

```
scala> val i = 12345678901234567890
<console>:1: error: integer number too large
val i = 12345678901234567890
scala> val b: Byte = 128
<console>:4: error: type mismatch;
found   : Int(128)
required: Byte
val b: Byte = 128
^
```

```
scala> val b: Byte = 127
b: Byte = 127
```

Floating-Point Literals

```
)ORDWLQJ SRLQW OLWHUDOV DUH H[SUHVVLRQV ZLWK ]HUR RU
PRUH GLJLWV IROORZHG E\ D SHULRG
. IROORZHG E\ ]HUR RU PRUH GLJLWV ,I WKHUH DUH QR GLJLWV
EHIRUH WKH SHULRG L H WKH QXPEHU LV OHVV WKDQ WKHQ
WKHUH PXVW EH RQH RU PRUH GLJLWV DIWHU WKH SHULRG )RU
Float OLWHUDOV DSSHQG WKH F RU f FKDUDFWHU DW WKH HQG
RI WKH OLWHUDO 2WKHUZLVH D Double LV DVVXPHG <RX FDQ
RSWLRQDOO\ DSSHQG D D RU d IRU D Double
```

```
)ORDWLQJ SRLQW OLWHUDOV FDQ EH H[SUHVVHG ZLWK RU
ZLWKRXW H[SRQHQLDOV 7KH IRUPDW RI WKH H[SRQHQLDO
SDUW LV e RU E IROORZHG E\ DQ RSWLRQDO + RU - IROORZHG E\
RQH RU PRUH GLJLWV
```

```
+HUH DUH VRPH H[DPSOH IORDWLQJ SRLQW OLWHUDOV
```

```
0.
.0
0.0
3.
3.14
.14
0.14
3e5
3E5
3.E5
3.e5
3.e+5
```

3.e-5
3.14e-5
3.14e-5f
3.14e-5F
3.14e-5d
3.14e-5D

Float FRQVLVWV RI DOO ,(((ELW VLQJOH SUHFLVLRQ ELQDU\
IORDWLQJ SRLQW YDOXH Double FRQVLVWV RI DOO ,(((ELW
GRXEOH SUHFLVLRQ ELQDU\ IORDWLQJ SRLQW YDOXH

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7R DYRLG SDUVLQJ DPXLJLWLHV \RX PXVW KDYL DW OHDVW RQH
VSDFH DIWHU D
IORDWLQJ SRLQW OLWHUDO LI LW LV IROORZHG E\ D WRNHQ WKDW
VWDUWV ZLWK D OHWWHU
\$OVR WKH H[SUHVVLRQ 1.toString UHWXUQV WKH LQWHJHU YDOXH 1 DV D VWULQJ
ZKLOH 1. toString XVHV WKH *RSHUDWRU QRWDWRQ* WR LQYRNH toString RQ WKH
IORDWLQJ SRLQW OLWHUDO 1.

Boolean Literals

7KH ERROHDQ OLWHUDOV DUH true DQG false 7KH \SH RI WKH
YDULDEOH WR ZKLFK WKH\ DUH DVLJQHG ZLOO EH LQIHUUHG WR
EH Boolean

```
scala> val b1 = true  
b1: Boolean = true
```

```
scala> val b2 = false  
b2: Boolean = false
```

Character Literals

\$ FKDUDFWHU OLWHUDO LV HLWKHU D SULQWDEOH 8QLFRGH
FKDUDFWHU RU DQ HVFDSH VHTXHQFH ZULWWHQ EHWZHHQ
VLQJOH TXRWHV \$ FKDUDFWHU ZLWK D 8QLFRGH YDOXH EHWZHHQ
DQG PD\ DOVR EH UHSUHVHQWHG E\ DQ RFWDO HVFDSH L H D
EDFNVDVK
\ IROORZHG E\ D VHTXHQFH RI XS WR WKUHH RFWDO FKDUDFWHUV
,W LV D FRPSLOH WLPH HUURU LI D EDFNVODVK FKDUDFWHU LQ D
FKDUDFWHU RU VWULQJ OLWHUDO GRHV QRW VWDUW D YDOLG
HVFDSH VHTXHQFH

+HUH DUH VRPH H[DPSOHV

```
'A'  
'\u0041' // 'A' in Unicode  
'n'  
'\012' // 'n' in octal  
't'
```

7KH YDOLG HVFDSH VHTXHQFHV DUH VKRZQ LQ **7DEOH**

7DEOH &KDUDFWHU HVFDSH VHTXHQFHV

Sequence Unicode Meaning

\b \u0008 Backspace (BS)
 \t \u0009 Horizontal tab (HT)
 \n \u000a Line feed (LF)
 \f \u000c Form feed (FF)
 \r \u000d Carriage return (CR)
 \" \u0022 Double quote (")
 \' \u0027 Single quote (')
 \\ \u005c Backslash (\)

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String Literals

\$ VWULQJ OLWHUDO LV D VHTXHQFH RI FKDUDFWHUV HQFORVHG LQ GRXEOH TXRWHV RU *WULSOHV* RI GRXEOH TXRWHV L H "" "" .. ""

)RU VWULQJ OLWHUDOV LQ GRXEOH TXRWHV WKH DOORZHG FKDUDFWHUV DUH WKH VDPH DV WKH FKDUDFWHU OLWHUDOV +RZHYHU LI D GRXEOH TXRWH "FKDUDFWHU DSSHDUV LQ WKH VWULQJ LW PXVW EH £HVFDShG ZLWK D \ FKDUDFWHU +HUH DUH VRPH H[DPSOHV

```

"Programming\nScala"
"He exclaimed, \"Scala is great!\""
"First\tSecond"

```

7KH VWULQJ OLWHUDOV ERXQG HG E\ WULSOHV RI GRXEOH TXRWHV DUH DOVR FDOOHG *PXOWL OLQH* VWULQJ OLWHUDOV 7KHVH VWULQJV FDQ FRYHU VHYHUDO OLQHV WKH OLQH IHHGV ZLOO EH SDUW RI WKH VWULQJ 7KH\ FDQ LQFOXGH DQ\ FKDUDFWHUV LQFOXGLQJ RQH RU WZR GRXEOH TXRWHV WRJHWKHU EXW QRW WKUHH WRJHWKHU 7KH\ DUH XVHIXO IRU VWULQJV ZLWK \ FKDUDFWHUV WKDW GRQ\W IRUP YDOLG 8QL FRGH RU HVFDSh VHTXHQFHV OLNH WKH YDOLG VHTXHQFHV OLVWHG LQ **7DEOH 5HJXODU** H[SUH VLRQV DUH D \SLFDO H[DPSOH ZKLFK ZH\OO GLVFXVV LQ **&KDSWHU 5RZHYHU** LI HVFDSh VHTXHQFHV DSSHDU WKH\ DUHQ\W LQWHUSUHHWG

+HUH DUH WKUHH H[DPSOH VWULQJV

```

""""Programming\nScala""""
""""He exclaimed, "Scala is great!" """"
""""First line\n
Second line\t

Fourth line""""

```

1RWH WKDW ZH KDG WR DGG D VSDFH EHIRUH WKH WUDLOLQJ "" "" LQ WKH VHFRQG H[DPSOH WR SUHYHQW D SDUVH HUURU 7ULQJ WR HVFDSh WKH VHFRQG " WKDW HQGV WKH "Scala is great!" TXRWH L H "Scala is great!\ " GRHVQ\W ZRUN

&RS\ DQG SDVWH WKHVH VWULQJV LQWR WKH scala LQWHUSUHHWHU 'R WKH VDPH IRU WKH SUHYLRXV VWULQJ H[DPSOHV +RZ DUH WKH\

Symbol Literals

```
6FDOD VXSSRUWV \PEROV ZKLFK DUH LQWHUQHVG VWULQJV
PHDQLQJ WKDW WZR \PEROV ZLWK WKH VDPH £QDPH
L H WKH VDPH FKDUFWHU VHTXHQFH ZLOO DFWXDOO\ UHIHU WR
WKH VDPH REMHFW LQ PHPRU\ 6PEROV DUH XVHG OHVV RIWHQ LQ
6FDOD WKDQ LQ VRPH RWKHU ODQJXDJHV OLNH 5XE\ 6PDOOWDON
DQG /LVS 7KH\ DUH XVHIXO DV PDS NHV LQVWHDG RI VWULQJV
```

```
$ \PERO OLWHUDO LV D VLQJOH TXRWH
```

```
' IROORZHG E\ D OHWWHU IROORZHG E\ JHUR RU PRUH GLJLWV DQG
OHWWHUV 1RWH WKDW DQ H[SUHVVLRQ OLNH '1 LV LQYDOLG
EHFDXVH WKH FRPSLOHU WKLQNV LW LV DQ LQFRPSOHWH
FKDUFWHU OLWHUDO
```

```
$ \PERO OLWHUDO 'id LV D VKRUWKDQG IRU WKH H[SUHVVLRQ scala.Symbol("id")
```

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```
,I \RX ZDQW WR FUHDWH D \PERO WKDW FRQWDLQV ZKLWHVSDFH
XVH H J
scala.Symbol(" Programming Scala ") $OO WKH ZKLWHVSDFH LV SUHVHUYHG
```

Tuples

```
+RZ PDQ\ WLPHV KDYH \RX ZDQWHG WR UHWXUQ WZR RU PRUH
YDOXH V IURP D PHWKRG" ,Q PDQ\ ODQJXDJHV OLNH -DYG \RX RQO\
KDYH D IHZ RSWLRQV QRQH RI ZKLFK LV YHU\ DSSHOLQJ <RX
FRXOG SDVV LQ SDUDPHWHUV WR WKH PHWKRG WKDW ZLOO EH
PRGLILHG IRU DOO RU VRPH RI WKH £UHWXUQ YDOXH ZKLFK LV
XJO\ 2U \RX FRXOG GHFODUH VRPH VPDOO £VWUXFWXUDO FODVV
WKDW KROGV WKH WZR RU PRUH YDOXH WKHQ UHWXUQ DQ
LQVWDQFH RI WKDW FODVV
```

```
6FDOD VXSSRUWV WXSOHV D JURXSLQJ RI WZR RU PRUH LWHPV
XVXDOO\ FUHDWHG ZLWK WKH OLWHUDO \QWD[ RI D FRPPD
VHSDUDWHG OLVW RI WKH LWHPV LQVLGH SDUHQWKHVVH H J (x1,
x2, ...) 7KH WSHV RI WKH xL HOHPHQWV DUH XQUHODWHG WR HDFK
RWKHU \RX FDQ PL[ DQG PDWFK WSHV 7KHVH OLWHUDO
£JURXSLQJV DUH LQVWDQGLDWHG DV scala.TupleN LQVWDQFHV
ZKHUH N LV WKH QXPEHU RI LWHPV LQ WKH WXS OH 7KH 6FDOD $3,
GHILQH VHSUDWH TupleN FODVVHV IRU N EHWZHHQ DQG
LQFOXVLYH 7XSOH LQVWDQFHV DUH LPPXWDEOH ILUVW FODVV
YDOXH V \RX FDQ DVVLJQ WKHP WR YDULDEOHV SDVV WKHP DV
YDOXH V DQG UHWXUQ WKHP IURP PHWKRGV
```

```
7KH IROORZLQJ H[DPSOH GHPRQVUDWHV WKH XVH RI WXS OHV
```

// code-examples/TypeLessDoMore/tuple-example-script.scala

```
def tupleator(x1: Any, x2: Any, x3: Any) = (x1, x2, x3)
```

```
val t = tupleator("Hello", 1, 2.3)
println( "Print the whole tuple: " + t )
println( "Print the first item: " + t._1 )
println( "Print the second item: " + t._2 )
println( "Print the third item: " + t._3 )
```

```
val (t1, t2, t3) = tupleator("World", '!', 0x22)
println( t1 + " " + t2 + " " + t3 )
```

5XQQLQJ WKLV VFULSW ZLWK scala SURGXFHV WKH IROORZLQJ RXWSXW

```
Print the whole tuple: (Hello,1,2.3)
Print the first item: Hello
Print the second item: 1
Print the third item: 2.3
World ! 34
```

7KH tupleator PHWKRG VLPSO\ UHWXUQV D £ WXSÖH ZLWK WKH LQSWX DUJXPHQWV 7KH ILUVW VWDWHPHQW WKDW XVHV WKLV PHWKRG DVVLJQV WKH UHWXUQHJ WXSÖH WR D VLQJOH YDULDEOH t 7KH QH[W IRXU VWDWHPHQWV SULQW t LQ YDULRXV ZD\V 7KH ILUVW SULQW VWDWHPHQW FDOOV Tuple3.toString ZKLFK ZUDSV SDUHQWKHVV DURXQG WKH LWHP OLWV 7KH IROORZLQJ WKUHH VWDWHPHQWV SULQW HDFK LWHP LQ t VHSUDWHO\ 7KH H[SUHVVLRQ t._N UHWULHYHV WKH N LWHP VWDUWLQJ DW QRW WKLV FKRLFH IROORZV IXQFWLRQDO SURJUDPPLQJ FRQYHQWLRQV

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7KH ODVW WZR OLQHV VKRZ WKDW ZH FDQ XVH D WXSÖH H[SUHVVLRQ RQ WKH OHIWKDQG VLGH RI WKH DVVLJQPHQW :H GHFODUH WKUHH val t1 t2 DQG t3 WR KROG WKH LQGLYLGXDO LWHPV LQ WKH WXSÖH ,Q HVVHQFH WKH WXSÖH LWHPV DUH H[WUDFWHG DXWRPDWLFDO\

1RWLFH KRZ ZH PL[HG WSHV LQ WKH WXSÖHV <RX FDQ VHH WKH WSHV PRUH FOHDU\ LI \RX XVH WKH LQWHUFWLYH PRGH RI WKH scala FRPPDQG ZKLFK ZH LQWURGXFHG LQ &KDSWHU

,QYRNH WKH scala FRPPDQG ZLWK QR VFULSW DUJXPHQW \$W WKH scala> SURPSW HQWHU val t = ("Hello",1,2.3) DQG VHH WKDW \RX JHW WKH IROORZLQJ UHVXOW ZKLFK VKRZV \RX WKH WSH RI HDFK HOHPHQW LQ WKH WXSÖH

```
scala> val t = ("Hello",1,2.3)
t: (java.lang.String, Int, Double) = (Hello,1,2.3)
```

,W\V ZRUWK QRWLQJ WKDW WKHUH\ PRUH WKDQ RQH ZD\ WR GHILQH D WXSÖH :H\YH EHHQ XVLQJ WKH PRUH FRPPRQ SDUHQWKHVLJHG \QWD[EXW \RX FDQ DOVR XVH WKH DUURZ RSHUDWRU EH WZHHQ WZR YDOXHV DV ZHOO DV VSHFLDO IDWRU\ PHWKRGV RQ WKH WXSÖH UHODWHG FODVVHV

```
scala> 1 -> 2
res0: (Int, Int) = (1,2)
```

```
scala> Tuple2(1, 2)
res1: (Int, Int) = (1,2)
```

```
scala> Pair(1, 2)
res2: (Int, Int) = (1,2)
```

Option, Some, and None: Avoiding nulls

```
:H|OO GLVFXVV WKH VWDQGDUG W\SH KLHUDUFK\ IRU 6FDOD LQ
£7KH 6FDOD 7\SH +LHUDU FK\ RQ SDJH +RZHYHU WKUHH
XVHIXO FODVVHV WR XQGHUVWDQG QRZ DUH WKH Option FODV
DQG LWV WZR VXEFOVVHV Some DQG None
```

```
0RVW ODQJXDJHV KDYH D VSHFLDO NH\ZRUG RU REMHFW WKDW|V
DVVLJQH WR UHIHUHQFH YDULDEOHV ZKHQ WKHUH|V QRWKLQJ
HOVH IRU WKHP WR UHIHU WR ,Q -DYD WKLV LV null LQ 5XE\ LW|V nil
,Q -DYD null LV D NH\ZRUG QRW DQ REMHFW DQG WKXV LW|V
LOOHJDO WR FDOO DQ\ PHWKRGV RQ LW %XW WKLV LV D FRQIXVLQJ
FKRLFH RQ WKH ODQJXDJH GHVLJQUH|V SDUW :K\ UHWXUQ D
NH\ZRUG ZKHQ WKH SURJUDPPHU H[SHFWV DQ REMHFW"
```

```
7R EH PRUH FRQLVWHQW ZLWK WKH JRDO RI PDNLQJ HYHU\WKLQJ
DQ REMHFW DV ZHOO DV WR FRQ IRUP ZLWK IXQFWLRQDO
SURJUDPPLQJ FRQYHQWLRQV 6FDOD HQFRXUDJHV \RX WR XVH
WKH Option W\SH IRU YDULDEOHV DQG IXQFWLRQ UHWXUQ YDOXH
ZKHQ WKH\ PD\ RU PD\ QRW UHIHU WR D YDOXH :KHQ WKHUH LV QR
YDOXH XVH None DQ object WKDW LV D VXEFOVV RI Option :KHQ
WKHUH LV D YDOXH XVH Some ZKLFK ZUDSV WKH YDOXH Some LV
DOVR D VXEFOVV RI Option
```

```
None LV GHFODUHG DV DQ object QRW D class EHFDXVH ZH UHDOO\ RQO\
QHHG
RQH LQVWDQFH RI LW ,Q WKDW VHQVH LW|V OLNH WKH null NH\ZRUG EXW LW LV D
UHDO
REMHFW ZLWK PHWKRGV
```

Option, Some, and None: Avoiding nulls | 41

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```
<RX FDQ VHH Option Some DQG None LQ DFWLRQ LQ WKH IROORZLQJ
H[DPSOH ZKHUH ZH FUHDWH D PDS RI VWDWH FDSLWDOV LQ WKH
8QLWHG 6WDWHV
```

```
// code-examples/TypeLessDoMore/state-capitals-subset-script.scala
```

```
val stateCapitals = Map(
  "Alabama" -> "Montgomery",
  "Alaska" -> "Juneau",
  // ...
  "Wyoming" -> "Cheyenne")
```

```
println( "Get the capitals wrapped in Options:" )
println( "Alabama: " + stateCapitals.get("Alabama") )
println( "Wyoming: " + stateCapitals.get("Wyoming") )
println( "Unknown: " + stateCapitals.get("Unknown") )
```

```
println( "Get the capitals themselves out of the Options:" )
println( "Alabama: " + stateCapitals.get("Alabama").get )
println( "Wyoming: " + stateCapitals.get("Wyoming").getOrElse("Oops!") ) println(
  "Unknown: " + stateCapitals.get("Unknown").getOrElse("Oops2!") ) )
```

```

7KH FRQYHQLHQW -> V\QWD[ IRU GHILQLQJ QDPH YDOXH SDLUV WR
LQLWLDOL]H D Map ZLOO EH GLV FXVVG LQ £7KH 3UHGHI 2EMHFW=
RQ SDJH )RU QRZ ZH ZDQW WR IRFXV RQ WKH WZR JURXS RI
println VWDWHPHQWV ZKHUH ZH VKRZ ZKDW KDSSHQV ZKHQ \RX
UHWULHYH WKH YDOXHV IURP WKH PDS ,I \RX UXQ WKLW VFULSW
ZLWK WKH scala FRPPDQG \RX\OO JHW WKH IROORZLQJ RXWSXW

```

Get the capitals wrapped in Options:

Alabama: Some(Montgomery)

Wyoming: Some(Cheyenne)

Unknown: None

Get the capitals themselves out of the Options:

Alabama: Montgomery

Wyoming: Cheyenne

Unknown: Oops2!

```

7KH ILUVW JURXS RI println VWDWHPHQWV LQYRNH toString
LPSOLFLWO\ RQ WKH LQVWDQFHV UH WXUQH E\ get :H DUH
FDOOLQJ toString RQ Some RU None LQVWDQFHV EHFDXVH WKH
YDOXHV UHWXUQH E\ Map.get DUH DXWRPDWLFDOO\ ZUDSSHG LQ D
Some ZKHQ WKHUH LV D YDOXH LQ WKH PDS IRU WKH VSHFLILHG NH\
1RWH WKDW WKH 6FDOD OLEUDU\ GRHVQ\W VWRUH WKH Some LQ
WKH PDS LW ZUDSV WKH YDOXH LQ D Some XSRQ UHWULHYDO
&RQYHUVHO\ ZKHQ ZH DVN IRU D PDS HQWU\ WKDW GRHVQ\W H[LWV
WKH None REMHFW LV UHWXUQH UDWKHU WKDQ null 7KLW
RFFXUUHG LQ WKH ODVW println RI WKH WKUHH

```

```

7KH VHFRQG JURXS RI println VWDWHPHQWV JRHV D VWHS IXUWKHU
$IWHU FDOOLQJ Map.get WKH\ FDOO get RU getOrElse RQ HDFK Option
LQVWDQFH WR UHWULHYH WKH YDOXH LW FRQWDLQV Option.get
UHTXLUV WKDW WKH Option LV QRW HPSW\WKDW LV WKH Option
LQVWDQFH PXVW DFWXDOO\ EH D Some ,Q WKLW FDVH get UHWXUQV
WKH YDOXH ZUDSSHG E\ WKH Some DV GHPRQ VWUDWHG LQ WKH
println ZKHUH ZH SULQW WKH FDSLWDO RI $ODEDPD +RZHYHU LI WKH
Option LV DFWXDOO\ None WKHQ None.get WKURZV D
NoSuchElementException

```

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```

:H DOVR VKRZ WKH DOWHUQDWLYH PHWKRQ getOrElse LQ WKH
ODVW WZR println VWDWHPHQWV 7KLW PHWKRQ UHWXUQV HLWKHU
WKH YDOXH LQ WKH Option LI LW LV D Some LQVWDQFH RU LW
UHWXUQV WKH VHFRQG DUJXPHQW ZH SDVVHG WR getOrElse LI LW LV
D None LQVWDQFH ,Q RWKHU ZRUGV WKH VHFRQG DUJXPHQW WR
getOrElse IXQFWLRQV DV WKH GHIDXOW UHWXUQ YDOXH

```

```

6R getOrElse LV WKH PRUH GHIHQVLYH RI WKH WZR PHWKRGV ,W
DYRLGV D SRWHQWLDO WKURZQ H[FHSLRQ :H\OO GLVFXVV WKH
PHULWV RI DOWHUQDWLYHV OLNH get YHUVV getOrElse LQ £(FHS
WLRQV DQG WKH $OWHUQDWLYHV= RQ SDJH

```

```

1RWH WKDW EHFDXVH WKH Map.get PHWKRQ UHWXUQV DQ Option LW
DXWRPDWLFDOO\ GRFXPHQWV WKH IDFW WKDW WKHUH PD\ QRW
EH DQ LWHP PDWFKLQJ WKH VSHFLILHG NH\ 7KH PDS KDQGOHV

```

```

WKL\ VLWXDWLRQ E\ UHWXUQLQJ D None 0RVW ODQJXDJHV ZRXOG
UHWXUQ null
RU WKH HTXLYDOHQW ZKHQ WKHUH LV QR £UHDO $\alpha$  YDOXH WR
UHWXUQ <RX OHDUQ IURP H[SHULHQFH WR H[SHFW D SRVVLEOH null
8VLQJ Option PDNH WKH EHKDYLRU PRUH H[SOLFLW LQ WKH
PHWKRG VLJQDWXUH VR LW\ PRUH VHOI GRFXPHQLQJ

$OVR WKDQNV WR 6FDOD\ VWDWLF W\SLQJ \RX FDQ\W PDNH WKH
PLVWDNH RI DWWHPSWLQJ WR FDOO D PHWKRG RQ D YDOXH WKDW
PLJKW DFWXDOO\ EH null :KLOH WKL\ PLVWDNH LV HD\ WR GR LQ
-DYD LW ZRQ\W FRPSLOH LQ 6FDOD EHFDXVH \RX PXVW ILUVW
H[WUDFW WKH YDOXH IURP WKH Option 6R WKH XVH RI Option
VWURQJO\ HQFRXUDJHV PRUH UHVLOLHQW SURJUDPPHQJ

%HFDXVH 6FDOD UXQV RQ WKH -90 DQG 1(7 DQG EHFDXVH LW PXVW
LQWHURSHUDWH ZLWK RWKHU OLEUDULHV 6FDOD KDV WR
VXSSRUW null 6WLOO \RX VKRXOG DYRLG XVLQJ null LQ \RXU FRGH
7RQ\ +RDUH ZKR LQYHQWHG WKH QXOO UHIHQFH LQ ZKLOH
ZRUNLQJ RQ DQ REMHFW RULHQWHG ODQJXDJH FDOOHG $/*2/ :
FDOOHG LWV LQYHQWLRQ KLV £ELOOLRQ GROODU PLVWDNH $\alpha$ 
VHH >+RDUH @ 'RQ\W FRQWULEXWH WR WKDW ILJXH

```

```

6R KRZ ZRXOG \RX ZULWH D PHWKRG WKDW UHWXUQV DQ Option"
+HUH LV D SRVVLEOH LPSOH PHQDWRQ RI get WKDW FRXOG EH
XVHG E\ D FRQFUHWH VXEFOVV RI Map
Map.get LWVHOI LV DEVUDFW )RU D PRUH VRSKLVWDFDWHG
YHUVLRQ VHH WKH LPSOHPHQDWRQ RI get LQ
scala.collection.immutable.HashMap LQ WKH 6FDOD OLEUDU\ VRXUFH FRGH
GLVWULEXWLRQ

```

```

def get(key: A): Option[B] = {
  if (contains(key))
    new Some(getValue(key))
  else
    None
}

```

```

7KH contains PHWKRG LV DOVR GHILQH IRU Map ,W UHWXUQV true LI
WKH PDS FRQWDLQV D YDOXH IRU WKH VSHFLILHG NH\ 7KH getValue
PHWKRG LV LQWHQGHG WR EH DQ LQWHUQDO PHWKRG WKDW
UHWULHYHV WKH YDOXH IURP WKH XQGHUOLQJ VWRUDJH
ZKDWHYHU LW LV

```

```

1RWH KRZ WKH YDOXH UHWXUQHG E\ getValue LV ZUDSSHG LQ D
Some[B] ZKHU WKH W\SH B LV LQIHUHG +RZHYHU LI WKH FDOO WR
contains(key) UHWXUQV false WKHQ WKH object None LV UHWXUQHG

```

```

<RX FDQ XVH WKL\ VDPH LGLRP ZKHQ \RXU PHWKRGV UHWXUQ DQ
Option :H\OO H[SORUH RWKHU XVHV IRU Option LQ VXEVHTXHQW
VHFWLRQV ,WV SHUYDVLYH XVH LQ 6FDOD FRGH PDNHV LW DQ
LPSRUWDQW FRQFHSW WR JUDVS

```

Option, Some, and None: Avoiding nulls | 43

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Organizing Code in Files and Namespaces

```

6FDOD DGRSWV WKH SDFNDJH FRQFHSW WKDW -DYD XVHV IRU

```

```
QDPHVSDFHV EXW 6FDOD RIIHUV D PRUH IOH[LEOH \QWD[ -XVW DV
ILOH QDPHV GRQ[W KDYH WR PDWFK WKH WSH QDPHV WKH
SDFNDJH VWUXFWXUH GRHV QRW KDYH WR PDWFK WKH
GLUHFWRU\ VWUXFWXUH 6R \RX FDQ GHILQH SDFNDJHV LQ ILOHV
LQGHSQGHQW RI WKHLU £SK\VLFDQ ORFDWLRQ
```

```
7KH IROORZLQJ H[DPSOH GHILQHV D FODVV MyClass LQ D SDFNDJH
com.example.mypkg XVLQJ WKH FRQYHQWLRQDO -DYD \QWD[
```

```
// code-examples/TypeLessDoMore/package-example1.scala
```

```
package com.example.mypkg
```

```
class MyClass {
  // ...
}
```

```
7KH QH[W H[DPSOH VKRZV D FRQWULYHG H[DPSOH WKDW GHILQHV
SDFNDJHV XVLQJ WKH QHVWHG SDFNDJH \QWD[ LQ 6FDOD ZKLFK
LV VLPLODU WR WKH namespace \QWD[ LQ & DQG WKH XVH RI modules
DV QDPHVSDFHV LQ 5XE\
```

```
// code-examples/TypeLessDoMore/package-example2.scala
```

```
package com {
  package example {
    package pkg1 {
      class Class11 {
        def m = "m11"
      }
      class Class12 {
        def m = "m12"
      }
    }
  }
}
```

```
package pkg2 {
  class Class21 {
    def m = "m21"
    def makeClass11 = {
      new pkg1.Class11
    }
    def makeClass12 = {
      new pkg1.Class12
    }
  }
}
```

```
package pkg3.pkg31.pkg311 {
  class Class311 {
    def m = "m21"
  }
}
```

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```
7ZR SDFNDJHV pkg1 DQG pkg2 DUH GHILQH XQGHU WKH
com.example SDFNDJH $ WRWDO RI WKUHH FODVVHV DUH GHILQH
EHWZHHQ WKH WZR SDFNDJHV 7KH makeClass11 DQG makeClass12
PHWKRGV LQ Class21 LOOXVUDWH KRZ WR UHIHUQFH D WSH
LQ WKH £VLEOLQJ SDFNDJH pkg1 <RX FDQ DOVR UHIHUQFH
```

WKHVV FODVVHV E\ WKHLU IXOO SDWKV com.example.pkg1.Class11
DQG com.example.pkg1.Class12 UHVSHFWLYHO\

7KH SDFNDJH pkg3.pkg31.pkg311 VKRZV WKDW \RX FDQ £FKDLQ
VHYHUDO SDFNDJHV WRJHWKHU LQ RQH FODXVH ,W LV QRW
QHFHVVDU\ WR XVH D VHSUDUDWH package FODXVH IRU HDFK
SDFNDJH

)ROORZLQJ WKH FRQYHQWLRQV RI -DYD WKH URRW SDFNDJH IRU
6FDOD,V OLEUDU\ FODVVHV LV QDPHG scala

6FDOD GRHV QRW DOORZ SDFNDJH GHFODUDWLRQV LQ VFULSWV WKDW
DUH H[HFVHG
GLUHFWO\ ZLWK WKH scala LQWHUSUHWHU 7KH UHDVRQ KDV WR GR ZLWK
WKH ZD\
WKH LQWHUSUHWHU FRQYHUWV VWDWHPHQWV LQ VFULSWV WR YDOLG
6FDOD FRGH
EHIRUH FRPSLOLQJ WR EWH FRGH 6HH £7KH VFDOD &RPPDQG /LQH
7RRO RQ SDJH IRU PRUH GHWDLOV

Importing Types and Their Members

7R XVH GHFODUDWLRQV LQ SDFNDJHV \RX KDYH WR LPSRUW WKHP
MXVW DV \RX GR LQ -DYD DQG VLPLODUO\ IRU RWKHU ODQJXDJHV
+RZHYHU FRPSDUHG WR -DYD 6FDOD JUHDWO\ H[SDQGV \RXU
RSWLRQV 7KH IROORZLQJ H[DPSOH LOOXVUDWHV VHYHUDO ZD\V
WR LPSRUW -DYD WSHV

[// code-examples/TypeLessDoMore/import-example1.scala](#)

```
import java.awt._  
import java.io.File  
import java.io.File._  
import java.util.{Map, HashMap}
```

<RX FDQ LPSRUW DOO WSHV LQ D SDFNDJH XVLQJ WKH
XQGHUVFRUH
_ DV D ZLOGFDUG DV VKRZQ RQ WKH ILUVW OLQH <RX FDQ DOVR
LPSRUW LQGLYLGXDO 6FDOD RU -DYD WSHV DV VKRZQ RQ WKH
VHFRQG OLQH

-DYD XVHV WKH £VWDU FKHUFWHU

* DV WKH ZLOGFDUG IRU PDWFKLQJ DOO WSHV LQ D SDFNDJH RU
DOO VWDWLF PHPEHUV RI D WSH ZKHQ GRLQJ £VWDWLF LPSRUW
 ,Q 6FDOD WKLV FKHUFWHU LV DOORZHG LQ PHWKRG QDPHV VR
_ LV XVHG DV D ZLOGFDUG DV ZH VDZ SUHYLRXVO\

\$V VKRZQ RQ WKH WKLUG OLQH \RX FDQ LPSRUW DOO WKH
VWDWLF PHWKRGV DQG ILHOGV LQ -DYD WSHV ,I java.io.File ZHUH
DFWXDOO\ D 6FDOD object DV GLVFXVVG SUHYLRXVO\ WKHQ WKLV
OLQH ZRXOG LPSRUW WKH ILHOGV DQG PHWKRGV IURP WKH
REMHFW

)LQDOO\ \RX FDQ VHOHFWLYHO\ LPSRUW MXVW WKH WSHV \RX
FDUH DERXW 2Q WKH IRXUWK OLQH ZH LPSRUW MXVW WKH
java.util.Map DQG java.util.HashMap WSHV IURP WKH java.util SDFN DJH
&RPSDUH WKLV RQH OLQH LPSRUW VWDWHPHQW ZLWK WKH WZR
OLQH LPSRUW VWDWHPHQWV ZH XVHG LQ RXU ILUVW H[DPSOH
LQ £,QIHUULQJ 7SH ,QIRUPDWLRQ RQ SDJH 7KH DUH
IXQFWLRQDOO\ HTXLDOHQW

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7KH QH[W H[DPSOH VKRZV PRUH DGYDQFHG RSWLRQV

IRU LPSRUW VWDWHPHQWV //

[code-examples/TypeLessDoMore/import-example2-script.scala](#)

```
def writeAboutBigInteger() = {
  import java.math.BigInteger.{
    ONE => _,
    TEN,
    ZERO => JAVAZERO }

  // ONE is effectively undefined
  // println( "ONE: "+ONE )
  println( "TEN: "+TEN )
  println( "ZERO: "+JAVAZERO )
}
```

writeAboutBigInteger()

7KLV H[DPSOH GHPRQVWUDWHV WZR IHDWXUHV)LUVW ZH FDQ
 SXW LPSRUW VWDWHPHQWV DOPRVW DQ\ZKHUH ZH ZDQW QRW
 MXVW DW WKH WRS RI WKH ILOH DV UHTXLUHG E\ -DYD 7KLV
 IHDWXUH DOORZV XV WR VFRSH WKH LPSRUWV PRUH QDUURZO\
)RU H[DPSOH ZH FDQ\W UHIHUHQFH WKH LPSRUWHG BigInteger
 GHILQLWRQV RXWVLGH WKH VFRSH RI WKH PHWKRG \$QRWKHU
 DGYDQWDJH RI WKLV IHD WXUH LV WKDW LW SXVV DQ LPSRUW
 VWDWHPHQW FORVHU WR ZKHUH WKH LPSRUWHG LWHPV DUH
 DFWXDOO\ XVHG

7KH VHFRQG IHDWXUH VKRZQ LV WKH DELOLW\ WR UHQDPH
 LPSRUWHG LWHPV)LUVW WKH java.math.BigInteger.ONE FRQVWDQW
 LV UHQDPHG WR WKH XQGHUVFRUH ZLOGFDUG 7KLV HIIHF WLYHO\
 PDNHV LW LQYLVLDOH DQG XQDYDLODEOH WR WKH LPSRUWLQJ
 VFRSH 7KLV LV D XVHIXO WHFK QLTXH ZKHQ \RX ZDQW WR LPSRUW
 HYHUWKLQJ H\FHSW D IHZ SDUWLFXODU LWHPV

1H[W WKH java.math.BigInteger.TEN FRQVWDQW LV LPSRUWHG ZLWKRXW
 UHQDPLQJ VR LW FDQ EH UHIHUHQFHG VLPSO\ DV TEN

)LQDOO\ WKH java.math.BigInteger.ZERO FRQVWDQW LV JLYHQ WKH £DOLDV¤ JAVAZERO
 \$OLDVLQJ LV XVHIXO LI \RX ZDQW WR JLYH WKH LWHP D PRUH
 FRQYHQLHQW QDPH RU \RX ZDQW WR DYRLG DPELJXLWLHV ZLWK
 RWKHU LWHPV LQ VFRSH WKDW KDYH WKH VDPH QDPH

Imports are Relative

7KHUH\ V RQH RWKHU LPSRUWDQW WKLQJ WR NQRZ DERXW
 LPSRUWV WKH\ DUH UHODWLYH 1RWH WKH FRPPHQWV IRU WKH
 IROORZLQJ LPSRUWV

```
// code-examples/TypeLessDoMore/relative-imports.scala
import scala.collection.mutable._
import collection.immutable._ // Since "scala" is already imported import
_root_.scala.collection.jcl._ // full path from real "root"
```

```
package scala.actors {
import remote._ // We're in the scope of "scala.actors" }
```

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```
1RWH WKDW WKH ODVW LPSRUW VWDWHPHQW QHVWHG LQ WKH
scala.actor SDFNDJH VFRSH LV UHODWLYH WR WKDW VFRSH
```

```
7KH >6FDOD:LNL@ KDV RWKHU H[DPSOHV DW KWWS VFDOD
VJQHFD FRP IDTV ODQJXDJH KRZ GR L LPSRUW
```

```
,W|V IDLUO\ UDUH WKDW \RX|OO KDYH SUREOHPV ZLWK UHODWLYH
LPSRUWV EXW WKH SUREOHP ZLWK WKLW FRQYHQWLRQ LV WKDW
WKH\ VRPHWLPHV FDXVH VXUSULVHV HVSHFLDOO\ LI \RX DUH
DFFXVWRPHG WR ODQJXDJHV OLNH -DYD ZKHUH LPSRUWV DUH
DEVROXWH ,I \RX JHW D P|VWL|LQJ FRPSLOHU HUURU WKDW D
SDFNDJH ZDVQ|W IRXQG FKHFN WKDW WKH VWDWHPHQW LV
SURSHUO\ UHODWLYH WR WKH ODVW LPSRUW VWDWHPHQW RU
DGG WKH _root_ SUHIL[ $OVR \RX PLJKW VHH DQ ,'( RU RWKHU WRRO
LQVHUW DQ import _root_... VWDWHPHQW LQ \RXU FRGH 1RZ \RX NQRZ
ZKDW LW PHDQV
```

```
5HPHPEHU WKDW LPSRUW VWDWHPHQWV DUH UHODWLYH QRW
DEVROXWH 7R FUHDWH
DQ DEVROXWH SDWK \VWUW ZLWK _root_
```

Abstract Types And Parameterized Types

```
:H PHQWLRQH G LQ £$ 7DVWH RI 6FDOD $\alpha$  RQ SDJH WKDW 6FDOD
VXSSRUWV SDUDPHWHULJHG WSHV ZKLFK DUH YHU\ VLPLODU WR
JHQHULFV LQ -DYD
```

```
:H FRXOG XVH WKH WZR WHUPV LQWHU FKDQJHDEO\ EXW LW|V
PRUH FRPPRQ WR XVH £SDUDPHWHULJHG WSHV $\alpha$  LQ WKH 6FDOD
FRP PXQLW DQG £JHQHULFV $\alpha$  LQ WKH -DYD FRPPXQLW 7KH PRVW
REYLXV GLIIHUHQFH LV LQ WKH \QWD[ ZKHUH 6FDOD XVHV
VTXDUH EUDFNHWV
```

```
[...] ZKLOH -DYD XVHV DQJOH EUDFNHWV
```

```
<...>
```

```
)RU H[DPSOH D OLWV RI VWULQJV ZRXOG EH GHFODUHG DV IROORZV
```

```
val languages: List[String] = ...
```

```
7KHUH DUH RWKHU LPSRUWDQW GLIIHUHQFHV ZLWK -DYD|V
JHQHULFV ZKLFK ZH|OO H[SORUH LQ £QGHUVWDQGLQJ
3DUDPHWHULJHG WSHV $\alpha$  RQ SDJH
```

```
)RU QRZ ZH|OO PHQWLRQ RQH RWKHU XVHIXO GHWDLO WKDW
\RX|OO HQFRXQWHU EHIRUH ZH FDQ H[SODLQ LW LQ GHSWK LQ
&KDSWHU ,I \RX ORRN DW WKH GHFODUDWLRQ RI scala.List LQ WKH
6FDODGRFV \RX|OO VHH WKDW WKH GHFODUDWLRQ LV ZULWWHQ
DV ... class List[+A] 7KH + LQ IURQW RI WKH A PHDQV WKDW List[B] LV D
VXEWSH RI List[A] IRU DQ B WKDW LV D VXEWSH RI A ,I WKUH LV D -
```

LQ IUHQW RI D WSH SDUDPHWHU WKHQ WKH UHODWLRQVKLS JRHV
WKH RWKHU ZD\ Foo[B] ZRXOG EHD VXSHUWISH RI Foo[A] LI WKH
GHFODUDWLRQ LV Foo[-A]

6FDOD VXSSRUWV DQRWKHU WISH DEVWUDFWLRQ PHFKDQLVP
FDOOHG DEVWUDFW WISHV XVHG LQ PDQ\ IXQFWLRQDO
SURJUDPPLQJ ODQJXDJHV VXFK DV +DVNHOO \$EVWUDFW WISHV
ZHUH DOVR FRQVLG HUHG IRU LQFOXVLRQ LQ -DYD ZKHQ JHQHULFV
ZHUH DGRSWHG :H ZDQW WR LQWURGXFH WKHP QRZ EHFDXVH
\RX\OO VHH PDQ\ H[DPSOHV RI WKHP EHIRUH ZH GLYH LQWR WKHLU
GHWDLOV LQ &KDSWHU)RU D YHU\ GHWDLOHG FRPSDULVRQ RI
WKHVVH WZR PHFKDQLVPV VHH >%UXFH @

\$EVWUDFW WISHV FDQ EH DSSOLHG WR PDQ\ RI WKH VDPH GHVLJQ
SUREOHPV IRU ZKLFK SDUD PHWHULJHG WISHV DUH XVHG +RZHYHU
ZKLOH WKH WZR PHFKDQLVPV RYHUODS WKH\ DUH QRW
UHGXQGDQW (DFK KDV VWUHQJWKV DQG ZHDNQHVHV IRU
FHUWDLQ GHVLJQ SUREOHPV

Abstract Types And Parameterized Types | 47

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+HUH LV DQ H[DPSOH WKDW XVHV DQ DEVWUDFW WISH

```
// code-examples/TypeLessDoMore/abstract-types-script.scala
```

```
import java.io._
```

```
abstract class BulkReader {  
  type In  
  val source: In  
  def read: String  
}
```

```
class StringBulkReader(val source: String) extends BulkReader {  
  type In = String  
  def read = source  
}
```

```
class FileBulkReader(val source: File) extends BulkReader {  
  type In = File  
  def read = {  
    val in = new BufferedInputStream(new FileInputStream(source))  
    val numBytes = in.available()  
    val bytes = new Array[Byte](numBytes)  
    in.read(bytes, 0, numBytes)  
    new String(bytes)  
  }  
}
```

```
println( new StringBulkReader("Hello Scala!").read )  
println( new FileBulkReader(new File("abstract-types-script.scala")).read )
```

5XQQLQJ WKLW VFULSW ZLWK scala SURGXFHV WKH

IROORZLQJ RXWSXW

```
Hello Scala!  
import java.io._
```

```
abstract class BulkReader {  
  ...
```

7KH BulkReader DEVWUDFW FODVV GHFODUHV WKUHH DEVWUDFW

PHPEHUV D type QDPHG In D val ILHOG source DQG D read PHWKRG
 \$V LQ -DYD LQVWDQFHV LQ 6FDOD FDQ RQO\ EH FUHDWHG IURP
FRQFUHWH FODVVHV ZKLFK PXVW KDYH GHILQLWLRQV IRU DOO
 PHPEHUV

7KH GHULYHG FODVVHV `StringBulkReader` `DQG` `FileBulkReader` `SURYLGH`
`FRQFUHWH` `GHILQL` `WLRQV` `IRU` `WKHVH` `DEVWUDFW` `PHPEHUV` :H\OO
`FRYHU` `WKH` `GHWDLOV` `RI` `FODVV` `GHFODUDWLRQV` `LQ` `&KDS` `WHU`
`DQG` `WKH` `SDUWLFXODUV` `RI` `RYHUULGLQJ` `PHPEHU` `GHFODUDWLRQV`
`LQ` `£2YHUULGLQJ` `0HPEHUV` `RI` `&ODVVHV` `DQG` `7UDLWV` `RQ` `SDJH` `LQ`
`&KDSWHU`

)RU QRZ QRWH WKDW WKH type ILHOG ZRUNV YHU\ PXFK OLNH D
 WSH SDUDPHWHU LQ D SDUDPH WHULJHG WSH ,Q IDFW ZH FRXOG
 UHZULWH WKLV H[DPSOH DV IROORZV ZKHUH ZH VKRZ RQO\ ZKDW
 ZRXOG EH GLIIHUHQW

```
abstract class BulkReader[In] {
  val source: In
  ...
}
```

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```
}

class StringBulkReader(val source: String) extends BulkReader[String] {...} class
FileBulkReader(val source: File) extends BulkReader[File] {...}
```

-XVW DV IRU SDUDPHWHULJHG WSHV LI ZH GHILQH WKH In WSH WR
 EH String WKHQ WKH source ILHOG PXVW DOVR EH GHILQHG DV D String
 1RWH WKDW WKH `StringBulkReader`’V `read` `PHWKRG` `VLPSO\` `UHWXUQV`
 WKH source ILHOG ZKLOH WKH `FileBulkReader`’V `read` `PHWKRG` `UHDGV`
 WKH `FRQ` `WHQWV` `RI` `WKH` `ILOH`

\$V GHPRQVWUDWHG E\ `>%UXFH` `@` `SDUDPHWHULJHG` `WSHV` `WHQG`
 WR EH EHVW IRU `FROOHFWLRQV` `ZKLFK` `LV` `KRZ` `WKH\` `DUH` `PRVW`
`RIWHQ` `XVHG` `LQ` `-DYD` `FRGH` `ZKHUHDV` `DEVWUDFW` `WSHV` `DUH` `PRVW`
`XVHIXO` `IRU` `WSH` `£IDPLOLHV` `DQG` `RWKHU` `WSH` `VFHQDULRV`

:H\OO H[SORUH WKH GHWDLOV RI 6FDOD’V `DEVWUDFW` `WSHV` `LQ`
`&KDSWHU`)RU H[DPSOH ZH\OO VHH KRZ WR `FRQVWUDLQ` `WKH`
`SRVVLEOH` `FRQFUHWH` `WSHV` `WKDW` `FDQ` `EH` `XVHG`

Reserved Words

`7DEOH` OLVVW WKH UHVHUYHG ZRUGV LQ 6FDOD ZKLFK ZH
 VRPHWLPHV FDOO £NH\ZRUGV `¤` `DQG` `EULHIO\` `GHVFULEHV` `KRZ` `WKH\`
`DUH` `XVHG`
`VHH` `>6FDOD6SHF` `@`

`7DEOH` `5HVHUYHG` `ZRUGV`

Word Description See ...

`abstract` Makes a declaration abstract.
 Unlike Java, the keyword is usually not

required for abstract members.
[“Class and Object Basics” on page 89](#)

case Start a case clause in a match expression. [“Pattern Matching” on page 63](#)

catch Start a clause for catching thrown exceptions. [“Using try, catch, and finally Clauses” on page 70](#)

class Start a class declaration. [“Class and Object Basics” on page 89](#)

def Start a method declaration. [“Method Declarations” on page 25](#)

do Start a do...while loop. [“Other Looping Constructs” on page 61](#)

else Start an else clause for an if clause. [“Scala if Statements” on page 58](#)

extends Indicates that the class or trait class or trait being declared.
that follows is the parent type of the [“Parent Classes” on page 91](#)

false Boolean *false*. [“The Scala Type Hierarchy” on page 155](#)

final Applied to a class or trait to prohibit deriving child types from it. Applied to a member to prohibit overriding it in a derived class or trait. [“Attempting to Override final Declarations” on page 112](#)

finally Start a clause that is executed after the corresponding try clause, whether or not an exception is thrown by the try clause. [“Using try, catch, and finally Clauses” on page 70](#)

Reserved Words | 49

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Word Description See ...

for Start a for comprehension (loop). [“Scala for Comprehensions” on page 59](#)

forSome Used in *existential type* declarations to constrain the allowed concrete types that can be used. [“Existential Types” on page 284](#)

if Start an if clause. [“Scala if Statements” on page 58](#)

implicit Marks a method as eligible to be used as an *implicit* type converter. Marks a method parameter as optional, as long as a type compatible substitute object is in the scope where the method is called. members of types into the current scope. [“Implicit Conversions” on page 186](#)

import Import one or more types or [“Importing Types and Their Members” on page 45](#)

lazy Defer evaluation of a val. [“Lazy Vals” on page 190](#)

match Start a pattern matching clause. [“Pattern Matching” on page 63](#)

new Create a new instance of a class. [“Class and Object Basics” on page 89](#)

null Value of a reference variable that has not been assigned a value. [“The Scala Type Hierarchy” on page 155](#)

object Start a *singleton* declaration: a class with only one instance. [“Classes and Objects: Where Are the Statics?” on page 148](#)

override Override a *concrete* member of a class or trait, as long as the original is not marked final. [“Overriding Members of Classes and Traits” on page 111](#)

package Start a package scope declaration. [“Organizing Code in Files and Namespaces” on page 44](#)

private Restrict visibility of a declaration. [“Visibility Rules” on page 96](#) protected

Restrict visibility of a declaration. [“Visibility Rules” on page 96](#)

requires Deprecated. Was used for *self typing*. [“The Scala Type Hierarchy” on page 155](#)

return Return from a function. [“A Taste of Scala” on page 10](#)

sealed Applied to a parent class to be declared in the same source file.
require all directly derived classes to [“Case Classes” on page 136](#)

super Analogous to this, but binds to the parent type. [“Overriding Abstract and Concrete Methods” on page 112](#)

this How an object refers to itself. The [“Class and Object Basics” on page 89](#)
method name for *auxiliary constructors*.

throw Throw an exception. [“Using try, catch, and finally Clauses” on page 70](#)

trait A *mixin module* that adds an instance of a class.
additional state and behavior to [Chapter 4](#)

try Start a block that may throw an exception. [“Using try, catch, and finally Clauses” on page 70](#)

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Word Description See ...

true Boolean *true*. [“The Scala Type Hierarchy” on page 155](#)

type Start a *type* declaration. [“Abstract Types And Parameterized Types” on page 47](#)

val Start a read-only “variable” declaration. [“Variable Declarations” on page 24](#) var

Start a read-write variable declaration. [“Variable Declarations” on page 24](#)

while Start a while loop. [“Other Looping Constructs” on page 61](#)

with Include the trait that follows in a sequence.
the class being declared or the object [Chapter 4](#)
being instantiated.

yield Return an element in a for [“Yielding” on page 60](#)
comprehension that becomes part of

_ A placeholder, used in imports, function literals, etc. *Many* : Separator between
identifiers and type annotations. [“A Taste of Scala” on page 10](#) = Assignment. [“A Taste of Scala” on page 10](#)

=> Used in *function literals* to separate [“Function Literals and Closures” on page 169](#)
the argument list from the function
body.

<- Used in for comprehensions in *generator* expressions. [“Scala for Comprehensions” on page 59](#)

<: Used in *parameterized* and *abstract type* declarations to constrain the

allowed types.

[Bounds” on page 259](#) [“Type Bounds”](#)

<% Used in *parameterized* and *abstract*
type “view bounds” declarations. [on page 259](#)

>: Used in *parameterized* and *abstract*
type declarations to constrain the
allowed types.
[“Type Bounds” on page 259](#) [“Type](#)

Used in *type projections*. [“Path-Dependent Types” on page 272](#) @ Marks an
annotation. [“Annotations” on page 289](#)

° (Unicode \u21D2) Same as =>. [“Function Literals and Clo sures” on page](#)
[169](#)

← (Unicode \u2190) Same as <-. [“Scala for Comprehen sions” on page 59](#)

1RWLFH WKDW break DQG continue DUH QRW OLVWHG 7KHVH
FRQWURO NHZRUGV GRQ|W H[LVW LQ 6FDOD ,QVWHDG 6FDOD
HQFRXUDJHV \RX WR XVH IXQFWLRQDO SURJUDPPLQJ LGLRPV
WKDW DUH XVXDOO\ PRUH VXFFLQFW DQG OHVV HUURU SURQH
:H|OO GLVFXVV DOWHUQDWLYH DSSURDFKHV ZKHQ ZH GLVFXVV for
ORRSV
VHH £*HQHUDWRU ([SUHVVLQRV= RQ SDJH

6RPH -DYG PHWKRGV XVH QDPHV WKDW DUH UHVHUYHG E\
6FDOD IRU H[DPSOH java.util.Scanner.match 7R DYRLG D
FRPSLODWLRQ HUURU VXUURXQG WKH QDPH ZLWK VLQJOH EDFN
TXRWHV H J java.util.Scanner.ÂmatchÂ

Reserved Words | 51

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Recap and What’s Next

:H FRYHUHG VHYHUDO ZD|V WKDW 6FDOD|V \QWD[LV FRQFLVH
IOH[LEOH DQG SURGXFWLYH :H DOVR GHVFULEHG PDQ\ 6FDOD
IHDWXUHV ,Q WKH QH[W FKDSWHU ZH ZLOO URXQG RXW VRPH
6FDOD HVVHQWLDV EHIRUH ZH GLYH LQWR 6FDOD|V VXSSRUW IRU
REMHFW RULHQWHG SURJUDPPLQJ DQG IXQFWLRQDO
SURJUDPPLQJ

CHAPTER 3 **Rounding Out the Essentials**

%HIRUH ZH GLYH LQWR 6FDOD\|V VXSSRUW IRU REMHFW RULHQWHG
DQG IXQFWLRQDO SURJUDPPLOJ OHW\|V ILQLVK RXU GLVFXVVLRRQ RI
WKH HVVHQWLDO IHDWXUHV \RX\|OO XVH LQ PRVW RI \RXU
SURJUDPV

Operator? Operator?

\$Q LPSRUWDQW IXQGDPHQWDO FRQFHSW LQ 6FDOD LV WKDW DOO
RSHUDWRUV DUH DFWXDOO\ PHWKRGV &RQVLGHU WKL V PRVW
EDVLF RI H[DPSOHV

[// code-examples/Rounding/one-plus-two-script.scala](#)

1 + 2

7KDW SOXV VLJQ EHWZHHQ WKH QXPEHUV" ,W\|V D PHWKRG
)LUVW 6FDOD DOORZV QRQ DOSKDQXPHULF PHWKRG QDPHV <RX
FDQ FDOO PHWKRGV + - \$ RU ZKDWHYHU \RX GHVLUH 6HFRQG
WKL V H[SUHVVLRQ LV LGHQWLFDQ WR 1 .+(2)

:H SXW D VSDFH DIWHU WKH 1 EHFDXVH 1. ZRXOG EH
LQWHUSUHWHG DV D Double :KHQ D PHWKRG WDNHV RQH
DUJXPHQW 6FDOD OHWV \RX GURS ERWK WKH SHULRG DQG WKH
SDUHQWKHVVH VR WKH PHWKRG LQYRFDWLRQ ORRN OLNH DQ
RSHUDWRU LQYRFDWLRQ 7KLV LV FDOOHG £LQIL[QRWDWLRQ
ZKHUH WKH RSHUDWRU LV EHWZHHQ WKH LQVWDQFH DQG WKH
DUJXPHQW :H\|OO ILQG RXW PRUH DERXW WKL V VKRUWO\

6LPLODUO\ D PHWKRG ZLWK QR DUJXPHQWV FDQ EH LQYRNHG
ZLWKRXW WKH SHULRG 7KLV LV FDOOHG £SRVWIL[QRWDWLRQ

5XE\ DQG 6PDOOWDON SURJUDPPHUV VKRXOG QRZ IHHO ULJKW DW
KRPH \$V XVHUV RI WKRVH ODQJXDJHV NQRZ WKHVVH VLPSOH
UXOHV KDYH IDU UHDFKLQJ EHQHILVV ZKHQ LW FRPHV WR
FUHDWLQJ SURJUDPV WKDW IORZ QDWXUDOO\ DQG HOHJDQWO\

6R ZKDW FKDUDFWHUV FDQ \RX XVH LQ LGHQWLILHUV" +HUH LV D
VXPPDU\ RI WKH UXOHV IRU LGHQ WLILHUV XVHG IRU PHWKRG DQG
W\SH QDPHV YDULDEOHV HWF)RU WKH SUHFLVH GHWDLOV VHH
>6FDOD6SHF @ 6FDOD DOORZV DOO WKH SULQWDEOH \$6&.,
FKDUDFWHUV VXFK DV OHWWHUV GLJLWV WKH XQGHUVFRUH

_ DQG WKH GROODU VLJQ
\$ ZLWK WKH H[FHSWLRQV RI WKH £SDUHQWKHWFDO $\varnothing$
FKDUDFWHUV ϕ () [] { DQG } ϕ DQG WKH £GHOLPLWHU $\varnothing$
FKDUDFWHUV ϕ ' ' " . ; DQG , 6FDOD DOORZV WKH RWKHU
FKDUDFWHUV EHWZHHQ ?X ;?X) WKDW DUH QRW LQ WKH VHWV
MXVV

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VKRZQ VXFk DV PDWKHPDWLFDQ \PEROV DQG £RWKHU\ \PEROV
7KHVH UHPDLQLQJ FKDUF WHUV DUH FDOOHG RSHUDWRU
FKDUDFWHUV DQG WKH\ LQFOXGH FKDUDFWHUV VXFk DV / < HWF

5HVHUYHG ZRUGV FDQ\W EH XVHG

\$V LQ PRVW ODQJXDJHV \RX FDQ\W UHXVH UHVHUYHG ZRUGV
IRU LGHQWLILHUV :H OLVWHG WKH UHVHUYHG ZRUGV LQ
£5HVHUYHG :RUGV\ RQ SDJH 5HFDOO WKDW VRPH RI WKHP
DUH FRPELQDWLRQV RI RSHUDWRU DQG SXQFWXDWRQ
FKDUDFWHUV)RU H[DPSOH D VLQJOH XQ GHUVFRUH
_ LV D UHVHUYHG ZRUG

3ODLQ LGHQWLILHUV\FRPELQDWLRQV RI OHWWHU GLJLWV \$ _
DQG RSHUDWRUV\LNH -DYD DQG PDQ\ ODQJXDJHV D SODLQ
LGHQWLILHU FDQ EHJLQ ZLWK D OHWWHU RU XQGHU VFRUH
IROORZHG E\ PRUH OHWWHU GLJLWV XQGHUVFRUHV DQG
GROODU VLJQV 8QLFRGH HTXLYDOHQW FKDUDFWHUV DUH
DOVR DOORZHG +RZHYHU OLNH -DYD 6FDOD UHVHUYHV WKH
GROODU VLJQ IRU LQWHUQDO XVH VR \RX VKRXOGQ\W XVH
LW LQ \RXU RZQ LGHQWLILHUV \$IWHU DQ XQGHUVFRUH \RX
FDQ KDYH HLWKHU OHWWHU DQG GLJLWV RU D VHTXHQFH RI
RSHUDWRU FKDU FWHUV 7KH XQGHUVFRUH LV LPSRUWDQW
,W WHOOV WKH FRPSLOHU WR WUHDW DOO WKH FKDUDFWHUV
XS WR WKH QH[W ZKLWHVDFH DV SDUW RI WKH LGHQWLILHU
)RU H[DPSOH val xyz_++ = 1 DVVLJQV WKH YDULDEOH xyz_++ WKH
YDOXH 1 ZKLOH WKH H[SUHVVLRQ val xyz++ = 1 ZRQ\W FRPSLOH
EHFDXVH WKH £LGHQWLILHU\ FRXOG DOVR EH LQWHUSUHWHG
DV xyz ++ ZKLFK ORRNV OLNH DQ DWWHPSW WR DSSHQG
VRPHWKLQJ WR xyz 6LPLODUO\ LI \RX KDYH RSHUDWRU FKDU
DFWHUV DIWHU WKH XQGHUVFRUH \RX FDQ\W PL[WKHP ZLWK
OHWWHU DQG GLJLWV 7KLV UH VWULFWLRQ SUHYHQWV
DPELJRXV H[SUHVVLRQV OLNH WKLW abc_=123 ,V WKDW DQ
LGHQWLILHU abc_=123 RU DQ DVVLJQPHQW RI WKH YDOXH 123 WR
abc_"

3ODLQ LGHQWLILHUV\RSHUDWRUV

,I DQ LGHQWLILHU EHJLQV ZLWK DQ RSHUDWRU FKDUDFWHU
WKH UHVW RI WKH FKDUDFWHUV PXVW EH RSHUDWRU
FKDUDFWHUV

£%DFN TXRWH\ OLWHUDOV

\$Q LGHQWLILHU FDQ DOVR EH DQ DUELWUDU\ VWULQJ
VXEMHFW WR SODWIRUP OLPLWDWRQV EH WZHHQ WZR EDFN
TXRWH FKDUDFWHUV H J val `this is a valid identifier` = "Hello World!"
5HFDOO WKDW WKLW \QWD[LV DOVR WKH ZD\ WR LQYRNH D
PHWKRG RQ D -DYD RU 1(7 FODVV ZKHQ WKH PHWKRG\V QDPH
LV LGHQWLFDQ WR D 6FDOD UHVHUYHG ZRUG H J
java.net.Proxy.ÂtypeÂ()

3DWWHUQ PDWFKLQJ LGHQWLILHUV

,Q SDWWHUQ PDWFKLQJ H[SUHVVLRQV WRNHQV WKDW EHJLQ

ZLWK D ORZHUFDVH OHWWHU DUH SDUVHG DV YDULDEOH
 LGHQWLILHUV ZKLOH WRNHQV WKDW EHJLQ ZLWK DQ
 XSSHUFDVH OHWWHU DUH SDUVHG DV FRQVWDQW
 LGHQWLILHUV 7KLV UHVWULFWLRQ SUHYHQWV VRPH
 DPELJXLWLHV EHFDXVH RI WKH YHU\ VXFFLQFW YDULDEOH
 \QWD\ WKDW LV XVHG H J QR val NHZRUG LV SUHVHQW

Syntactic Sugar

2QFH \RX NQRZ WKDW DOO RSHUDWRUV DUH PHWKRGV LW\V
 HDVLHU WR UHDVRQ DERXW XQIDPLOLDU 6FDOD FRGH <RX GRQ\W
 KDYH WR ZRUU\ DERXW VSHFLDO FDVHV ZKHQ \RX VHH QHZ
 RSHUDWRUV :KHQ ZRUNLQJ ZLWK \$FWRUV LQ £\$ 7DVWH RI
 &RQFXUUHQF\ RQ SDJH \RX PD\ KDYH QRWLFHG WKDW ZH XVHG
 DQ H[FODPDWLRQ SRLQW
 ! WR VHQG D PHVVDJH WR DQ \$FWRU 1RZ \RX

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NQRZ WKDW WKH ! LV MXVW DQRWKHU PHWKRG DV DUH WKH
 RWKHU KDQG\ VKRUWFXW RSHUDWRUV \RX FDQ XVH WR WDON WR
 \$FWRUV 6LPLODU\ 6FDOD\ ;0/ OLEUDU\ SURYLGHV WKH \ DQG \\
 RSHUDWRUV WR GLYH LQWR GRFXPHQW VWUXFWXUH 7KHVH DUH
 MXVW PHWKRGV RQ WKH scala.xml.NodeSeq FODV

7KLV IOH\LEOH PHWKRG QDPLQJ JLYHV \RX WKH SRZHU WR ZULWH
 OLEUDULHV WKDW IHHO OLNH D QDWXUDO H\WHQVLRQ RI 6FDOD
 LWVHOI <RX FRXOG ZULWH D QHZ PDWK OLEUDU\ ZLWK QXPHULF
 WSHV WKDW DFFHSW DOO WKH XVXDO PDWKHPDWLFDO
 RSHUDWRUV OLNH DGGLWLRQ DQG VXEWDUFWLRQ <RX FRXOG
 ZULWH D QHZ FRQFXUUHQW PHVVDJLQJ OD\HU WKDW EHKDYHV
 MXVW OLNH \$FWRUV 7KH SRVVLELOLWLHV DUH FRQVWUDLQHG
 RQO\ E\ 6FDOD\ PHWKRG QDPLQJ OLPLDWLRQV

-XVW EHFDXVH \RX FDQ GRHVQ\W PHDQ \RX VKRXOG :KHQ GHVLJQLQJ
 \RXU

RZQ OLEUDULHV DQG \$3,V LQ 6FDOD NHHS LQ PLQG WKDW REVFXUH
 SXQFWXDWRQDO

RSHUDWRUV DUH KDUG IRU SURJUDPPHUV WR UHPHPEHU 2YHUXVH RI WKHVH FDQ
 FRQWULEXWH D £OLQH QRLV\ TXDOLW\ RI XQUHDGDELOLW\ WR \RXU FRGH 6WLFN
 WR

FRQYHQWLRQV DQG HUU RQ WKH VLGH RI VSHOOLQJ PHWKRG QDPHV RXW ZKHQ D
 VKRUWFXW GRHVQ\W FRPH UHDGLO\ WR PLQG

Methods Without Parentheses and Dots

7R IDFLOLWDWH D YDULHW\ RI UHDGDEOH SURJUDPPLQJ VWOHV
 6FDOD LV IOH\LEOH DERXW WKH XVH RI SDUHQWKHVVHV LQ
 PHWKRGV ,I D PHWKRG WDNHV QR SDUDPHWHUV \RX FDQ GHILQH
 LW ZLWKRXW SDUHQWKHVVHV &DOOHUV PXVW LQYRNH WKH
 PHWKRG ZLWKRXW SDUHQWKHVVHV ,I \RX DGG HPSW\
 SDUHQWKHVVHV WKHQ FDOOHUV PD\ RSWLRQDOO\ DGG
 SDUHQWKHVVHV)RU H[DPSOH WKH size PHWKRG IRU List KDV QR

```
SDUHQWKHVHV VR \RX ZULWH List(1, 2, 3).size ,\RX WU\ List(1, 2, 3).size()
\RX\OO JHW DQ HUURU +RZHYHU WKH length PHWKRG IRU
java.lang.String GRHV KDYH SDUHQWKHVHV LQ LWV GHILQLWLRQ
EXW 6FDOD OHWV \RX ZULWH ERWK "hello".length() DQG "hello".length

7KH FRQYHQWLRQ LQ WKH 6FDOD FRPPXQLW\ LV WR RPLW
SDUHQWKHVHV ZKHQ FDOOLQJ D PHWKRG WKDW KDV QR VLGH
IIHFWV 6R DVNLQJ IRU WKH VLJH RI D VHTXHQFH LV ILQH ZLWKRXX
SDUHQWKHVHV EXW GHILQLQJ D PHWKRG WKDW WUDQVIRUPV WKH
HOHPHQWV LQ WKH VHTXHQFH VKRXOG EH ZULWWHQ ZLWK
SDUHQWKHVHV 7KLW FRQYHQWLRQ VLJQDOV D SRWHQWLDOO\
WULFN\ PHWKRG IRU XVHUV RI \RXU FRGH
```

```
,W\I\ DOVR SRVVLEOH WR RPLW WKH GRW
SHULRG ZKHQ FDOOLQJ D SDUDPHWHUOHVV PHWKRG RU RQH
WKDW WDNHV RQO\ RQH DUJXPHQW :LWK WKLW LQ PLQG RXU List(1,
2, 3).size H[DPSOH FRXOG EH ZULWWHQ DV
```

```
// code-examples/Rounding/no-dot-script.scala
```

```
List(1, 2, 3) size
```

```
1HDW EXW FRQIXVLQJ :KHQ GRHV WKLW \QWDFWLFDO
IOH[LELOLW\ EHFRPH XVHIXO" :KHQ FKDLQLQJ PHWKRG FDOOV
WRJHWKHU LQWR H[SUHVVLYH VHOI H[SODQDWRU\ £VHQWHQFHVα
RI FRGH
```

Methods Without Parentheses and Dots | 55

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```
// code-examples/Rounding/no-dot-better-script.scala
```

```
def isEven(n: Int) = (n % 2) == 0
```

```
List(1, 2, 3, 4) filter isEven foreach println
```

```
$V \RX PLJKW JXHV\ UXQQLQJ WKLW SURGXFHV WKH IROORZLQJ RXWSXW

2
4
```

```
6FDOD\I\ OLEHUDO DSSURDFK WR SDUHQWKHVHV DQG GRWV RQ
PHWKRGV SURYLGHV RQH EXLOGQLQJ EORFN IRU ZULWLQJ 'RPDLQ
6SHFLILF /DQJXDJHV :H\OO OHDUQ PRUH DERXW WKHP DIWHU D
EULHI GLVFXVVLQJ RI RSHUDWRU SUHFHGHQFH
```

Precedence Rules

```
6R LI DQ H[SUHVVLQJ OLNH 2.0 * 4.0 / 3.0 * 5.0 LV DFWXDOO\ D VHULHV
RI PHWKRG FDOOV RQ DoubleV ZKDW DUH WKH RSHUDWRU
SUHFHGHQFH UXOHV" +HUH WKH\ DUH LQ RUGHU IURP ORZHVW WR
KLJKHVW SUHFHGHQFH
VHH >6FDOD6SHF @
```

```
$OO OHWWHV
```

```
|
^
```

&
 < >
 = !
 :
 + -
 * / %

\$OO RWKHU VSHFLDO FKDUDFWHUV

&KDUDFWHUV RQ WKH VDPH OLQH KDYH WKH VDPH SUHFHGHQFH
 \$Q H[FHSWLRQ LV = ZKHQ XVHG IRU DVVLJQPHQW ZKHQ LW KDV
 WKH ORZHVW SUHFHGHQFH

6LQFH DQG KDYH WKH VDPH SUHFHGHQFH WKH WZR OLQHV LQ
 WKH IROORZLQJ scala VHVLRQ EHKDYH WKH VDPH

```
scala> 2.0 * 4.0 / 3.0 * 5.0
res2: Double = 13.333333333333332
```

```
scala> ((2.0 * 4.0) / 3.0) * 5.0
res3: Double = 13.333333333333332
```

,Q D VHTXHQFH RI OHIW DVVRFLDWLYH PHWKRG LQYRFDWLRQV
 WKH\ VLPSO\ ELQG LQ OHIW WR ULJKW RUGHU £/HIW
 DVVRFLDWLYH\RX VD" ,Q 6FDOD DQ\ PHWKRG ZLWK D QDPH WKDW
 HQGV ZLWK D FRORQ : DFWXDOO\ ELQGV WR WKH ULJKW ZKLOH
 DOO RWKHU PHWKRGV ELQG WR WKH OHIW)RU H[DPSOH

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